



Internal Use Only

MOBILE PHONE SERVICE MANUAL

CAUTION

BEFORE SERVICING THE UNIT, READ THE "SAFETY PRECAUTIONS" IN THIS MANUAL

MODEL : LG-X155

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1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

C. Attention

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.

Following information is ESD handling:

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

2.1 Band Specification

Support Band	TX Freq (MHz)	RX Freq (MHz)
WCDMA(FDD1)	1920 – 1980	2110 – 2170
WCDMA(FDD8)	880 – 915	925 – 960
EGSM	880 – 915	925 – 960
GSM850	824 – 849	869 – 894
DCS1800	1710 – 1785	1805 – 1880
PCS1900	1850 – 1910	1930 – 1990

2.2 HW Features

List	Type / Spec.	
1. Phone Type	BAR Type	
2. Size	140.8x71.6x9.6mm	
3. Weight	157.5g(with battery)	
4. Battery	2540mAh (Li-Ion)	
5. Chipset	MT6582V/W	
6. Memory	8GB eMMC+1GB LPDDR3, External Memory(SD Card) : Up to 32GB	
7. LCD	Size	5.0 inch
	Dot	480RGB X 854
	Color	16.7M colors
	Display Type	TFT/TRANSMISSIVE
	NTSC	Typ 50%
	Contrast Ratio	Typ. 350:1
	Pixel Pitch	0.1293(W) *0.1293(L)
	Color temperature	6000 ~ 8000(K) (typ. 7000K)
8. Main Camera (5M)	Type	CMOS, Samsung LSI S5K5E2YA03-FGX2 5M sensor
	Resolution	2,560 x 1,920 pixels.
	Frame Rate	Capture – 4:3 30fps FHD Video – 16:9 60fps
	View Angle	64.8(D), 53.9(H), 41.7(V)
	Image Scaling Down	VGA(640*480)
	Pixel x size	1.12um x 1.12um
	Format	1/5" optical format, MIPI RAW10 output format

2.2 HW Features

10. Audio	Receiver	15 X 06 X 2.2T Receiver
	Speaker	18 X 10 X 3.1T Speaker
	Format	MP3, AAC, MIDI, EAAC+, OGG, AMR
11. Bluetooth	Standard	Bluetooth 4.0
	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
12. WLAN	Standard	IEEE 802.11 b/g/n
	Throughput	Max 40Mbps (SDIO Driver performance)
	Depend on environment	0 ~ 50m (depend on environment)
13. GPS	type	A-GPS
14. FM	type	FM Radio, 3.5pi Ear-jack

2.3 RSSI Display

Antenna Bar indicator (3G)				
DBM (RF Antenna measured)		ASU (Unit used by Android for RF signal)		Notification Icon
By LG's specification		Target algorithm for android (Arima modify)	Display for measured RF signal	Antenna display
≥ -85	-85dBm ~	≥ 14	14, 15, 16, ...	5 
≥ -91 & < -85	-91dBm ~ -86dBm	≥ 11 & < 14	11, 12, 13	4 
≥ -97 & < -91	-97dBm ~ -92dBm	≥ 8 & < 11	8, 9, 10	3 
≥ -101 & < -97	-101dBm ~ -98dBm	≥ 6 & < 8	6, 7	2 
≥ -105 & < -101	-105dBm ~ -102dBm	≥ 4 & < 6	4, 5	1 
< -105	~ -106dBm	< 4 or Unknown	0, 1, 2, 3	0 
				OFF

* DBM = -113+2*ASU

Antenna Bar indicator (2G)				
DBM (RF Antenna measured)		ASU (Unit used by Android for RF signal)		Notification Icon
By LG's specification		Target algorithm for android (Arima modify)	Display for measured RF signal	Antenna display
≥ -91	-91dBm ~	≥ 11	11, 12, 13, ...	5 
≥ -95 & < -91	-95dBm ~ -92dBm	≥ 9 & < 11	9, 10	4 
> -99 & < -95	-98dBm ~ -96dBm	> 7 & < 9	8	3 
> -103 & ≤ -99	-102dBm ~ -99dBm	> 5 & ≤ 7	6, 7	2 
> -105 & ≤ -103	-104dBm ~ -103dBm	> 4 & ≤ 5	5	1 
≤ -105	~ -105dBm	≤ 4 or Unknown	0, 1, 2, 3, 4	0 
				OFF

2.4 Current consumption

Item	Specification	
	WCDMA	GSM
1. Sleep Mode	Under 6.0mA,@ BAT4V,DRX7	Under 6mA@ P.P 5
2. Sleep : connector Ear jack	Under 6.0mA,@ BAT4V,DRX7	Under 6mA@ P.P 5
3. Current(Sleep & Idle AVG)	Under 6.0mA,@ BAT4V,DRX7	Under 6mA @ P.P 5
4. BT Connected idle	Under 7.5mA	Under 7.5mA
5. Talk Mode	≤260mA @10dBm ≤270mA @12dBm	1) GSM900/GSM850 ≤340mA@PCL5
		2) DCS1800/PCS1900 ≤280mA@PCL0
6. No service	N/A	NA
7. Power Off Mode	Under 200 uA	Under 200 uA
8. Backup Battery	N/A	N/A

2.5 Battery bar

7	100%-91%	Blue bar	
6	90%-76%	Blue bar	
5	75%-61%	Blue bar	
4	60%-50%	Blue bar	
3	49%-36%	Blue bar	
2	35%-16%	Blue bar	
1	15%-5%	Orange bar	
0	4%-0%		
Low battery warning	15%		
Forced shutdown	3.4± 0.05 V	Power OFF	

2.6 SW Specification

Item	Feature	Comment
RSSI	0 ~ 5 Levels	Yes
Battery Charging	0 ~ 20 Levels	Percentage dispalying
Key Volume	0 ~ 7 Level	Not adjustable
Audio Volume	0 ~ 15 Level	Yes
Time / Date Display	Yes	Yes
Multi-Language	Yes	depending on build language
Quick Access Mode	Phone / Messaging / Browser/ Applications	Phone / Contact / Applications/ Bowser/ Messaging
PC Sync	No	No
Speed Dial	Yes	Voice mail center -> Long press 1 key
Profile	Yes	not same with feature phone setting
CLIP / CLIR	Yes	Yes
Phone Book	Name / Number / Email / Groups / Postal addresses / Organizations / IM / Note / Nickname / Website / Event /	There is no limitation on the number of items. It depends on available memory amount.
Last Dial Number	Yes	Yes
Last Received Number	Yes	Yes
Last Missed Number	Yes	Yes
Search by Number/Name	Yes	Yes
Group	Yes	There is no limitation on the number of items. It depends on available memory amount.
Fixed Dial Number	Yes	Yes
Service Dial Number	Yes	Depend on SIM
Own Number	Yes	My Profile (add/edit/delete are supported)

2.6 SW Specification

Voice Memo	Yes	Support voice recorder
Call Reminder	No	No
Network Selection	Automatic	Yes
Mute	Yes	Yes
Call Divert	Yes	Yes
Call Barring	Yes	Dependong operator
Call Charge (AoC)	No	No
Call Duration	Yes	Yes
SMS (EMS)	There is no limitation on the number of items. It depends on available memory amount.	No
SMS Over GPRS	Yes	Yes
EMS Melody / Picture Send / Receive / Save	No	No
MMS MPEG4 Send / Receive / Save	Yes	➤ Send / Receive : Yes ➤ Save : depends on content typeSupport video content type list 1. video/mp4 2. video/h263 3. video/3gpp2 - video/3gpp
Long Message	MAX 2000 characters	The standard of Open vender
Cell Broadcast	Yes	Yes
Download	Over the Web	Yes
Game	No	depend on CA preloading
Calendar	Yes	Yes
Memo	Yes	There is no limitation on the number of items. It depends on available memory amount.
World Clock	Yes	Yes

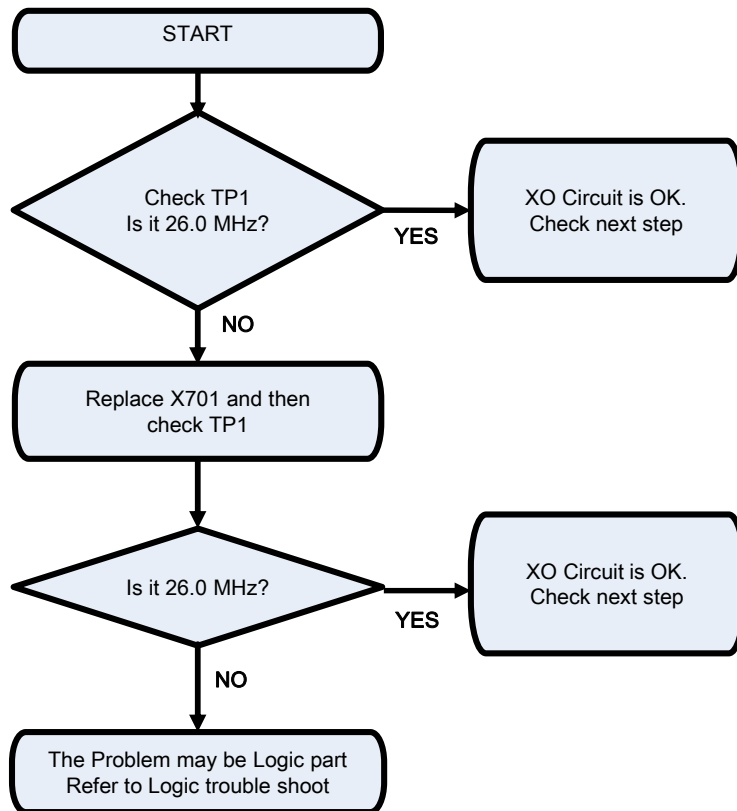
2.6 SW Specification

Unit Convert	No	No
Stop Watch	Yes	Yes
Wall Paper	Yes	Yes
WAP Browser	No	Support only web browser based on webkit. WAP stack and wml are not supported.
Download Melody / Wallpaper	Yes	Over web browser
SIM Lock	No	depend on operator
SIM Toolkit	Yes	Yes
MMS	Yes	Google MMS Client
EONS	Yes	Yes
CPHS	Yes	V4.2
ENS	No	No
Camera	Yes	5M AF / Digital Zoom : x4
JAVA	No	
Voice Dial	No	US English only
IrDa	No	No
Bluetooth	Yes	Ver. 4.0
FM radio	Yes	Yes
GPRS	Yes	Class 12
EDGE	Yes	Class 12(Rx only)
Hold / Retrieve	Yes	Yes
Conference Call	Yes	Max. 6
DTMF	Yes	Yes
Memo pad	No	No
TTY	No	No
AMR	Yes	Yes
SyncML	No	No
IM	Yes	Google Hangout
Email	Yes	Yes

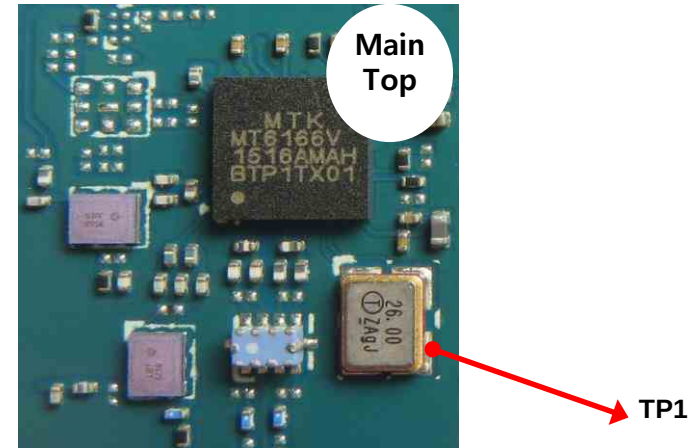
3.1 Checking XO Block

The out put frequency(26.0MHz) of XO(X701) is used as the reference one of MT6166V and MT6582V/W internal VCO

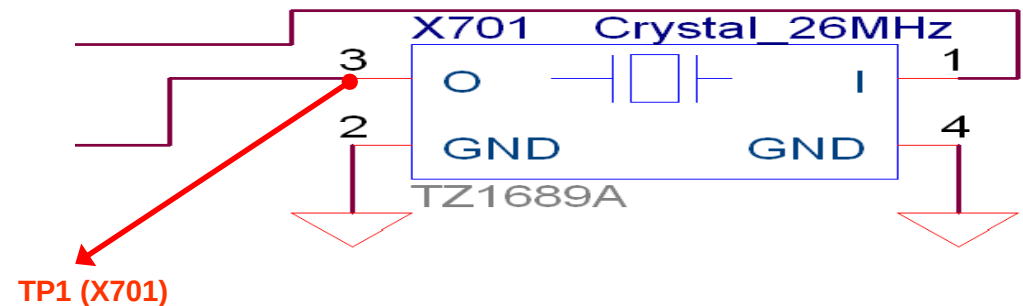
Checking Flow



Image



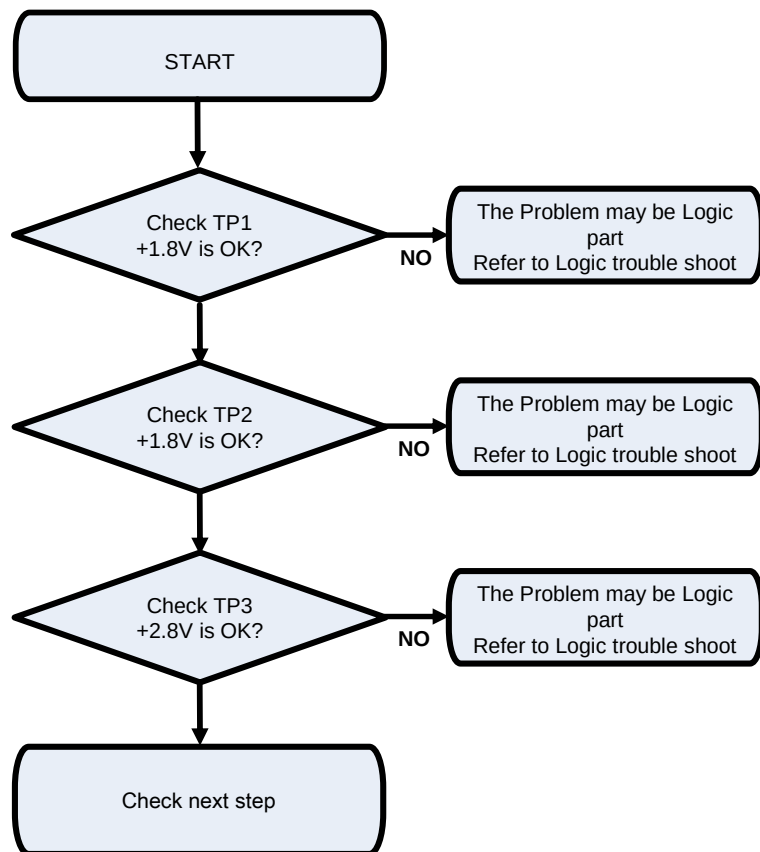
Circuit Diagram



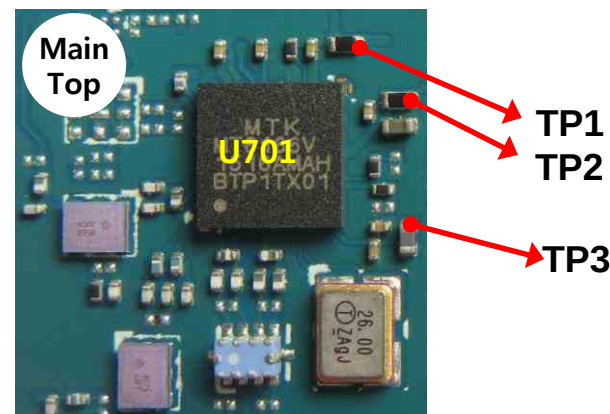
3.2 Checking Transceiver DC Power Supply Circuit Block

The MT6166V operating voltages used three voltage sources 1.8V and 1.8V and 2.8V

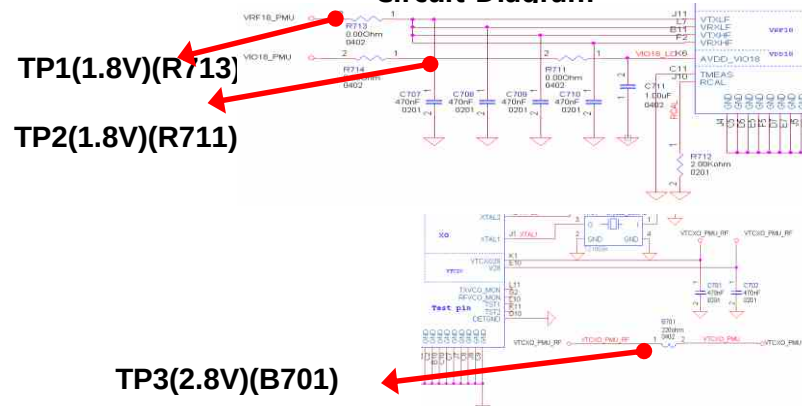
Checking Flow



Image



Circuit Diagram



3.3 FEM (Front-End Module) Block

3.3.1 . FEM Mode Control Logic Table

FEM Mode Control Logic Table

Table 3. SKY77589-11 Mode Control Logic

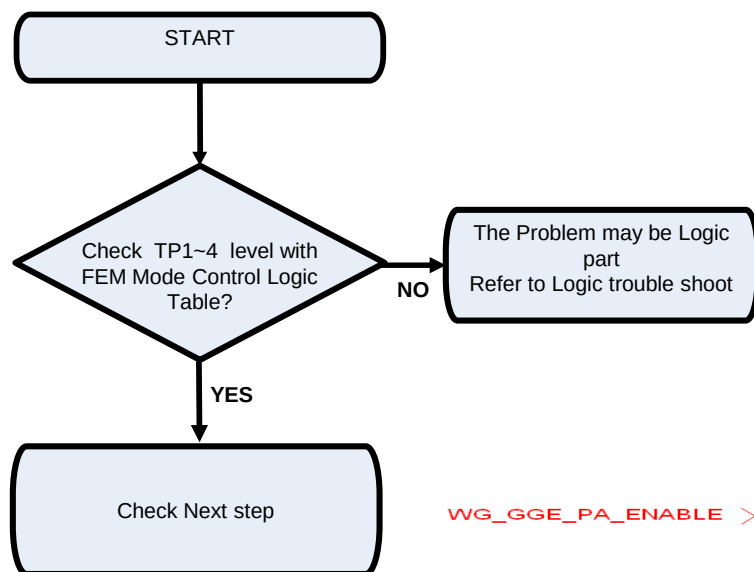
Mode	Input Control Bits			
	TxEN	MODE	BS1	BS2
Standby	0	0	0	0
LB_GMSK_Tx 2G_LB_TX	1	0	0	1
HB_GMSK_Tx 2G_HB_TX	1	0	1	1
TRx1 2G_HB_RX	0	1	0	0
TRx2 2G_LB_RX	0	1	1	0
TRx3	0	1	0	1
TRx4 3G_BAND_V	0	1	1	1
TRx5	0	0	1	0
TRx6 3G_BAND_I	0	0	0	1

3. Troubleshooting

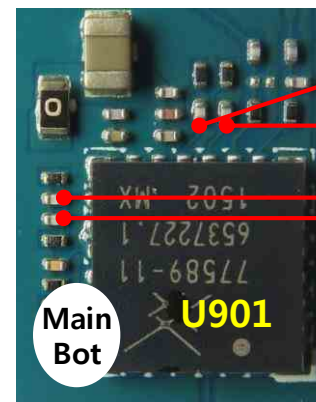
3.3 FEM (Front-End Module) Block

3.3.2 . All Band FEM Block (GSM 850/900/1800/1900, WCDMA B1/5)

Checking Flow

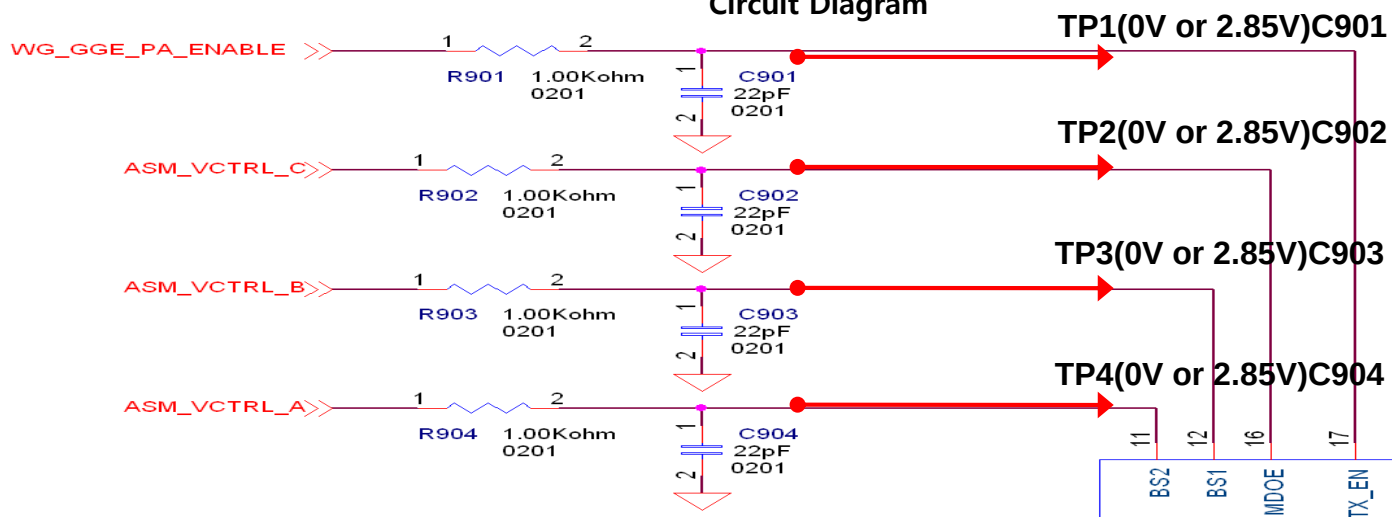


Image



TP3(0V or 2.85V)
TP4(0V or 2.85V)
TP2(0V or 2.85V)
TP1(0V or 2.85V)

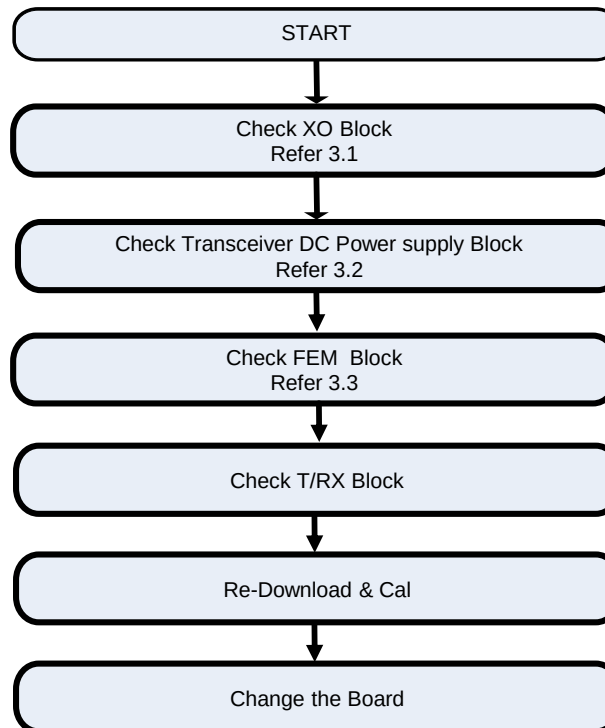
Circuit Diagram



3.4 GSM RF PART

GSM RF Part support GSM850/900/1800/1900 with ASM, PAM, Transceiver component

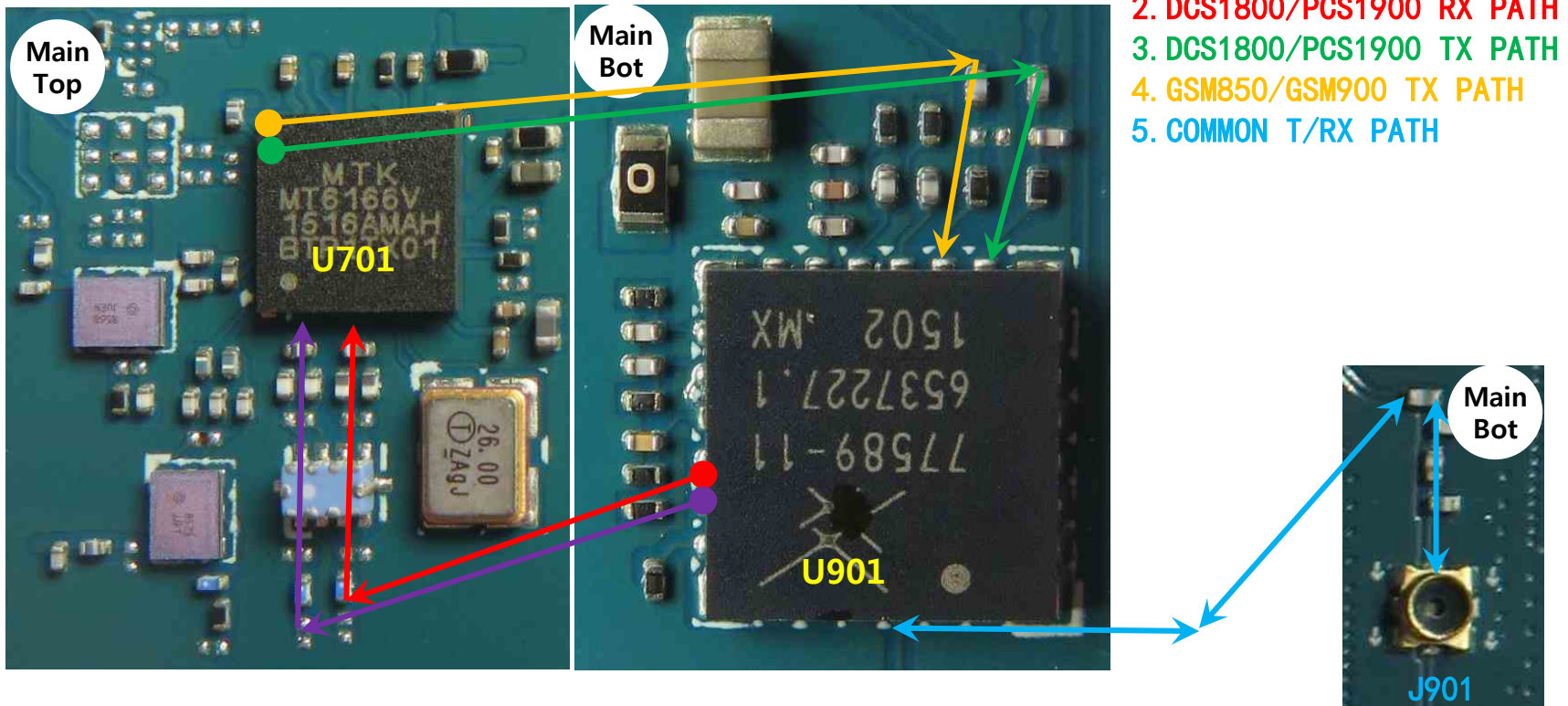
Checking Flow



3.4 GSM RF PART

3.4.1 GSM850/900/1800/1900 Rx

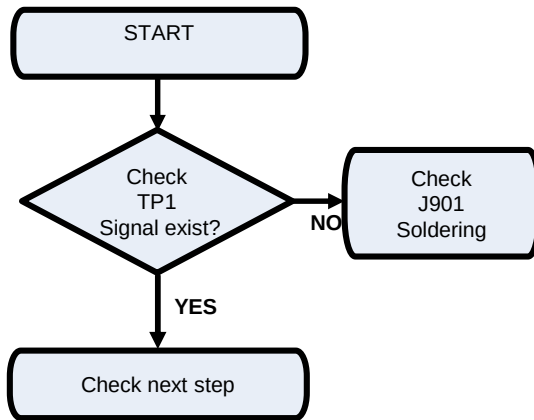
Image



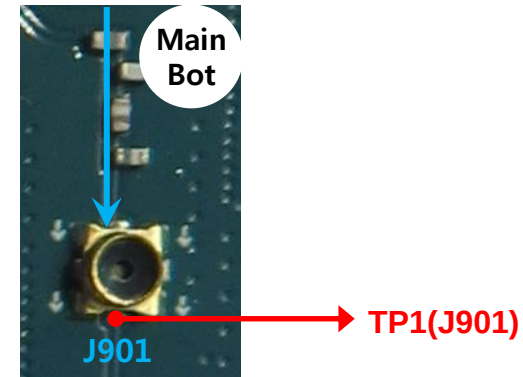
3.4 GSM RF PART

3.4.1.1 Checking RF signal path (SW)

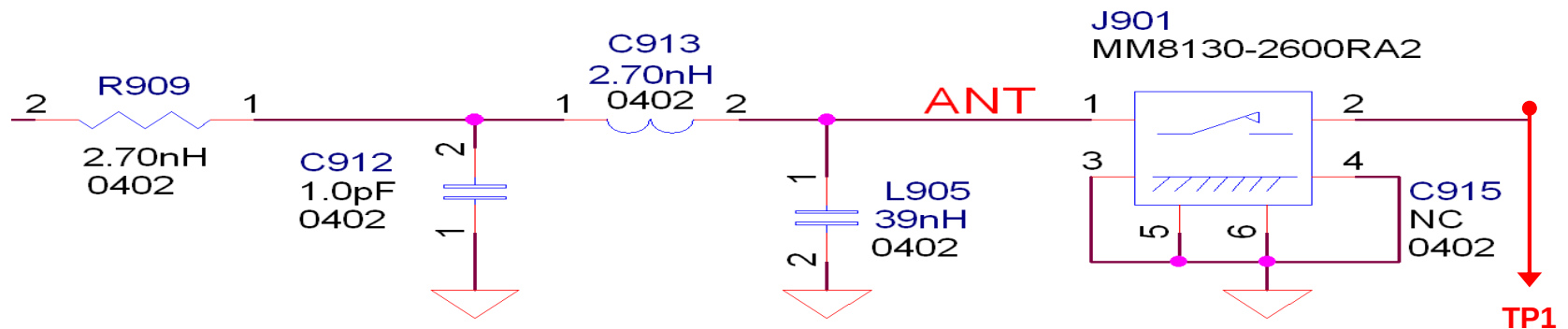
Checking Flow



Image



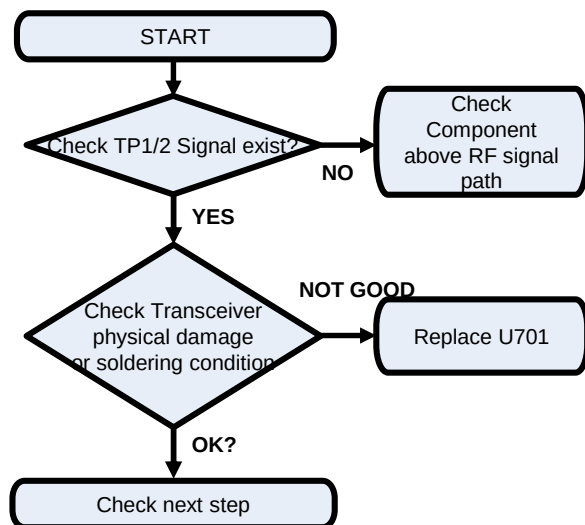
Circuit Diagram



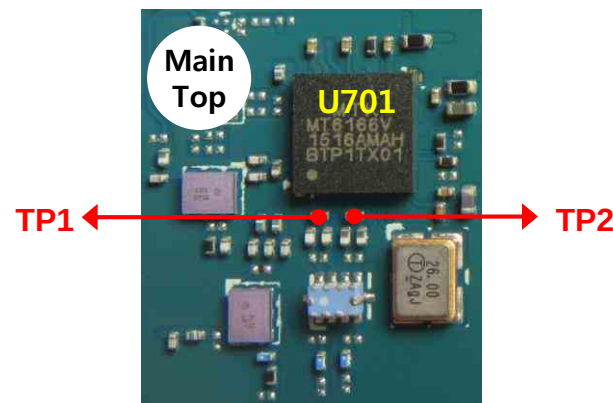
3.4 GSM RF PART

3.4.1.2 Checking RF Signal RX path(GSM850/GSM900/DCS1800/PCS1900)

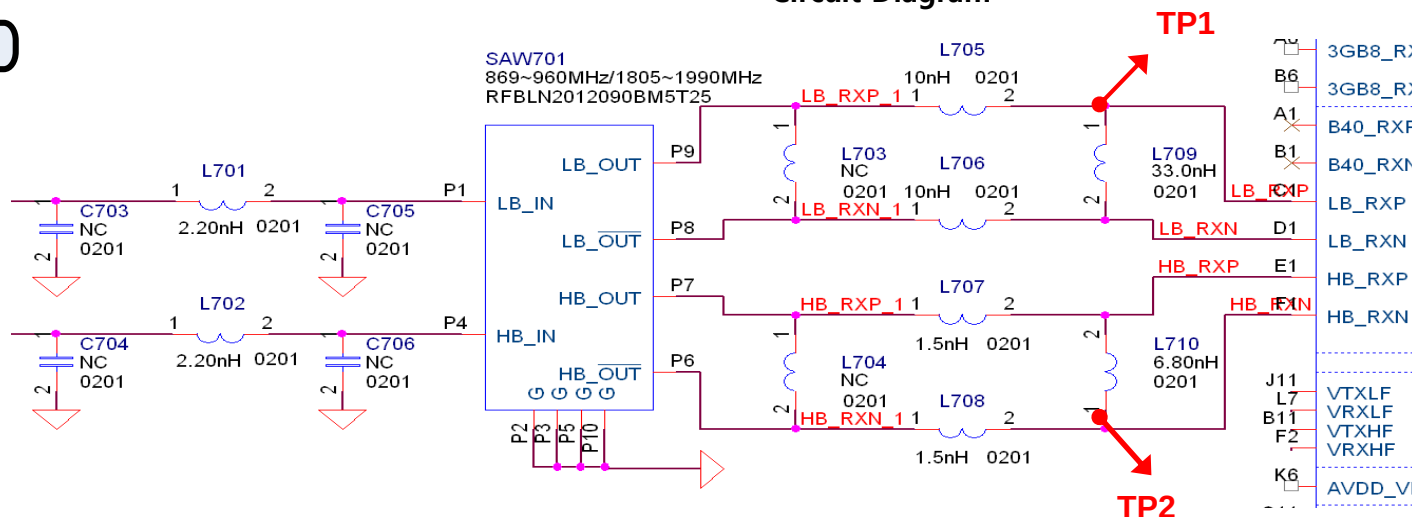
Checking Flow



Image



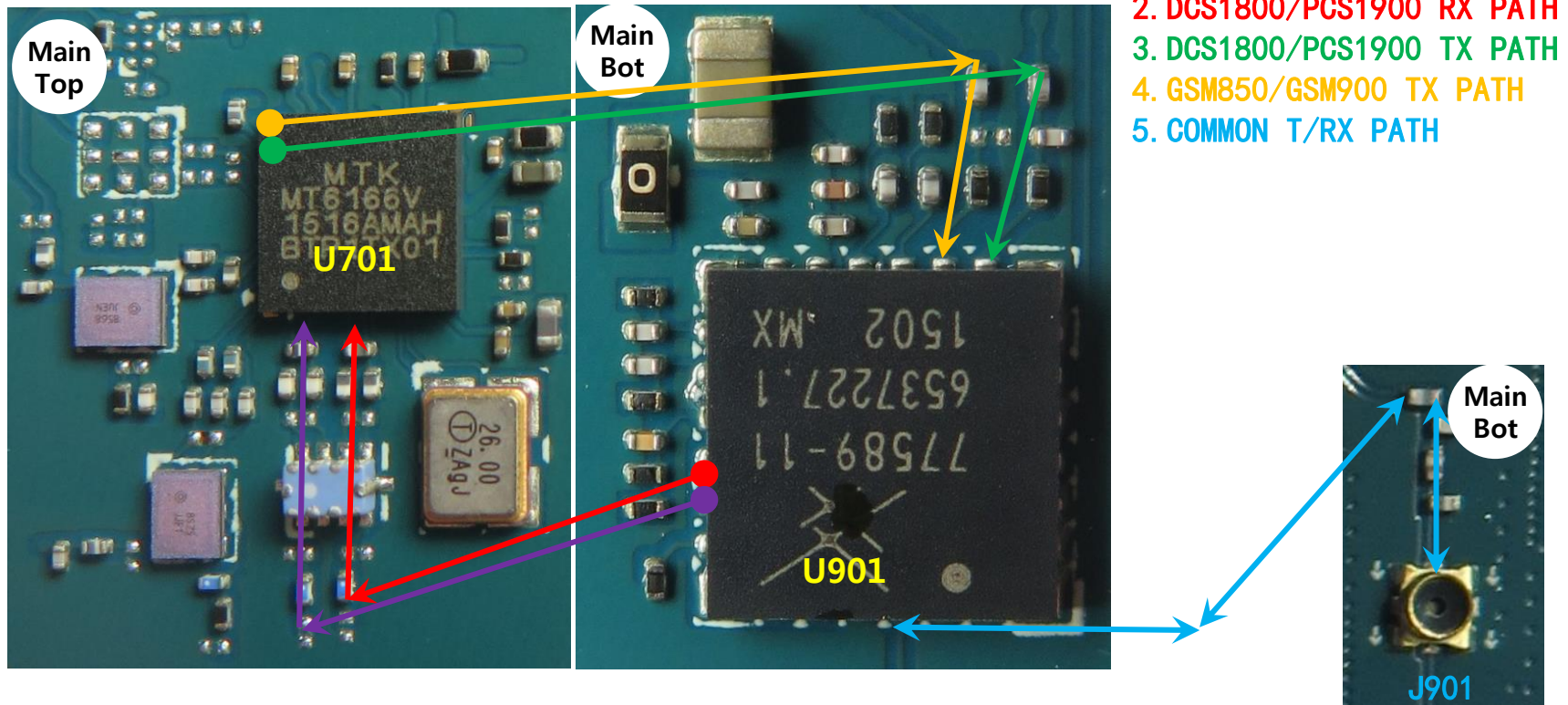
Circuit Diagram



3.4 GSM RF PART

3.4.2.1 GSM850/900/1800/1900 Tx

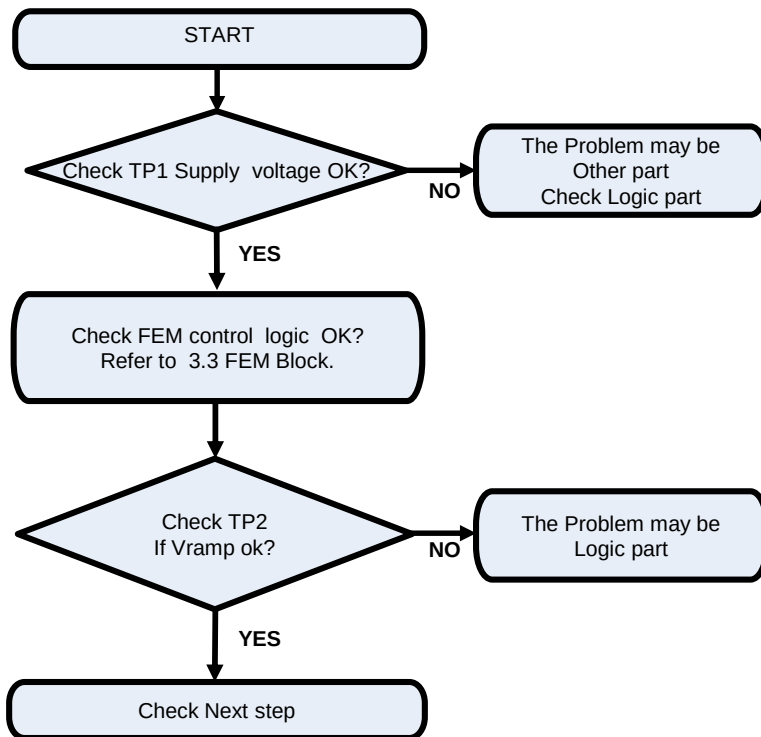
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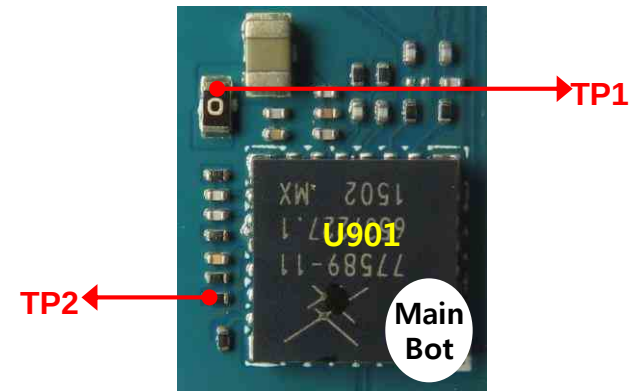
3.4 GSM RF PART

3.4.2.2 Checking GSM PAM DC Power Circuit

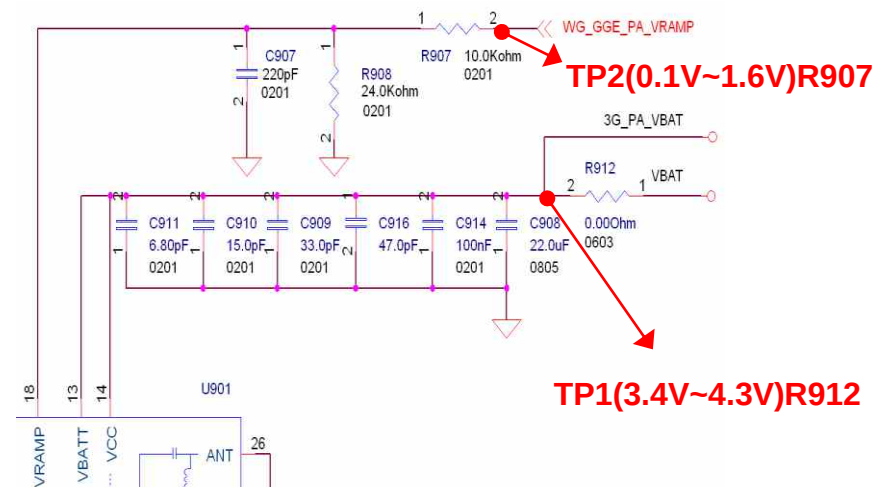
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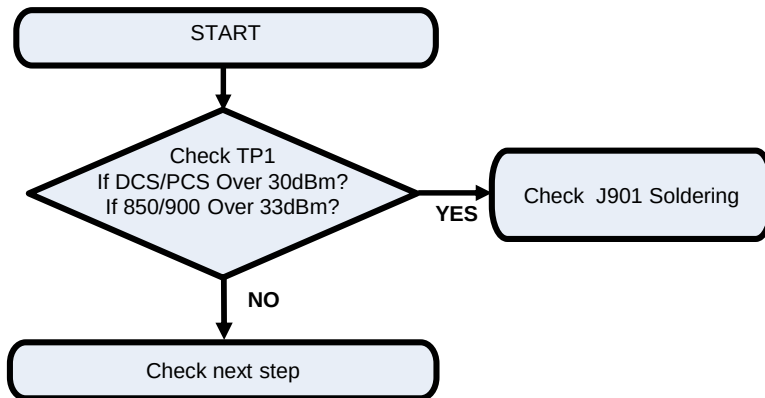
Circuit Diagram



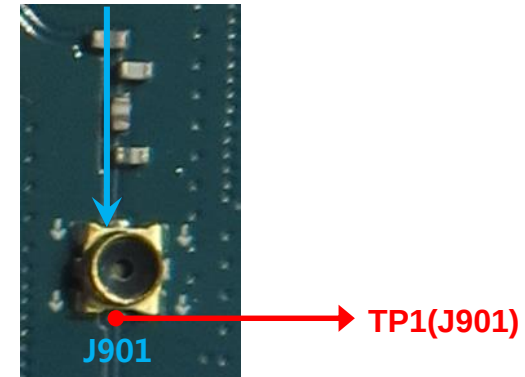
3.4 GSM RF PART

3.4.2.3 Checking RF signal path(SW)

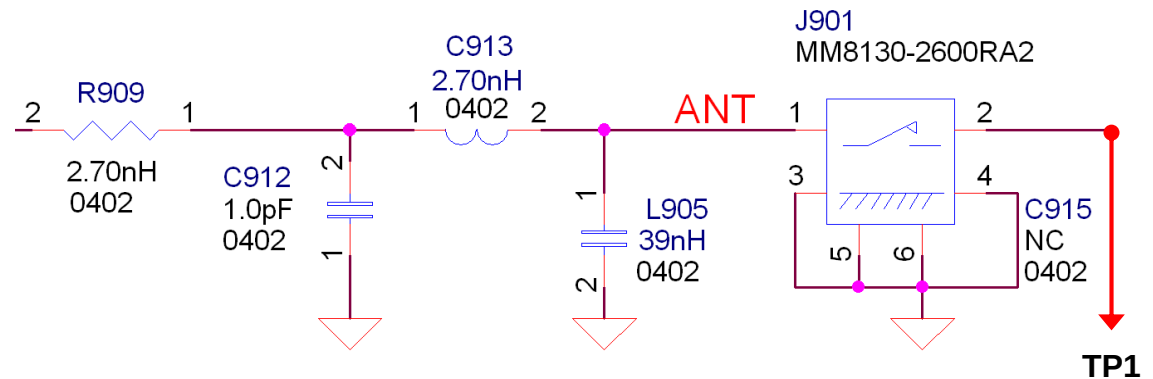
Checking Flow



Image



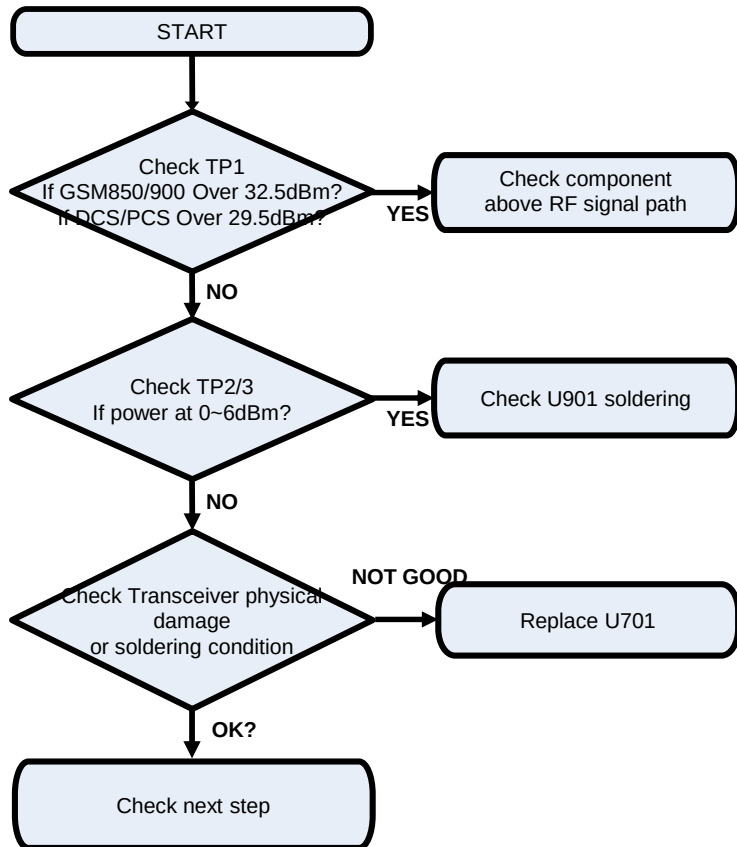
Circuit Diagram



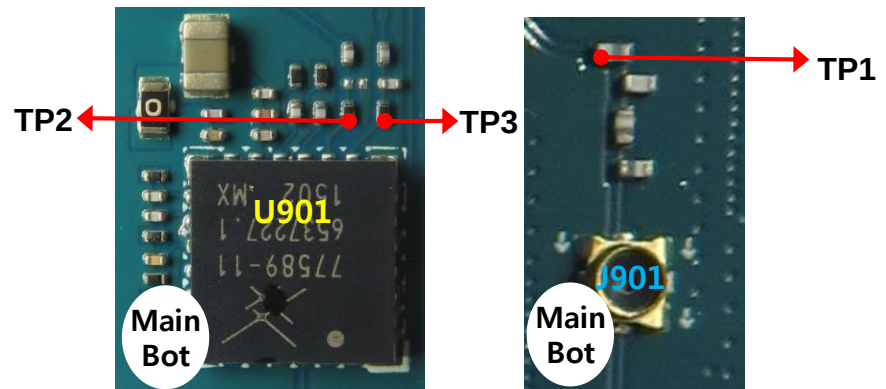
3.4 GSM RF PART

3.4.2.4 Checking RF signal path

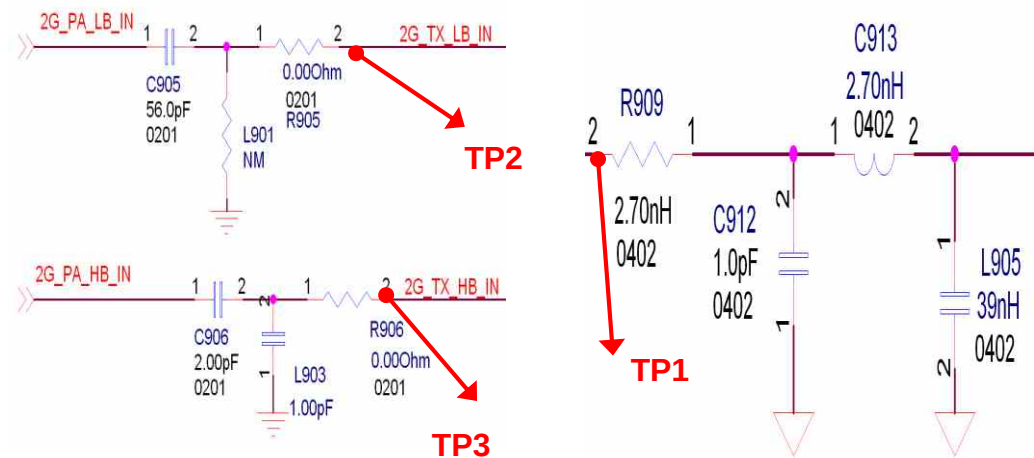
Checking Flow



Image



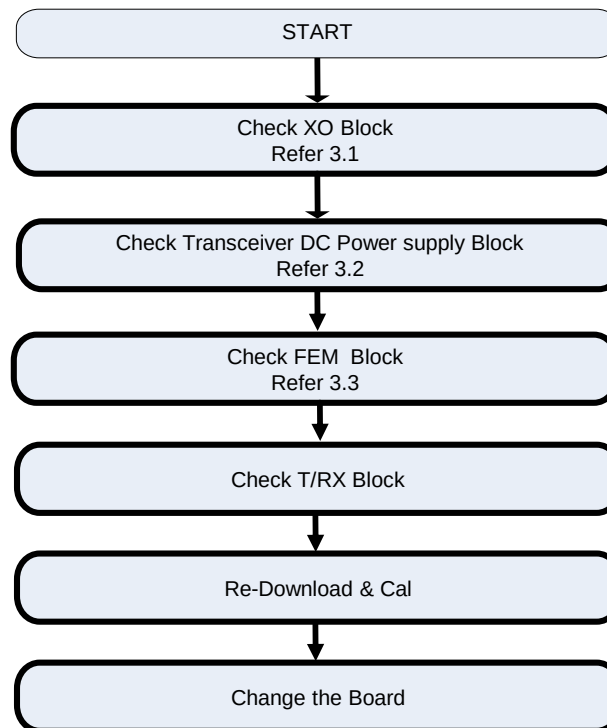
Circuit Diagram



3.5 WCDMA RF Part

WCDMA RF Part support WCDMA B1/B8 with FEM, PAM, Transceiver component

Checking Flow

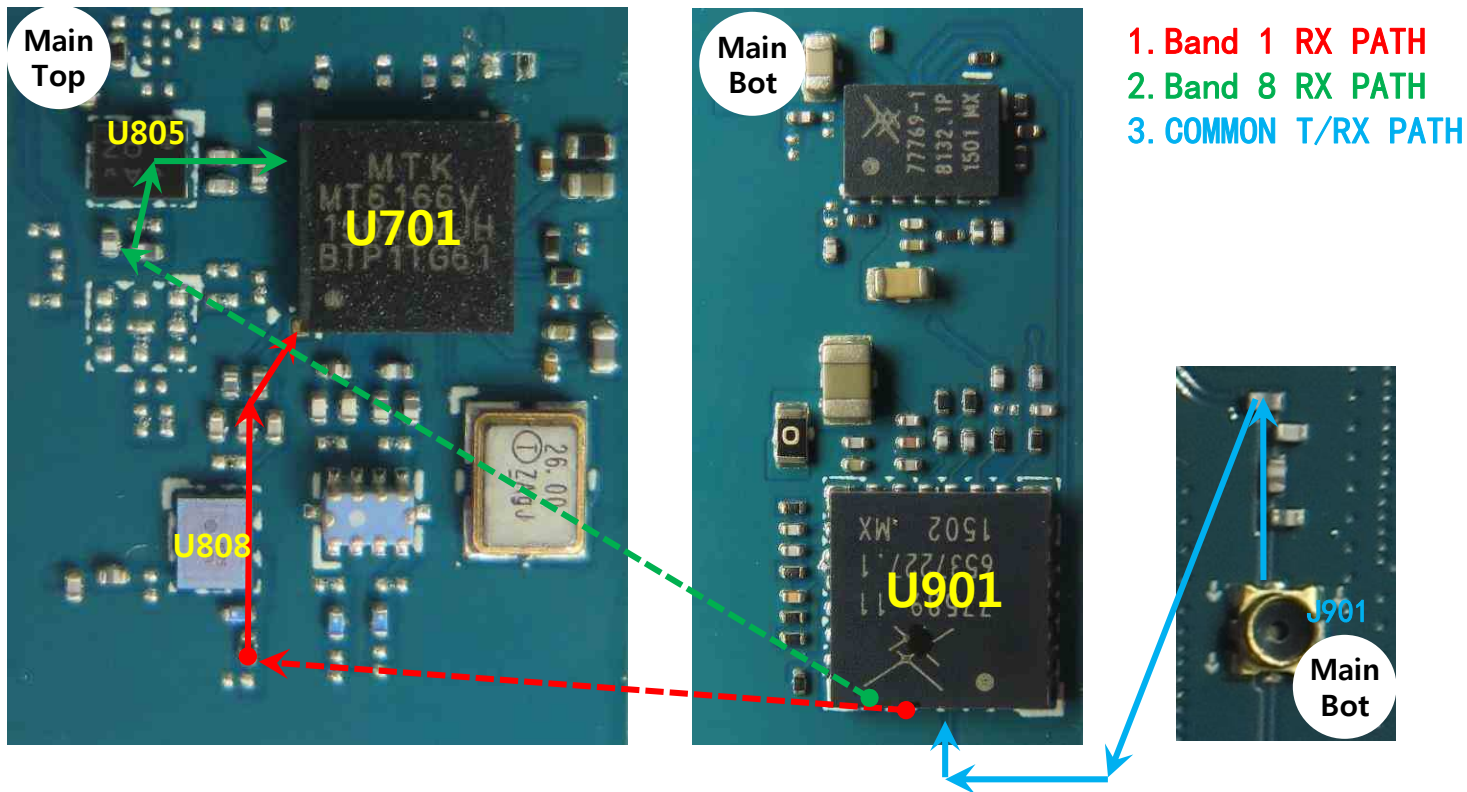


3.5 WCDMA RF PART

3.5.1 WCDMA RF Part RX PATH (B1/B8)

Image

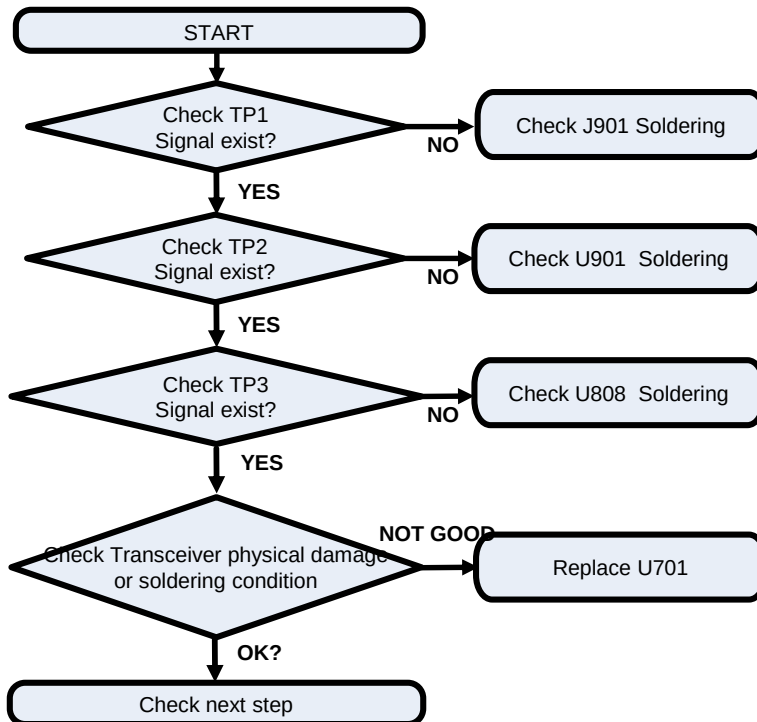
1) WCDMA RX Path



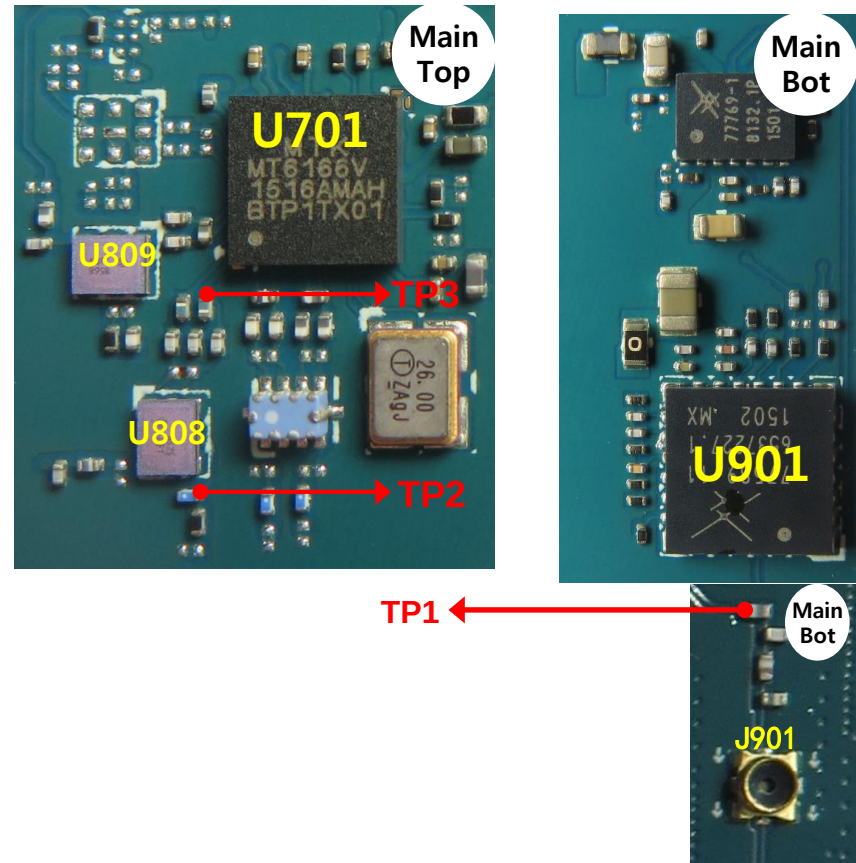
3.5 WCDMA RF PART

3.5.1.1 Checking RF signal path(Band 1)

Checking Flow



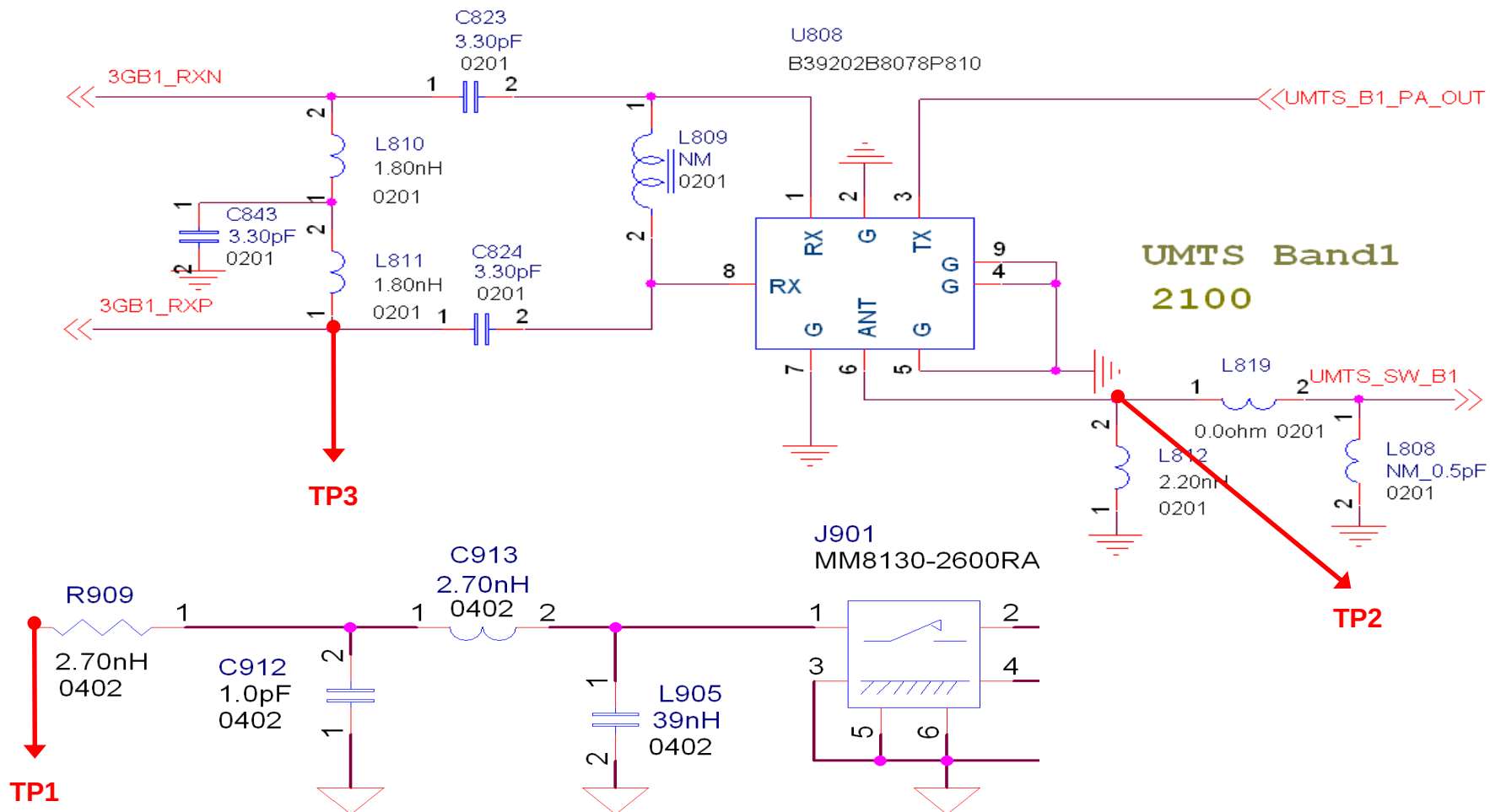
Image



3.5 WCDMA RF PART

3.5.1.1 Checking RF signal path(B1)

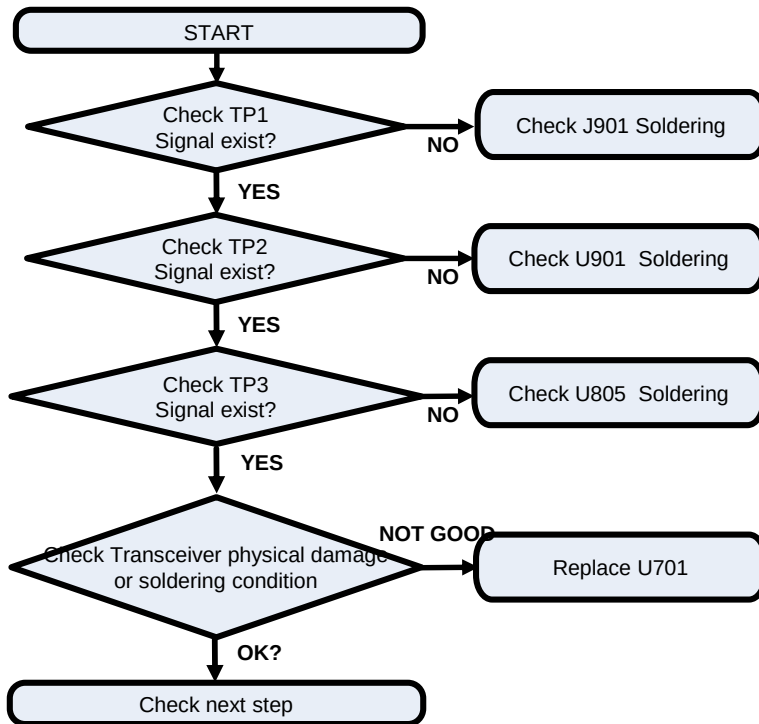
Circuit Diagram



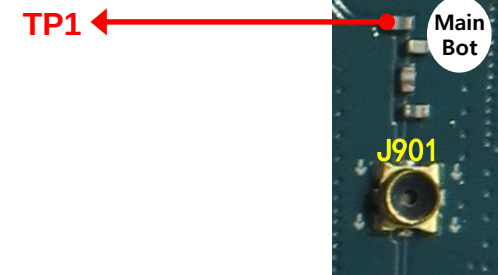
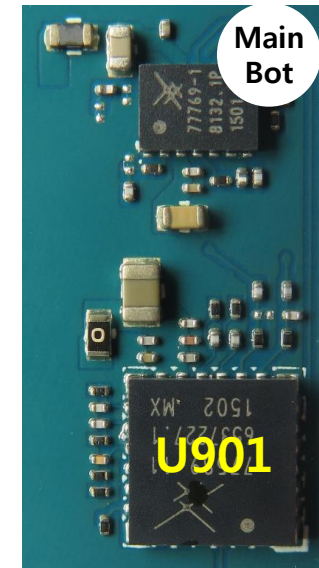
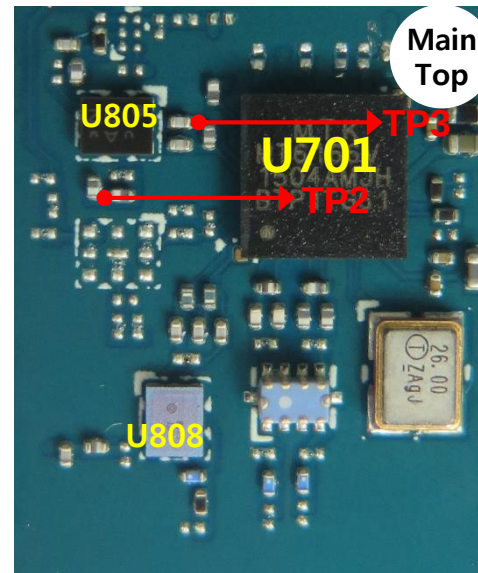
3.5 WCDMA RF PART

3.5.1.2 Checking RF signal path(Band 8)

Checking Flow



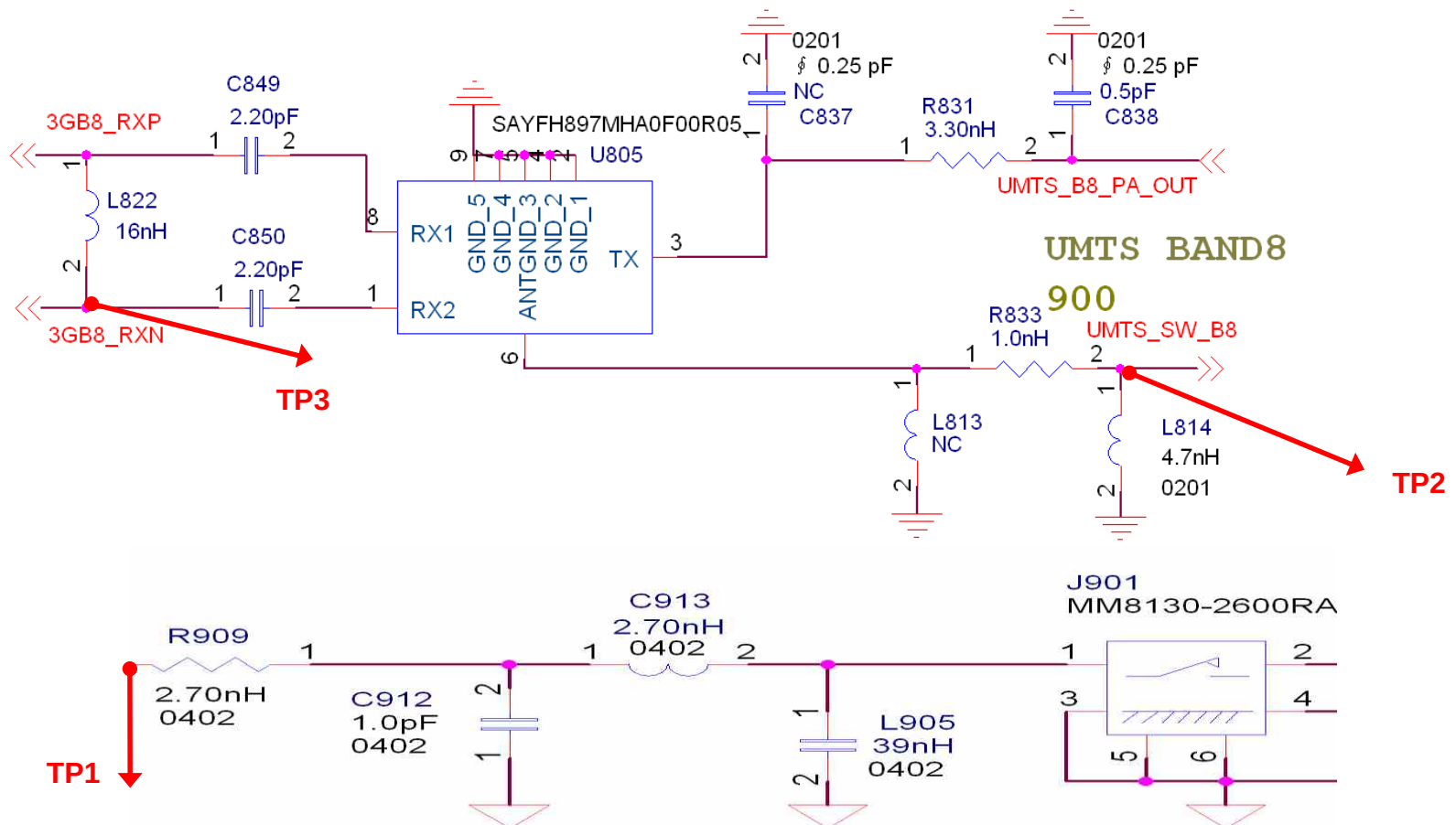
Image



3.5 WCDMA RF PART

3.5.1.2 Checking RF signal path(Band 8)

Circuit Diagram

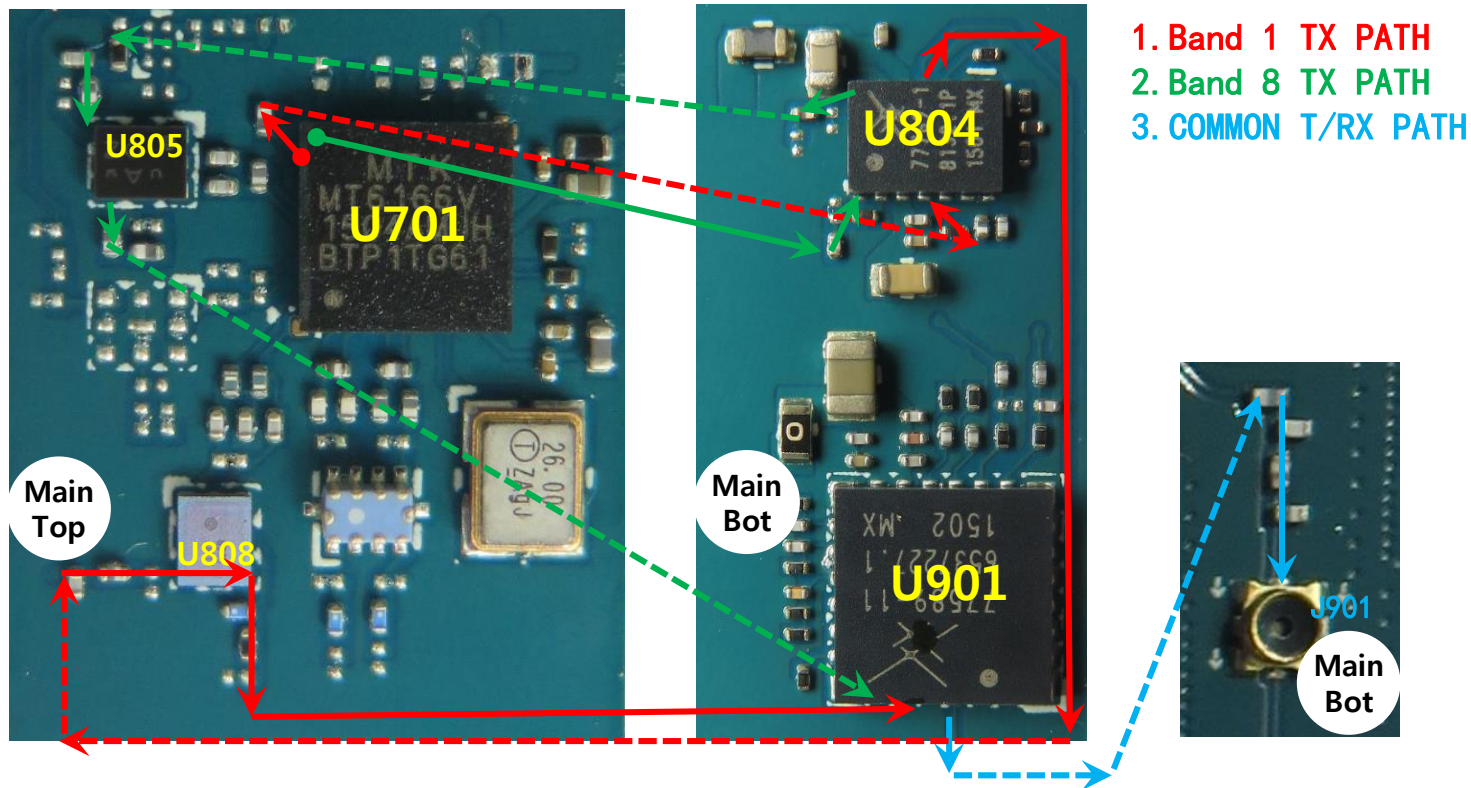


3.5 WCDMA RF PART

3.5.2 WCDMA RF Part TX PATH (B1/B8)

2) WCDMA TX Path

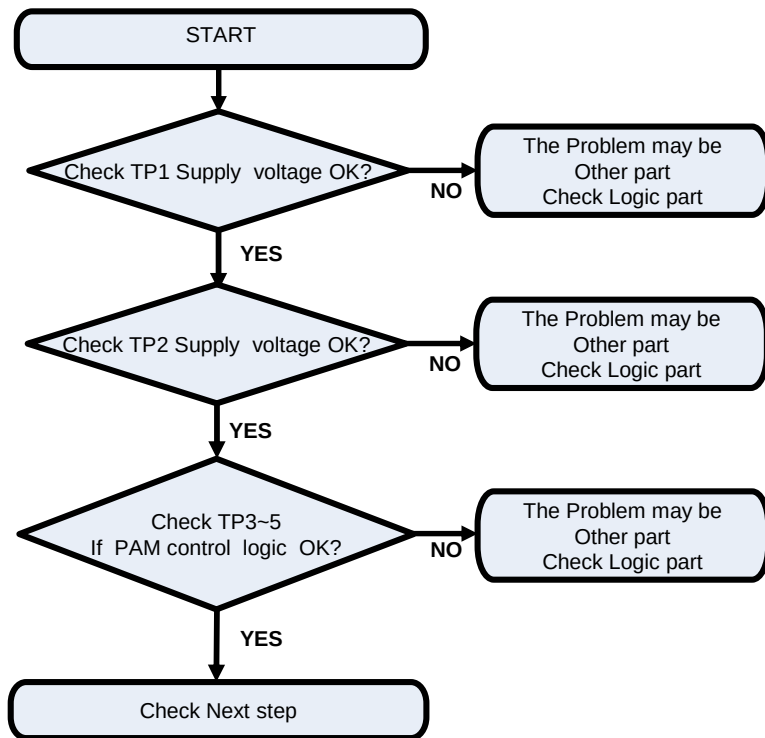
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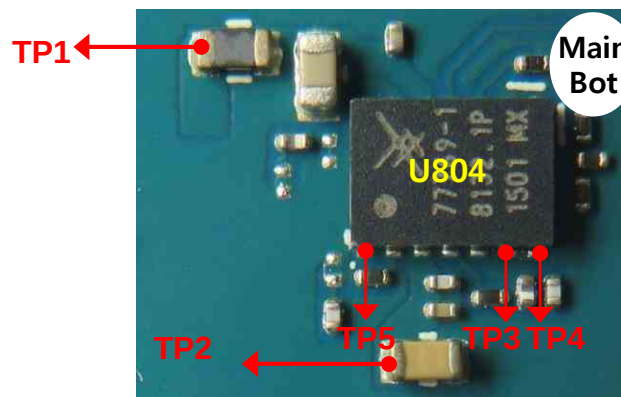
3.5 WCDMA RF PART

3.5.2.1 Checking WCDMA PAM DC Power & Control Circuit

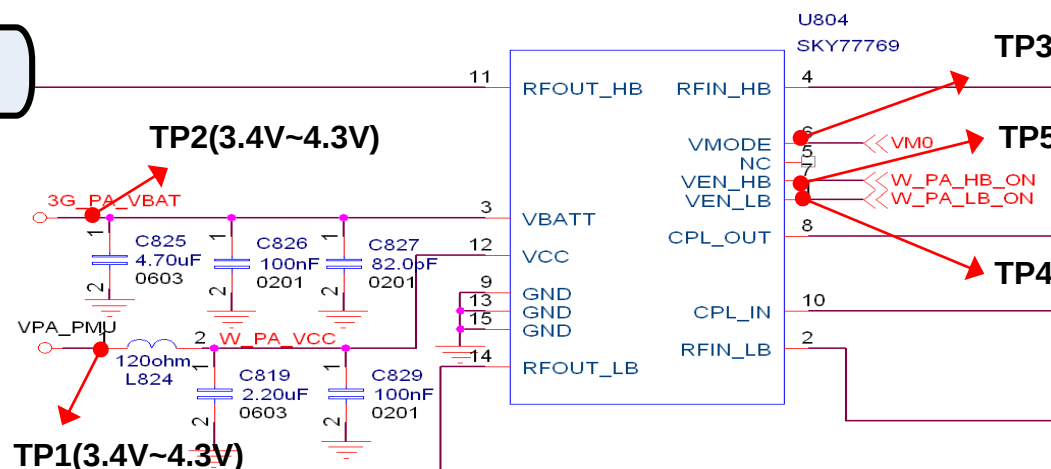
Checking Flow



Image



Circuit Diagram



3.5 WCDMA RF PART

3.5.2.1 Checking WCDMA PAM DC Power & Control Circuit

U901 PAM CONTROL LOGIC

TABLE 3. SKY77769 MODES OF OPERATION

Power Setting	Band	VENHB	VENLB	VMODE	VBATT
Power Down Mode		Low	Low	Low	On
Standby Mode		Low	Low	X	On
Low Power Mode ($P_{OUT} \leq 17 \text{ dBm}$)	I, II, IV	High	Low	High	On
High Power Mode ($P_{OUT} = 17 \text{ dBm to } P_{MAX}$)	I, II, IV	High	Low	Low	On
Low Power Mode ($P_{OUT} \leq 17 \text{ dBm}$)	V, VIII	Low	High	High	On
High Power Mode ($P_{OUT} = 17 \text{ dBm to } P_{MAX}$)	V, VIII	Low	High	Low	On

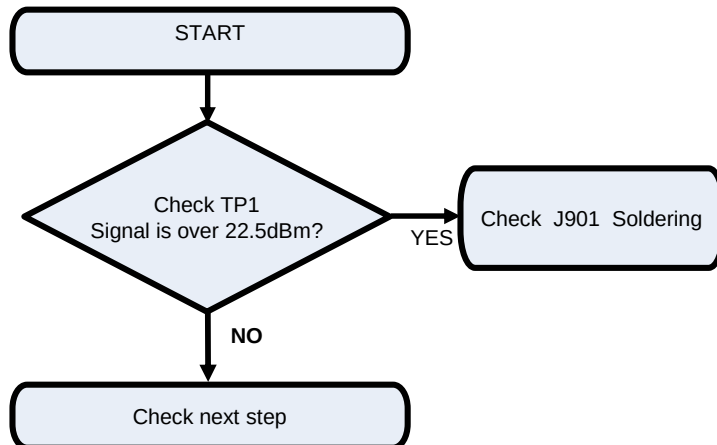
3.5 WCDMA RF PART

3.5.2.2 Checking RF signal path(SW)

Checking Flow

Image

Refer to 3.4.1.1



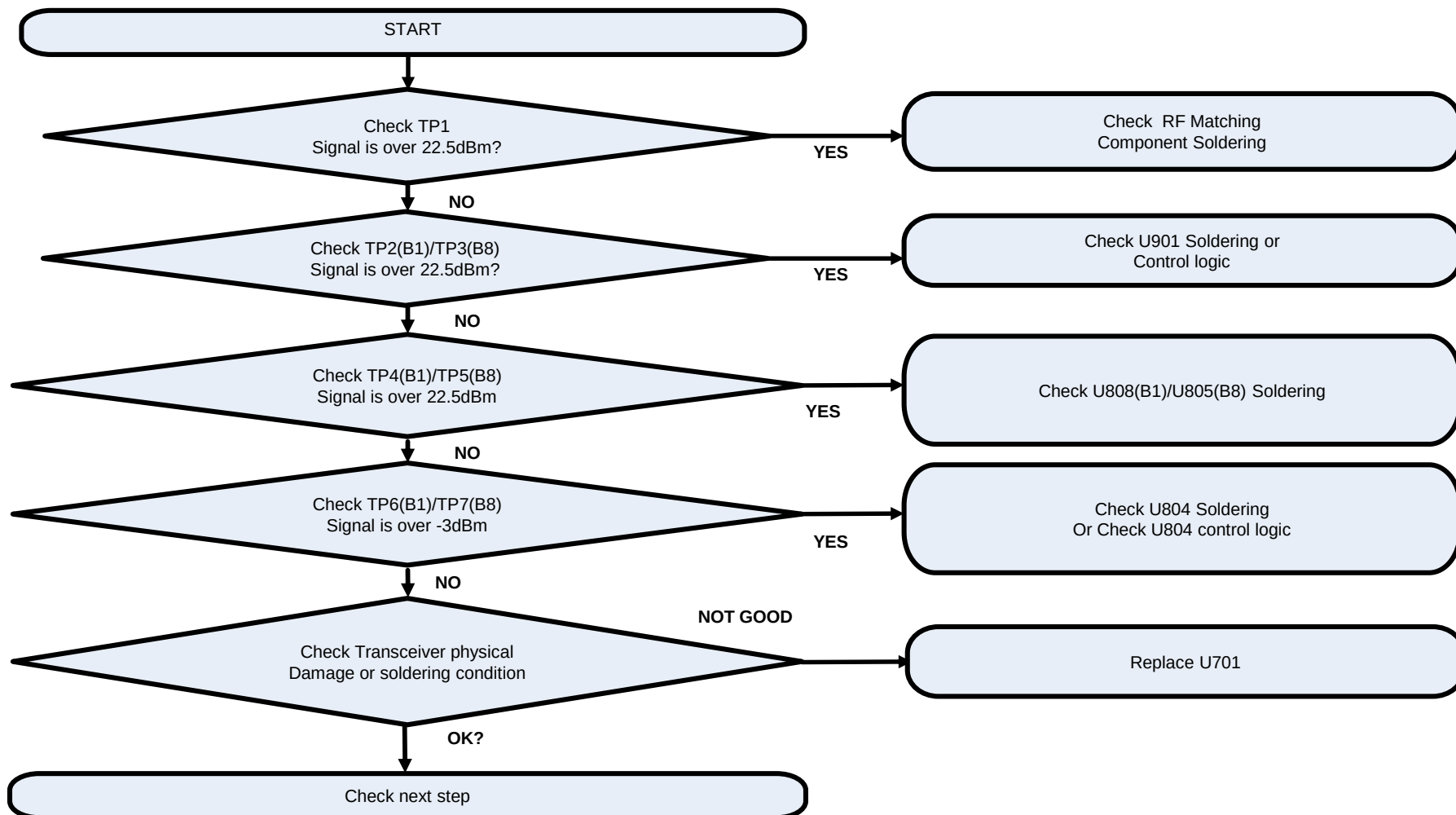
Circuit Diagram

Refer to 3.4.1.1

3.5 WCDMA RF PART

3.5.2.3 Checking RF signal path

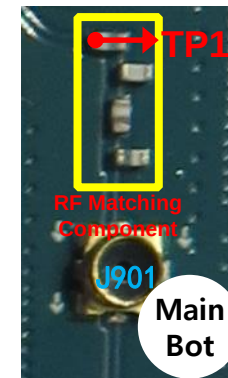
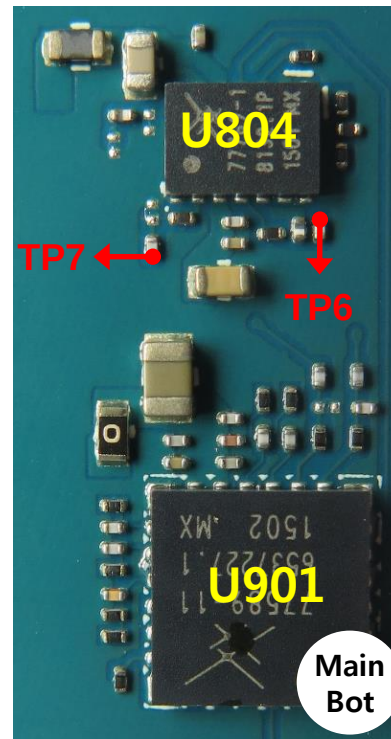
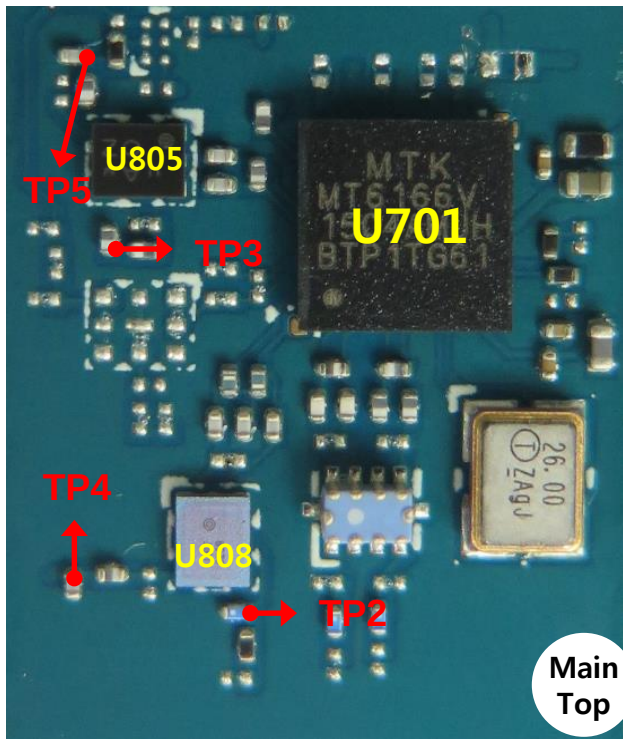
Checking Flow



3.5 WCDMA RF PART

3.5.2.3 Checking RF signal path

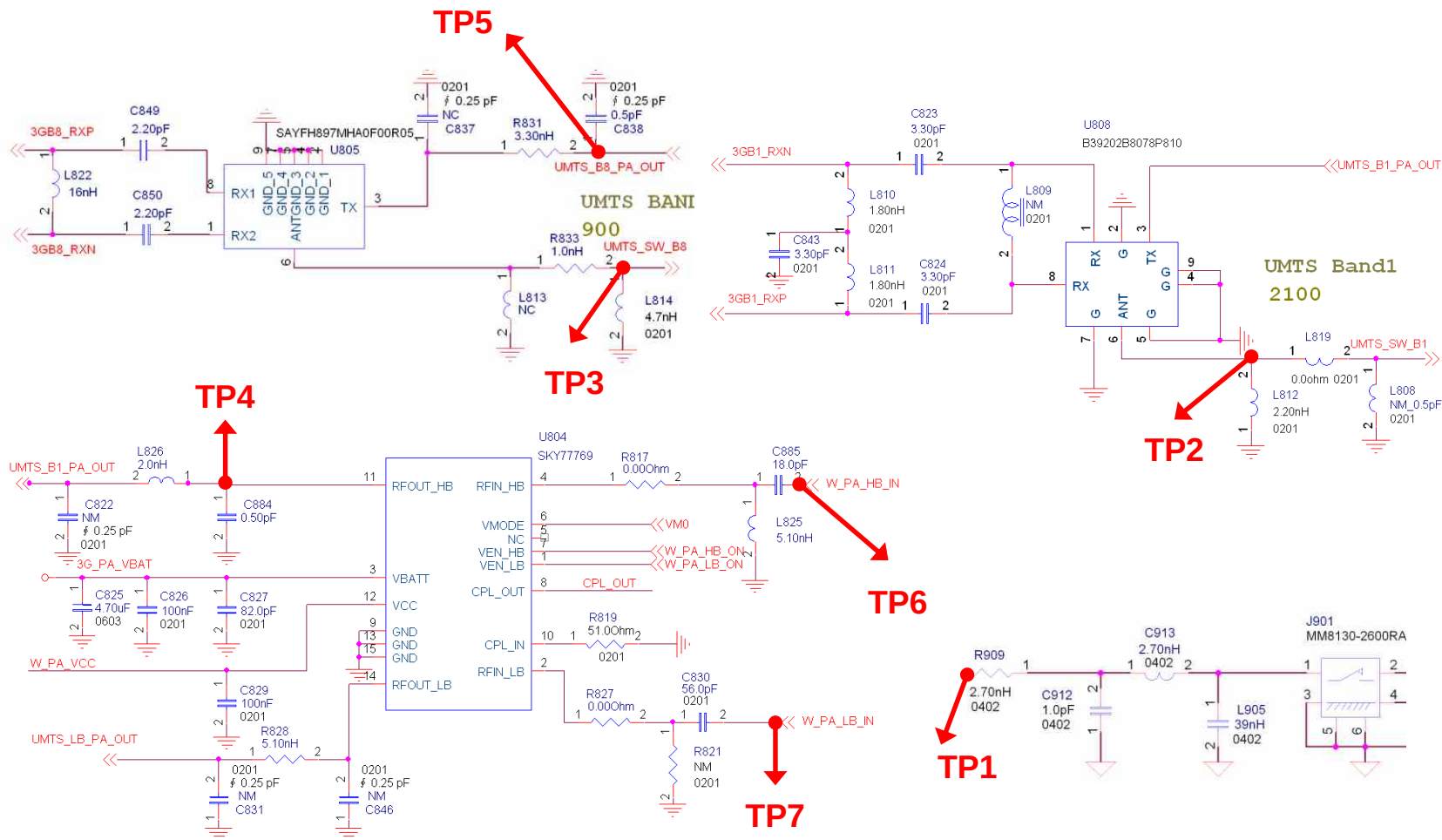
Image



3.5 WCDMA RF PART

3.5.2.3 Checking RF signal path

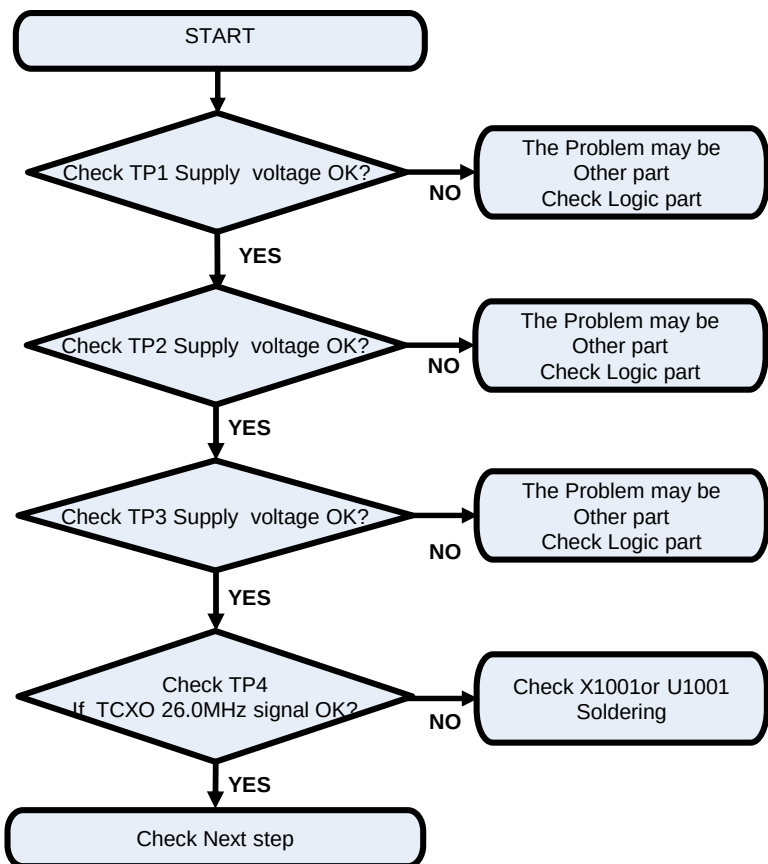
Circuit Diagram



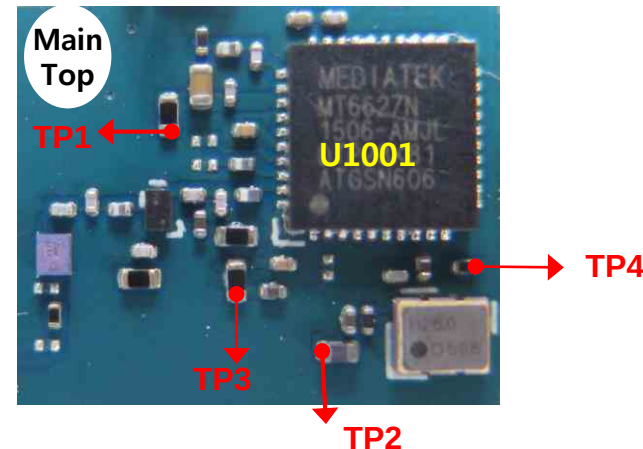
3.6 Connectivity PART

3.6. 1. Checking Connectivity DC Power & TCXO Circuit

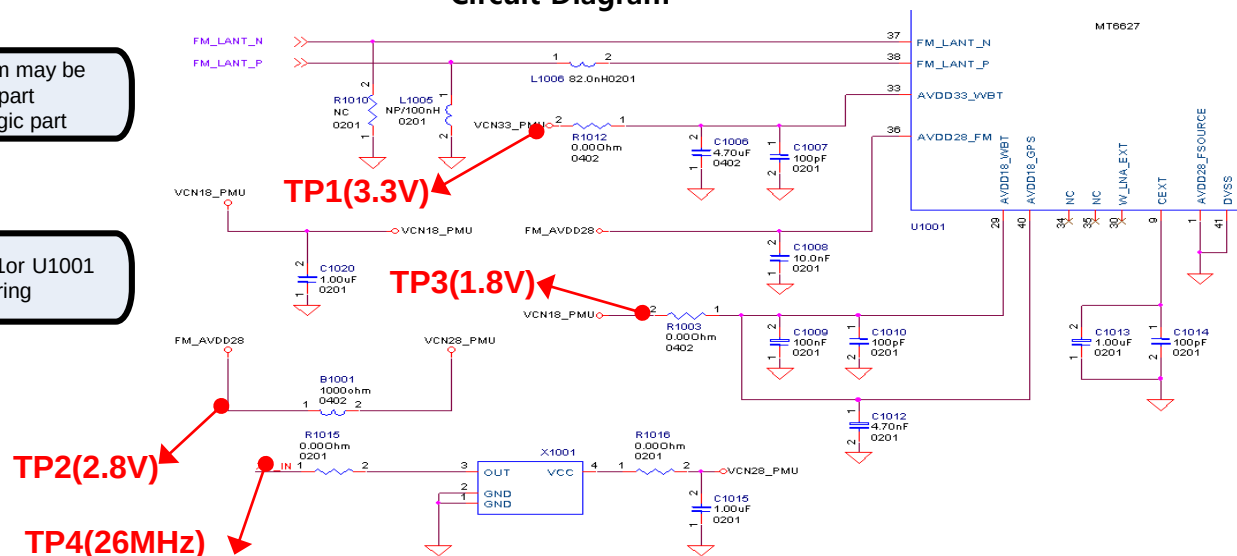
Checking Flow



Image



Circuit Diagram

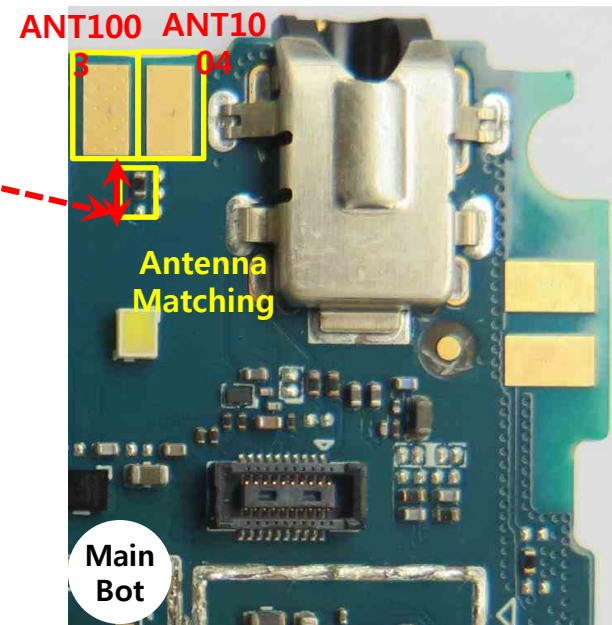
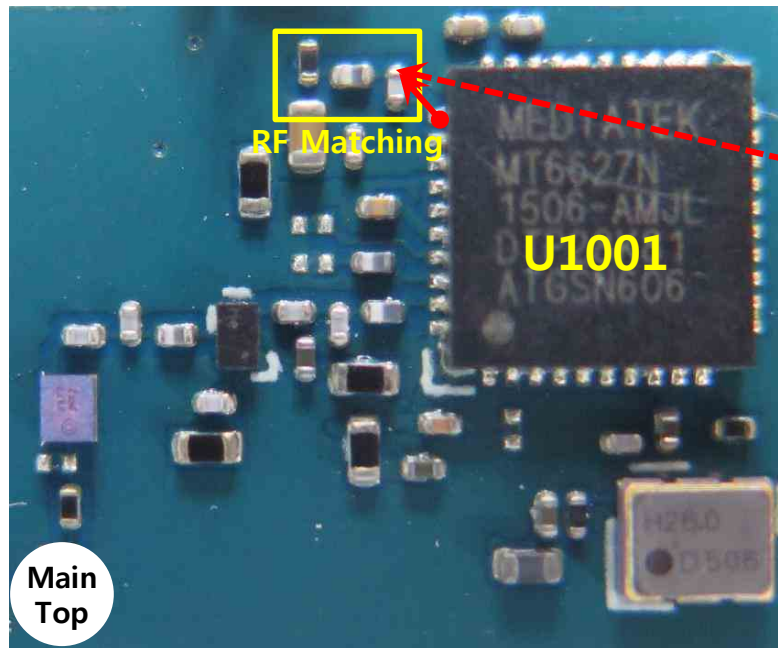


3.6 Connectivity PART

3.6. 2. 1. BT&WIFI Trouble Shooting

Image

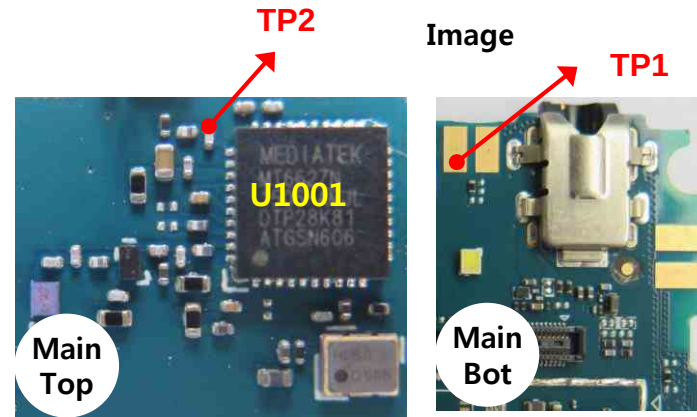
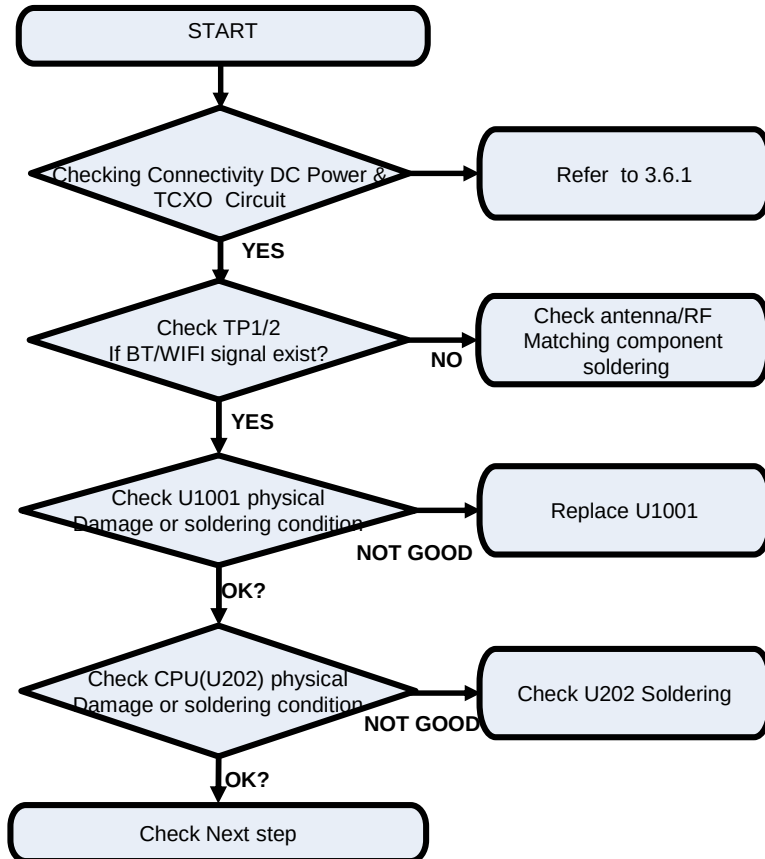
BT/WIFI Signal Path



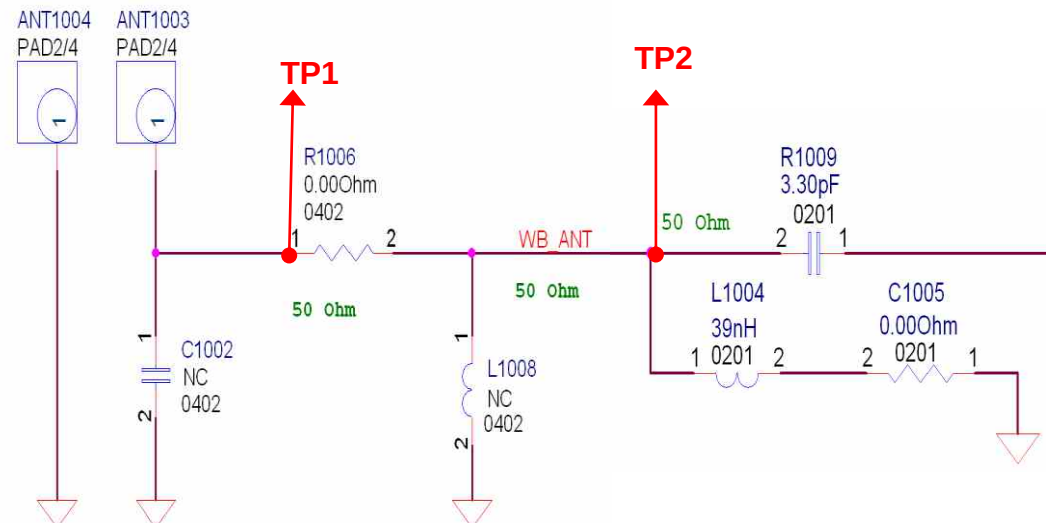
3.6. Connectivity PART

3.6. 2. 2. BT&WIFI Trouble Shooting

Checking Flow



Circuit Diagram

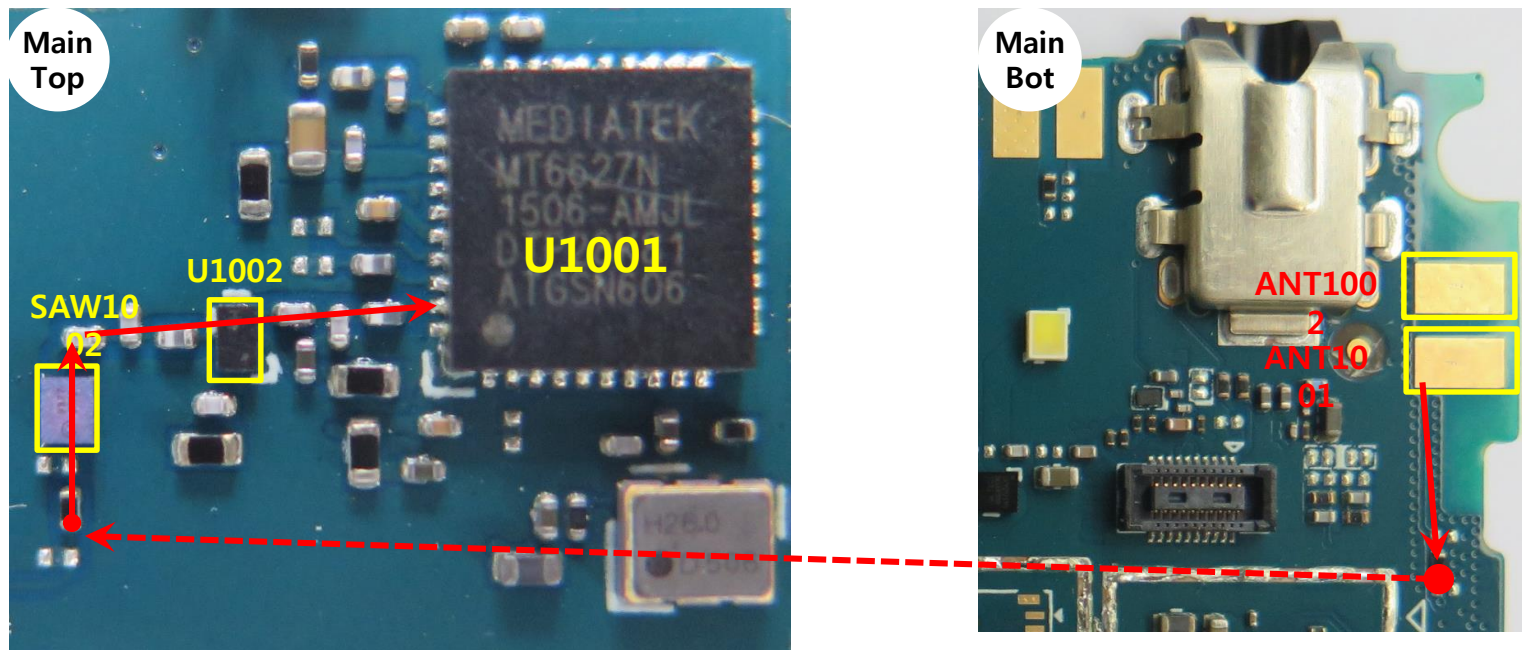


3.6 Connectivity PART

3.6. 3. 1. GPS Trouble Shooting

Image

GPS Signal Path

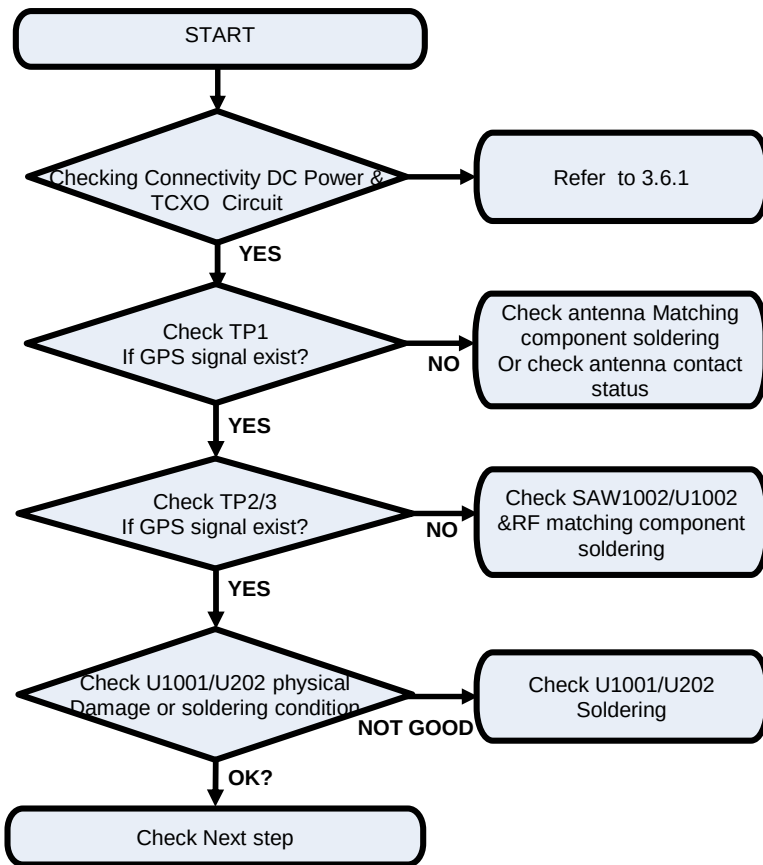


3. Troubleshooting

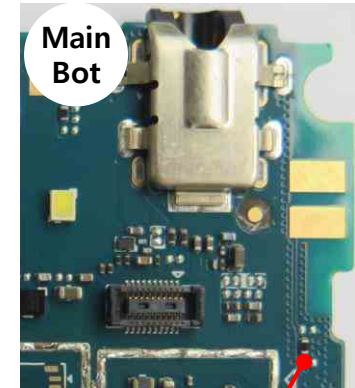
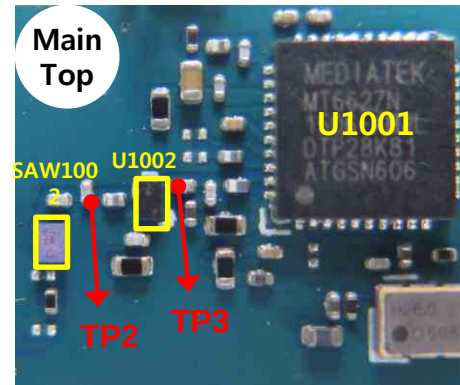
3.6 Connectivity PART

3.6. 3. 2. GPS Trouble Shooting

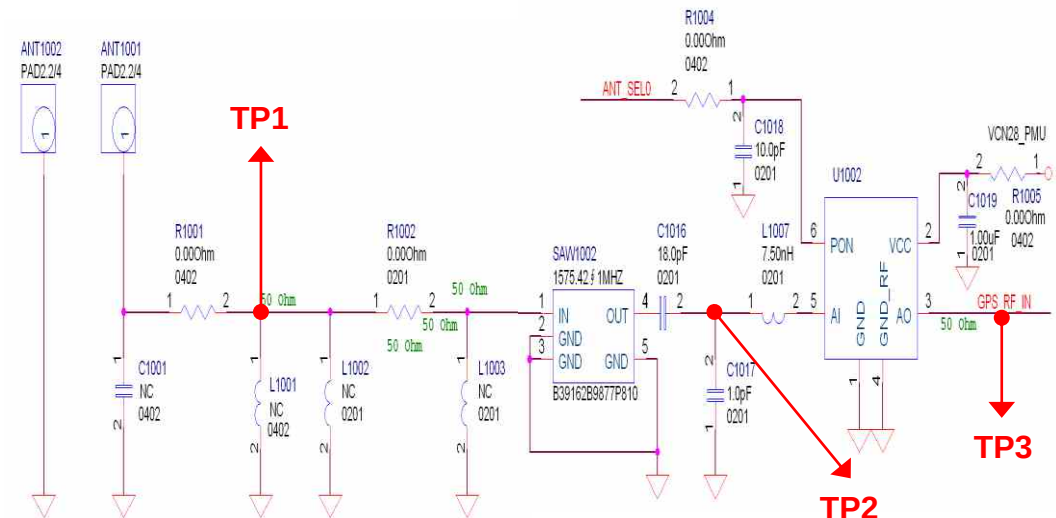
Checking Flow



Image



Circuit Diagram

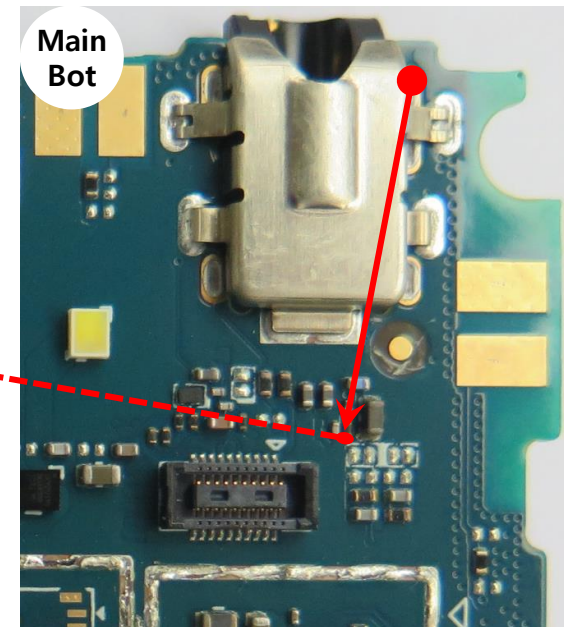
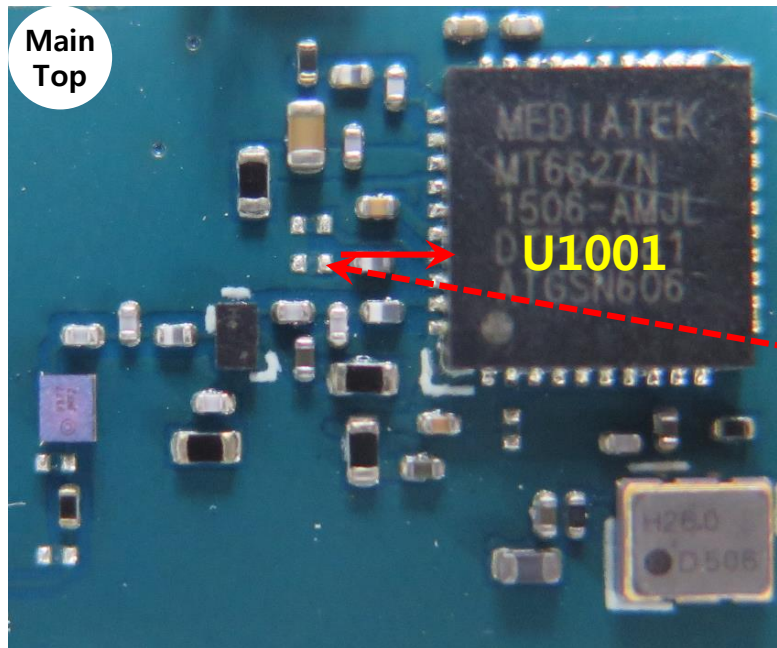


3.6 Connectivity PART

3.6. 4. 1. FM Trouble Shooting

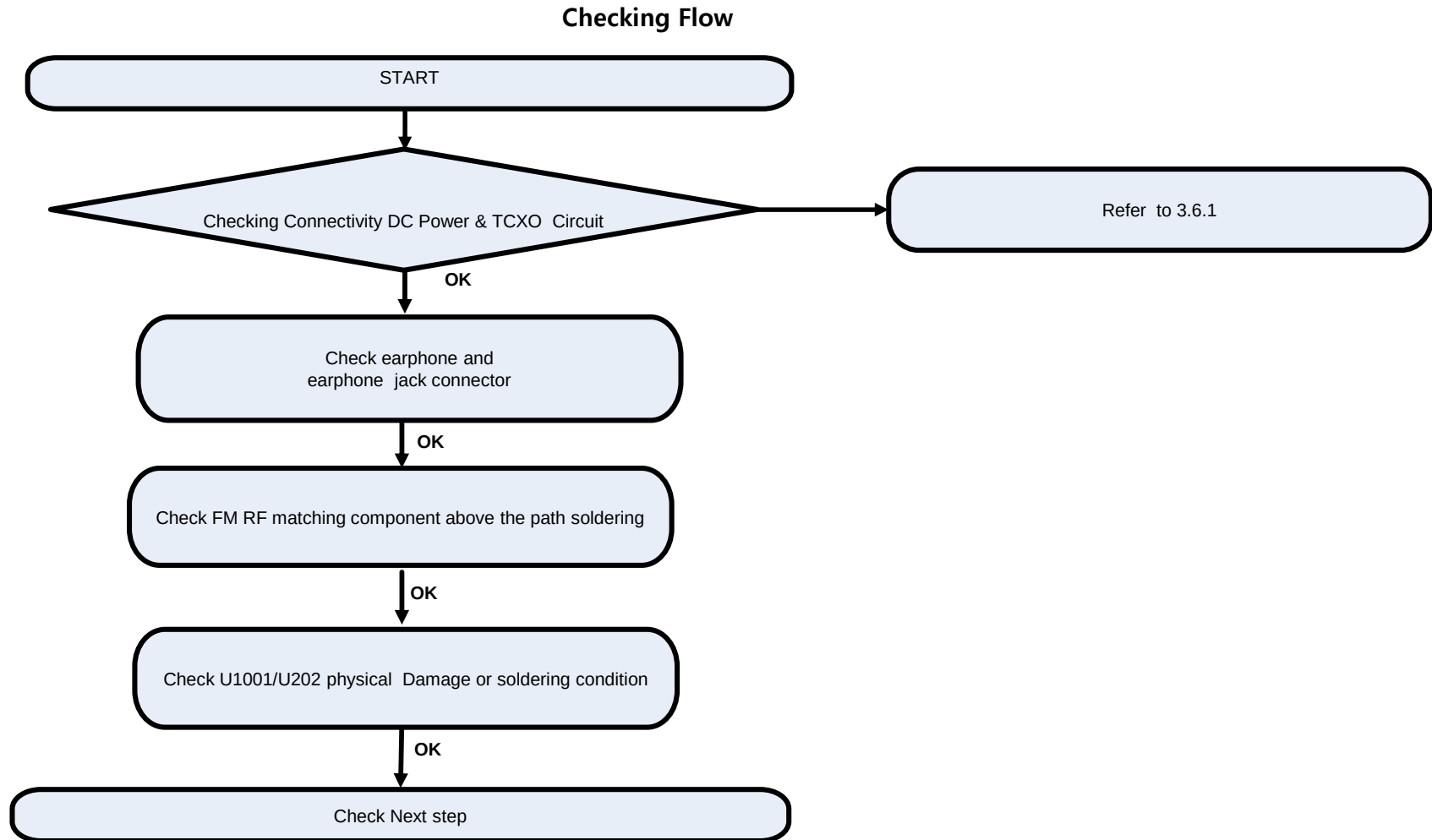
Image

FM Signal Path



3.6 Connectivity PART

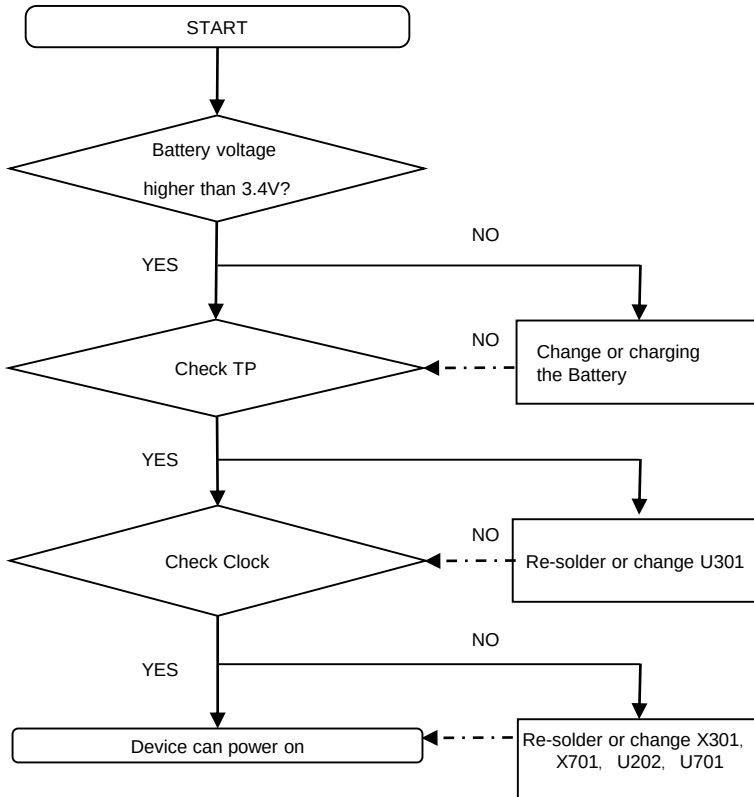
3.6. 4. 2.FM Trouble Shooting



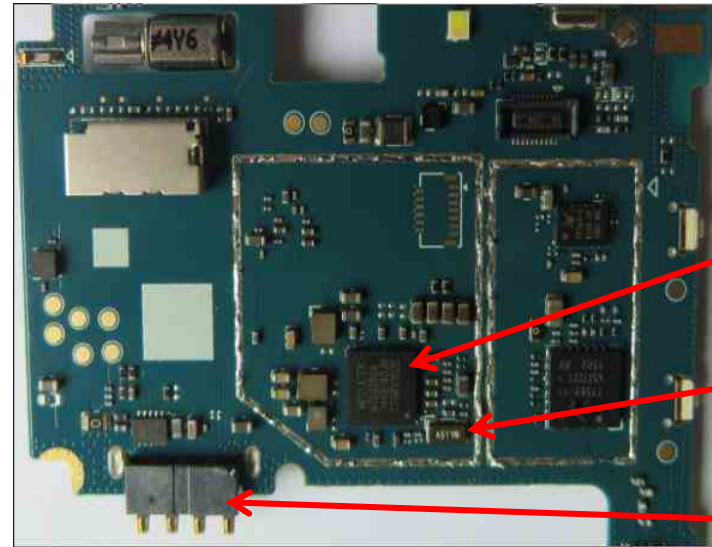
3.7 Power

Checking Power signal(Battery connector, PMU, Clock)

Checking Flow



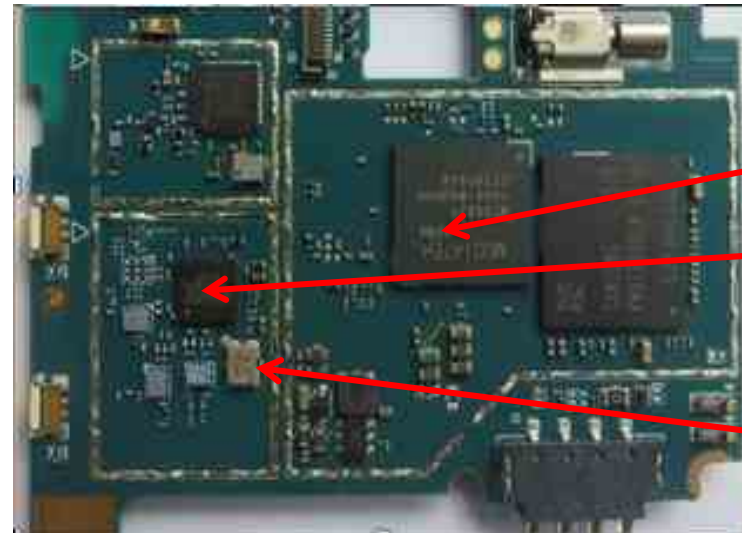
Image



PMU (U301)

32K(X301)

BATTERY CONNECTOR
(J1402)



CPU(U202)

Transceiver(U701)

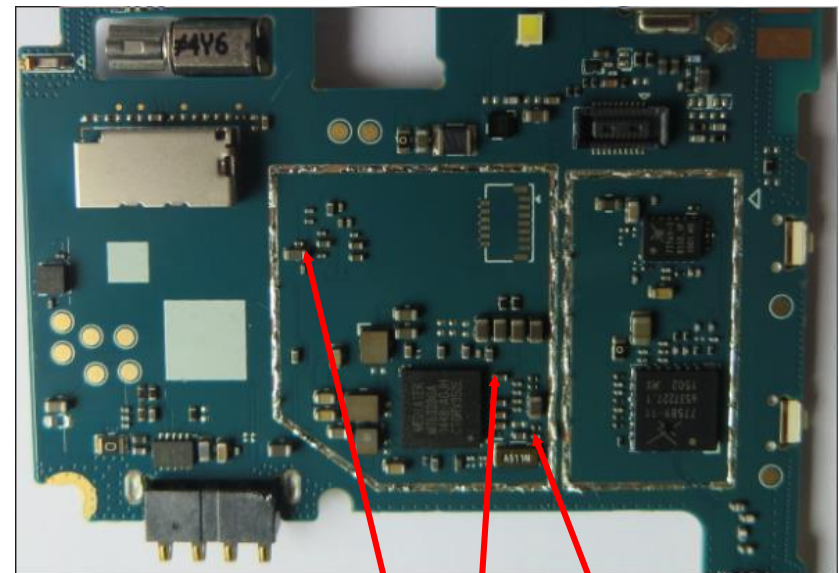
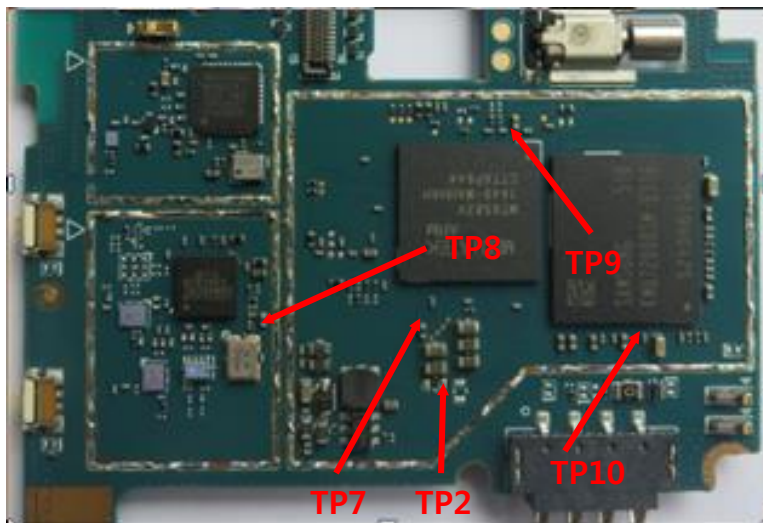
26M(X701)

3.7 Power

Checking Power signal(Battery connector, PMU, Clock)

Check Output Voltage of U301 (PMU)

Power on voltage check list				
Power do main	Configurable Voltage	Signal Name	Measurment Location	TP
VRTC	2.8 V	VRTC	C323	TP1
VPROC	0.75 ~ 1.3V (25mV/step)	VPROC_PMU	C333	TP2
VIO18	1.8 V	VIO18_PMU	C1706	TP3
VIO28	2.8 V	VIO28_PMU	C1702	TP4
VM	1.2 V	VM_PMU	C130	TP5
VA	2.8 V	VA_PMU	C353	TP6
VUSB	3.3V	VUSB_PMU	C119	TP7
VTCXO	2.8V	VTCXO_PMU_RF	B701	TP8
VMC	1.8 V/3.3 V	VMC_PMU	C115	TP9
VEMC_3V3	3.3 V	VEMC_3V3	C413	TP10



TP5 TP6 TP1

TP3 TP4

3.7 Power

Checking Power signal(Battery connector, PMU, Clock)

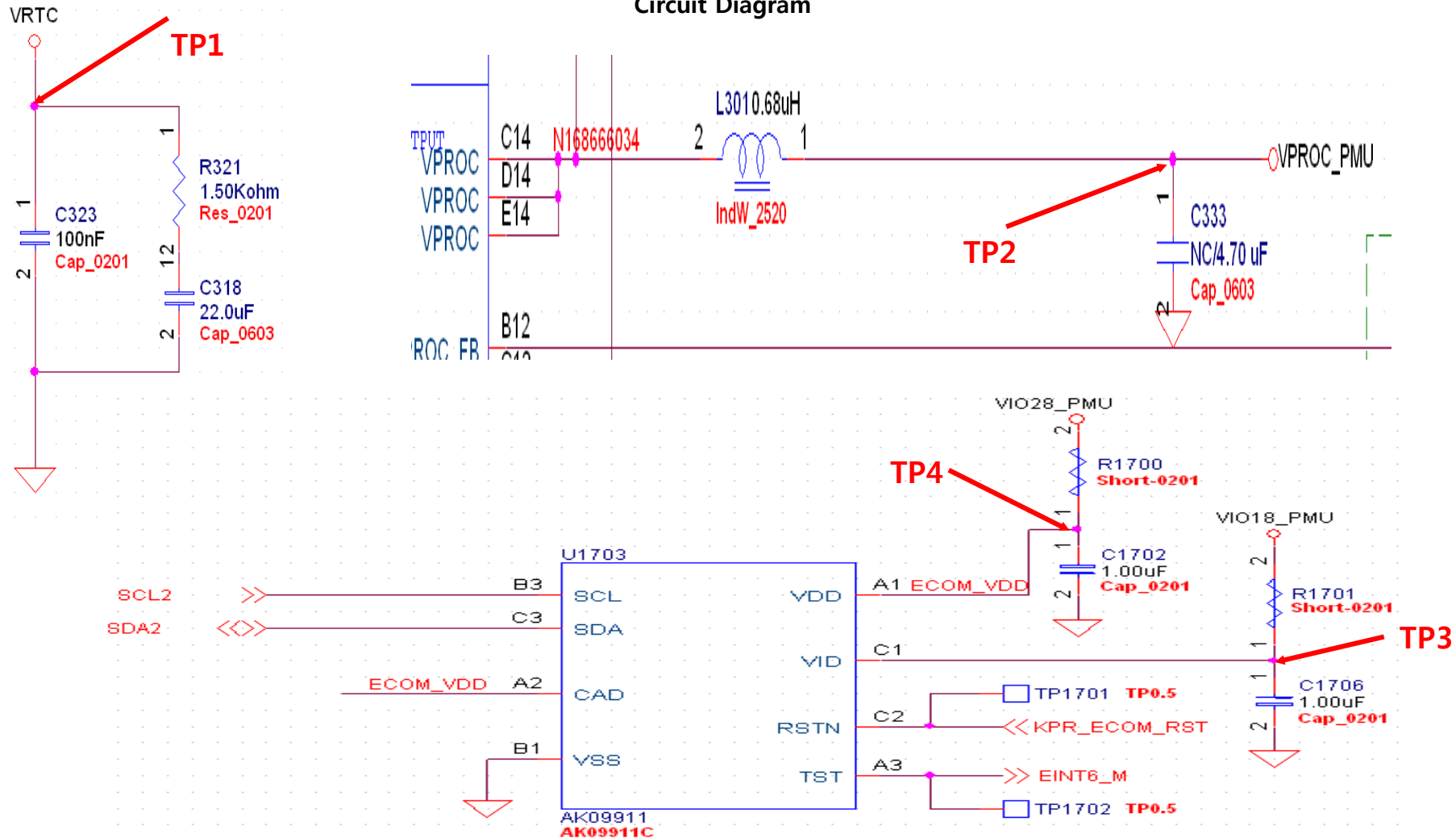
Oscillate frequency measurement

Signal name	Description	Frequency MHz	Measurment Location	Check Clock Result
X301 32.768 KHz crystal		32.768kHz	C319	
RTC32K_CK	32.768KHz for MT6582	32.768kHz	X	
X701 26MHz Crystal		26MHz	X	
CLK1_BB	26MHz back from MT6166 to MT6582	26MHz	C712	

3.7 Power

Checking Power signal(Battery connector, PMU, Clock)

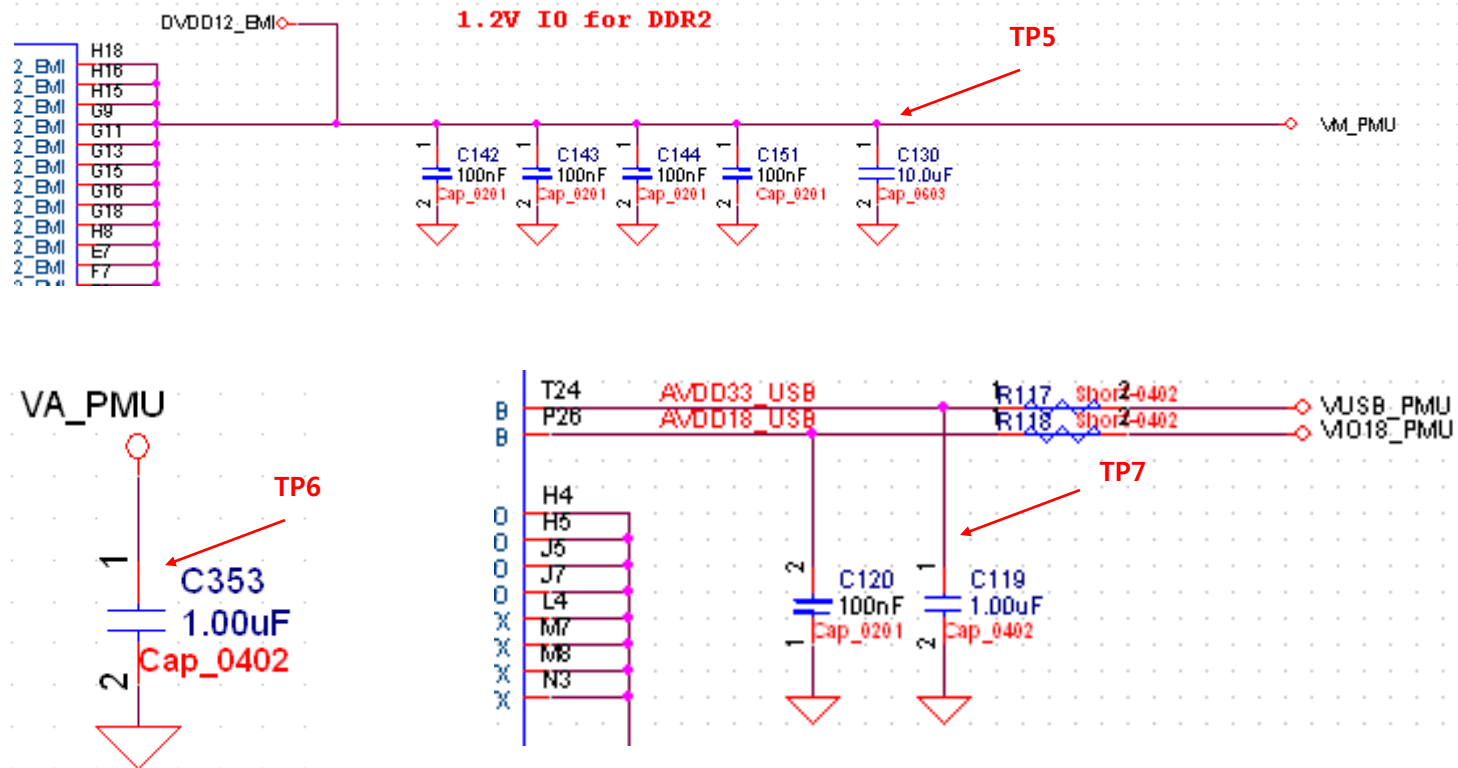
Circuit Diagram



3.7 Power

Checking Power signal(Battery connector, PMU, Clock)

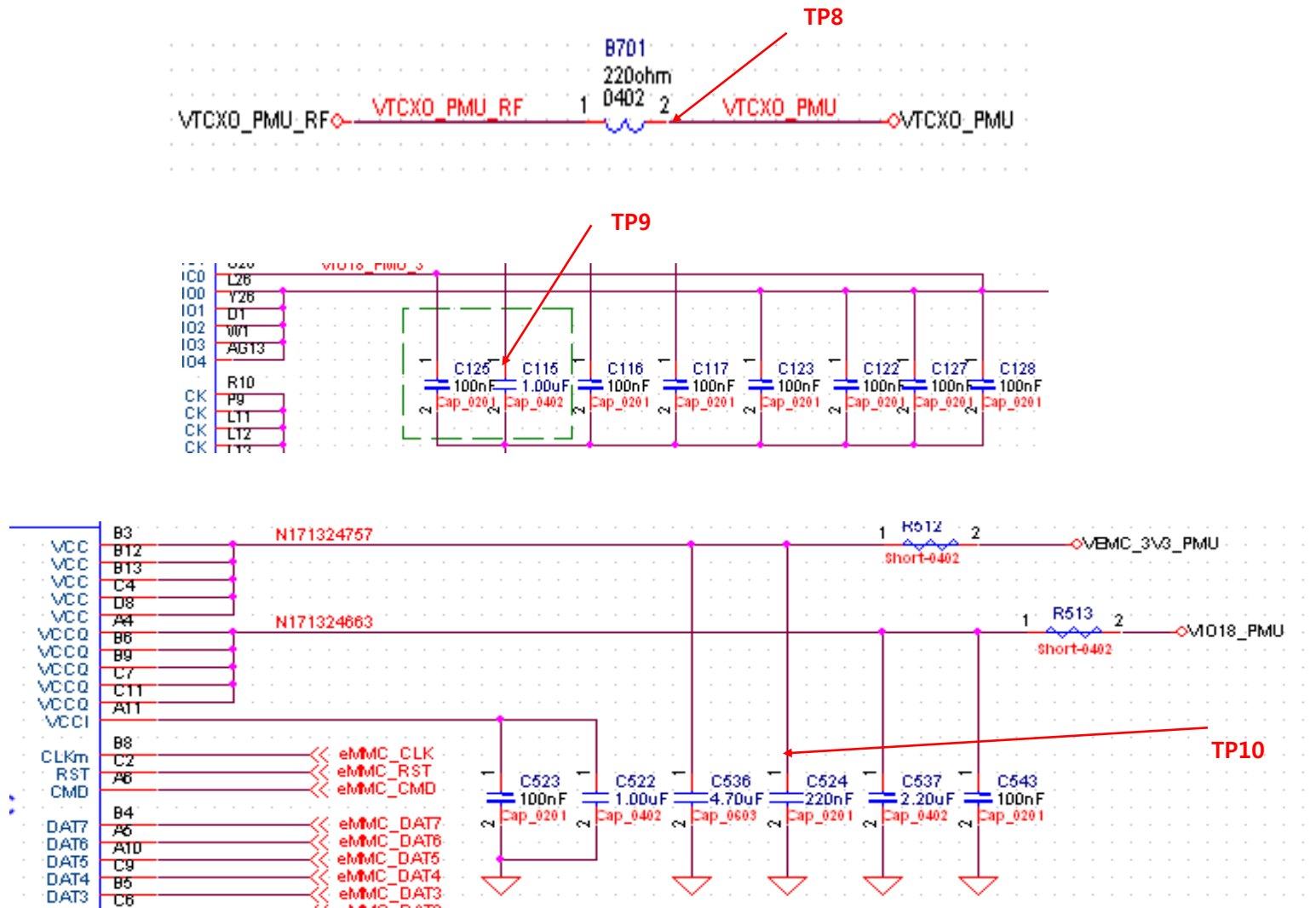
Circuit Diagram



3.7 Power

Checking Power signal(Battery connector, PMU, Clock)

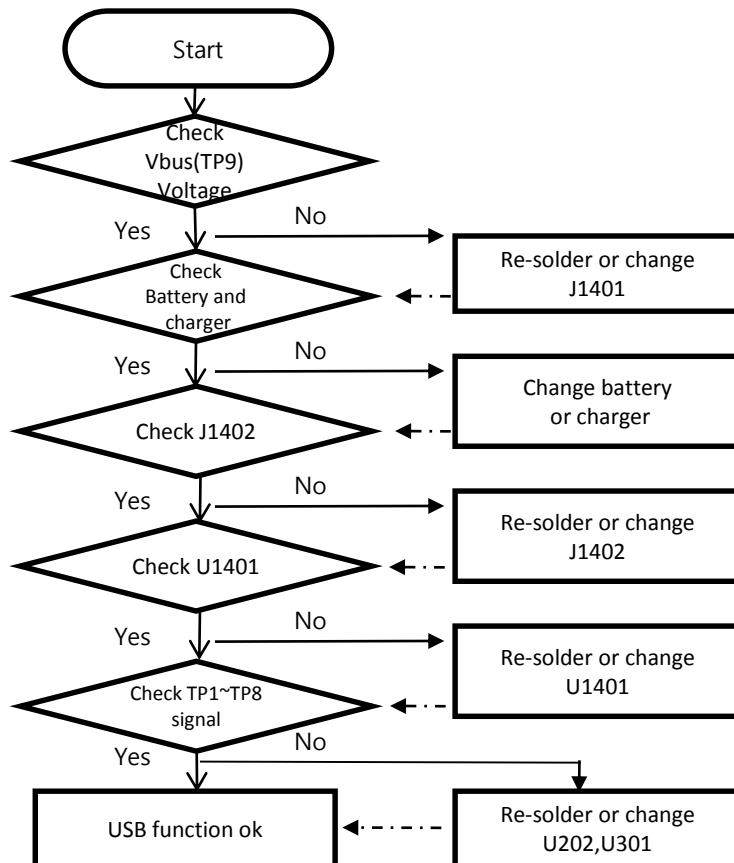
Circuit Diagram



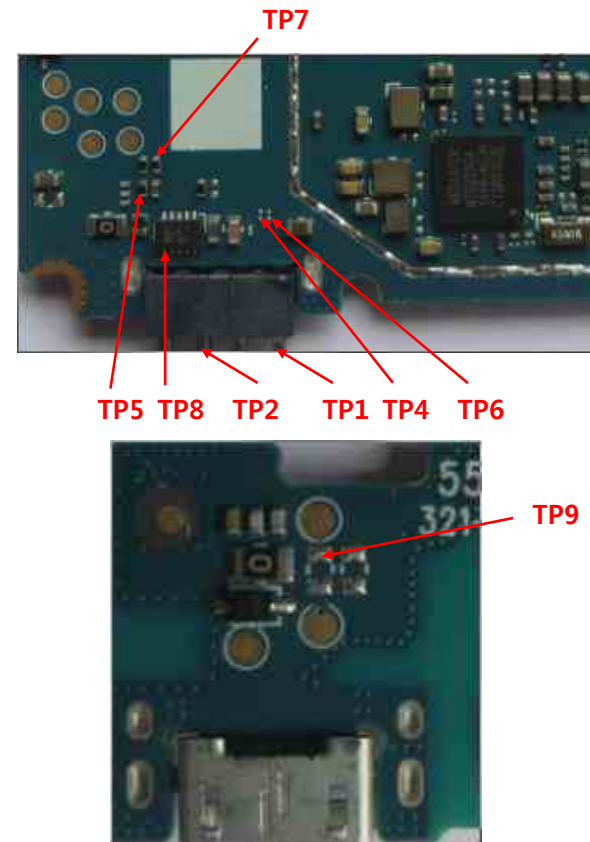
3.8 charger

The charging controlled by BB chip MT6582 U202),PMU chip MT6323(U301),charge IC RT9536(U1401).

Checking Flow



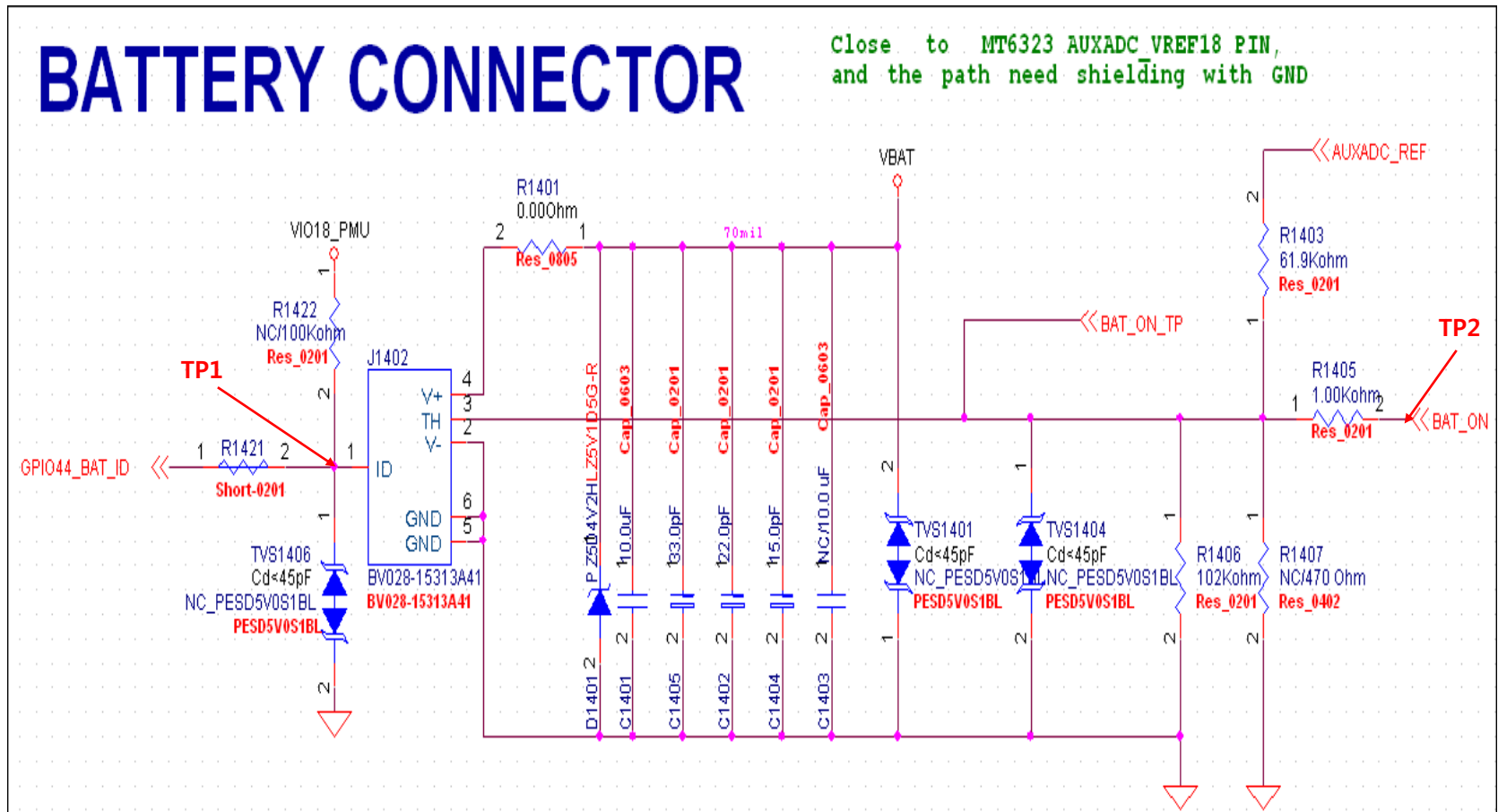
Image



3.8 Charger

The charging controlled by BB chip MT6582 U202),PMU chip MT6323(U301),charge IC RT9536(U1401).

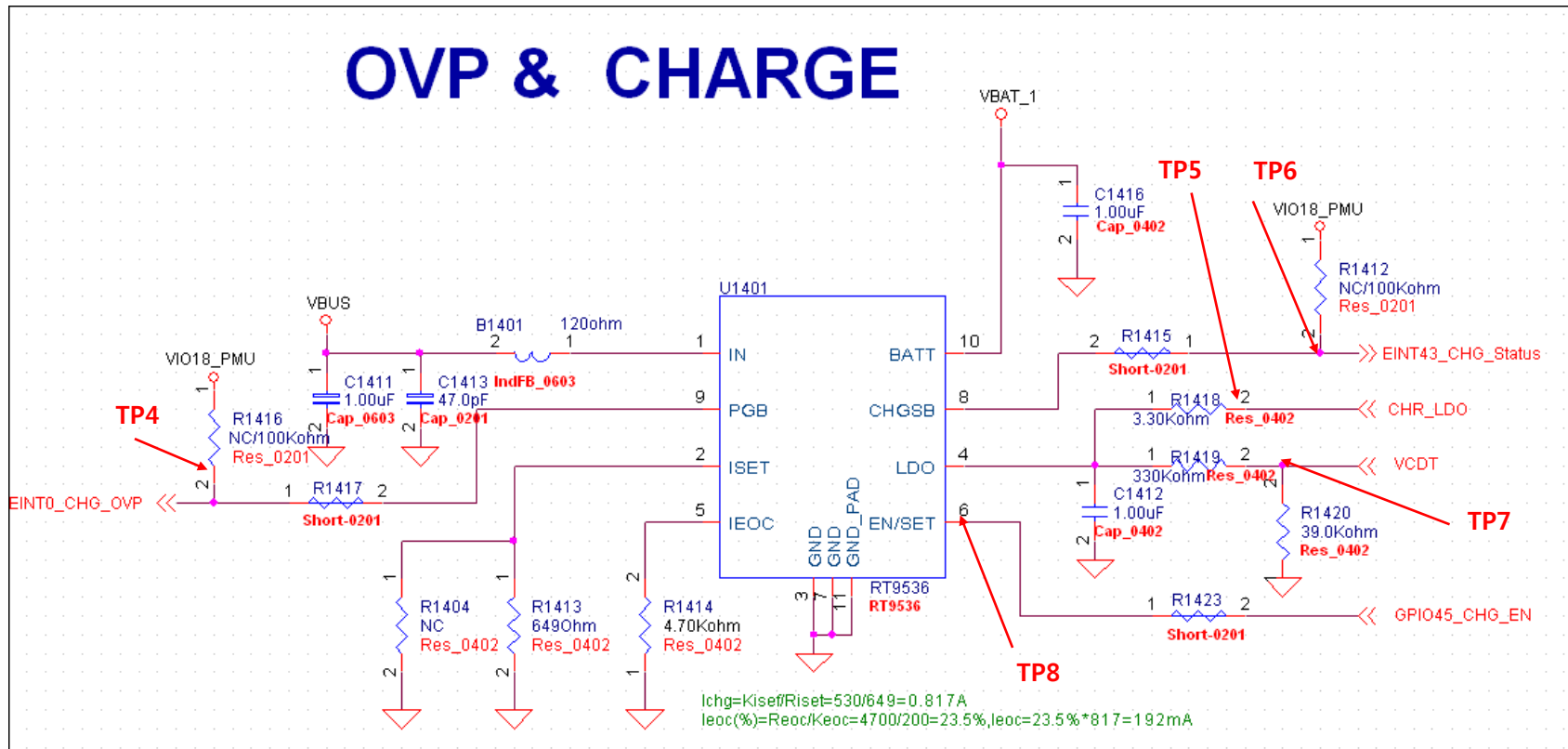
Circuit Diagram



3.8 Charger

The charging controlled by BB chip MT6582 U202),PMU chip MT6323(U301),charge IC RT9536(U1401).

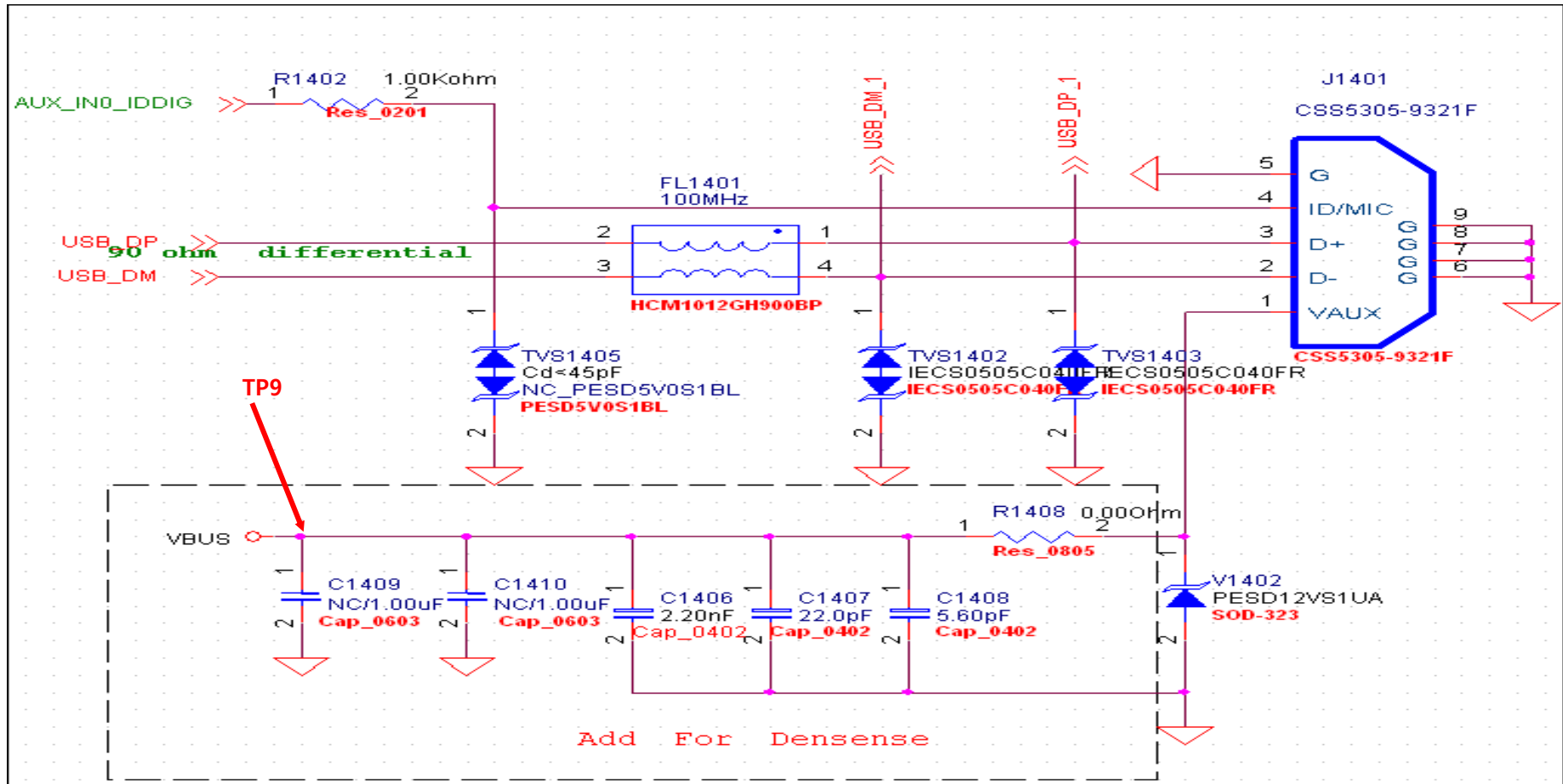
Circuit Diagram



3.8 Charger

The charging controlled by BB chip MT6582 U202),PMU chip MT6323(U301),charge IC RT9536(U1401).

Circuit Diagram

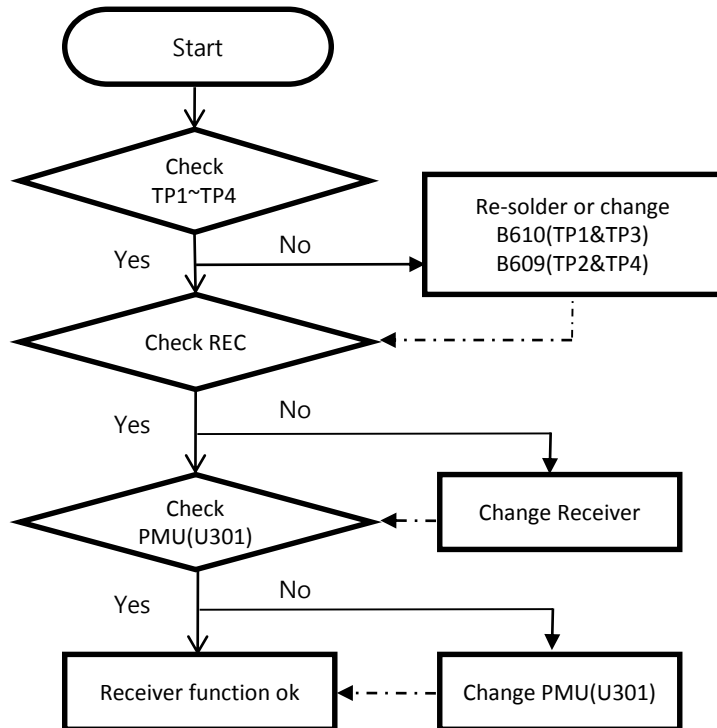


3. TROUBLE SHOOTING

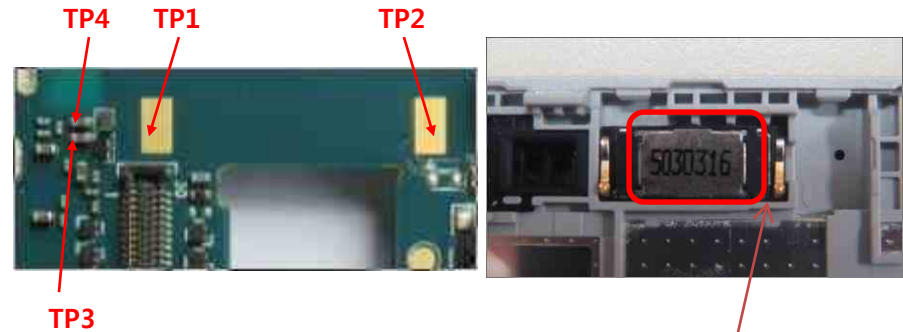
3.9 Checking Audio_receiver Block

The receiver control signals are generated by PMU chip MT6323(U301), the PMU chip and the receiver are to be checked out.

Checking Flow

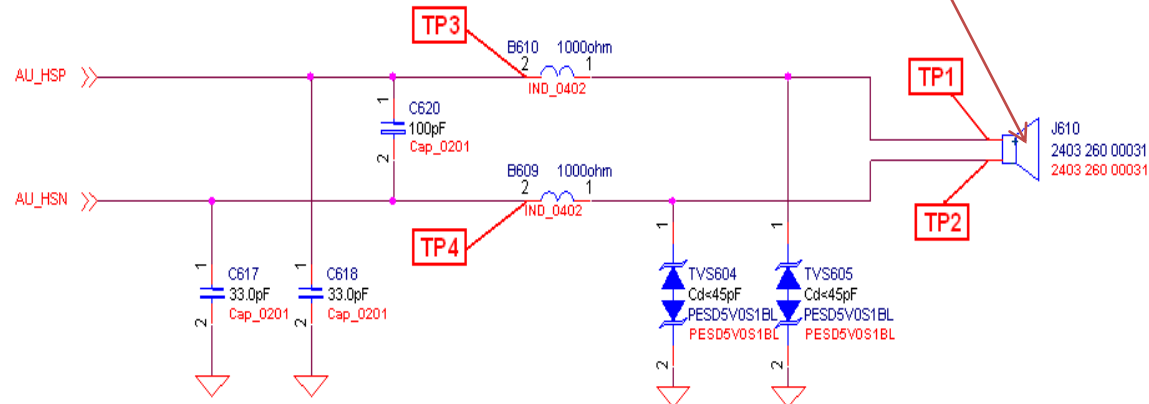


Image



Circuit Diagram

RECEIVER

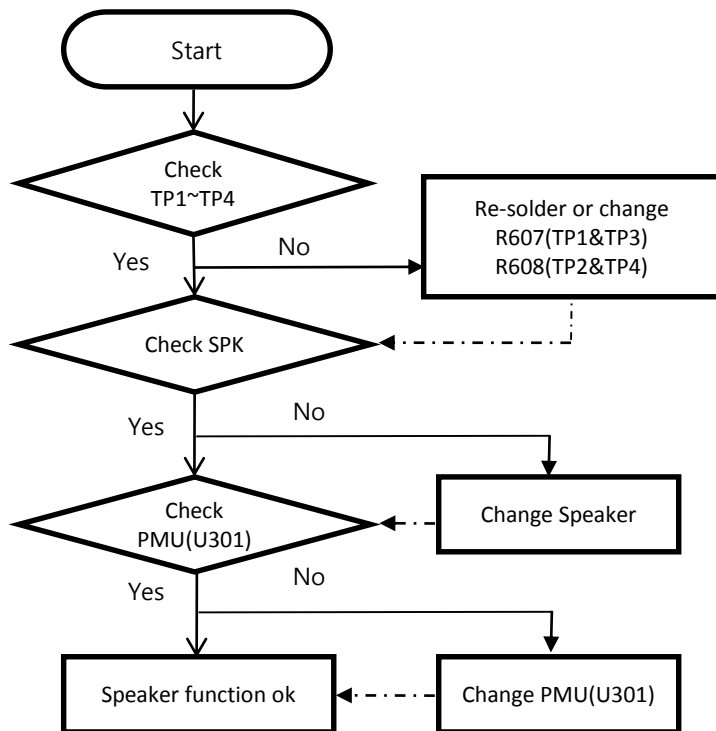


3. TROUBLE SHOOTING

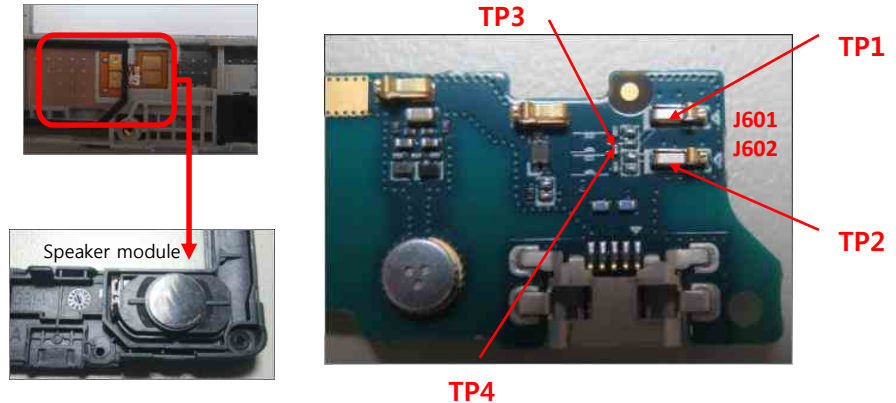
3.9.2 Audio_speaker

The Speaker control signals are generated by PMU chip MT6323(U301), the PMU chip and the speaker are to be checked out.

Checking Flow

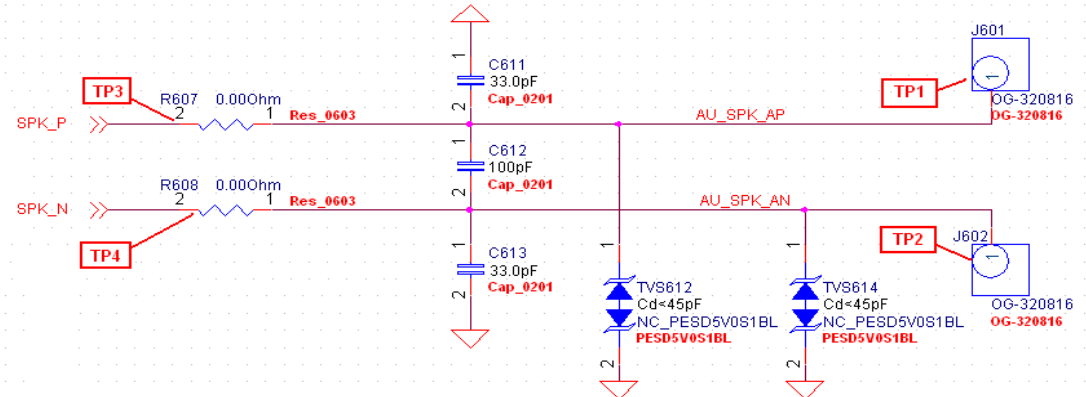


Image



Circuit Diagram

SPEAKER

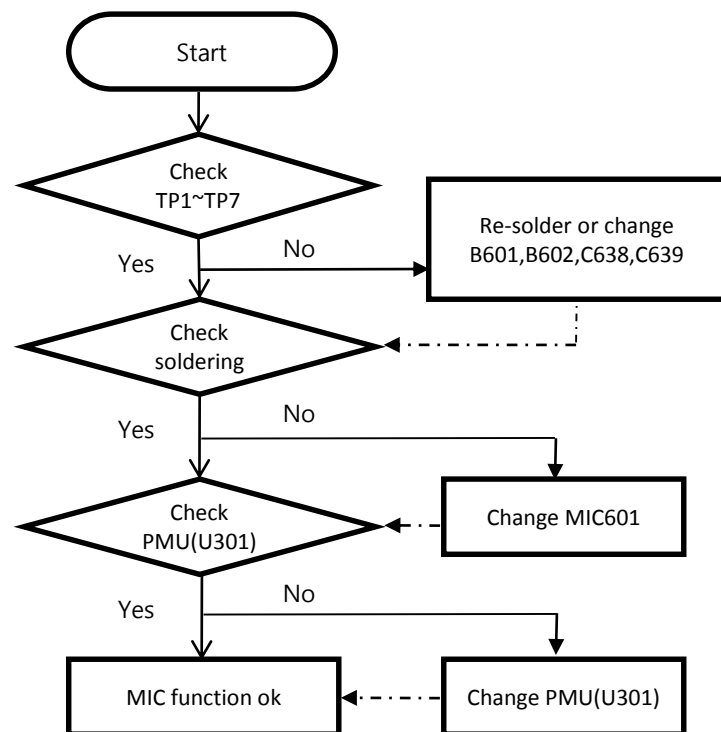


3. TROUBLE SHOOTING

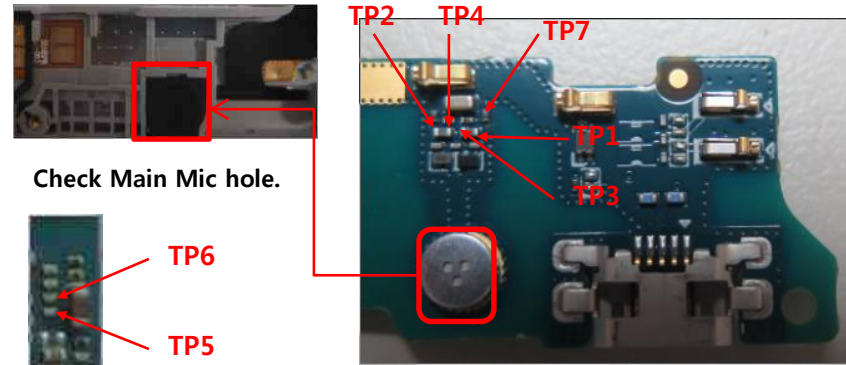
3.9.3 Audio_Main MIC

The MIC control signals are generated by PMU chip MT6323(U301), the PMU chip and the MIC are to be checked out.

Checking Flow

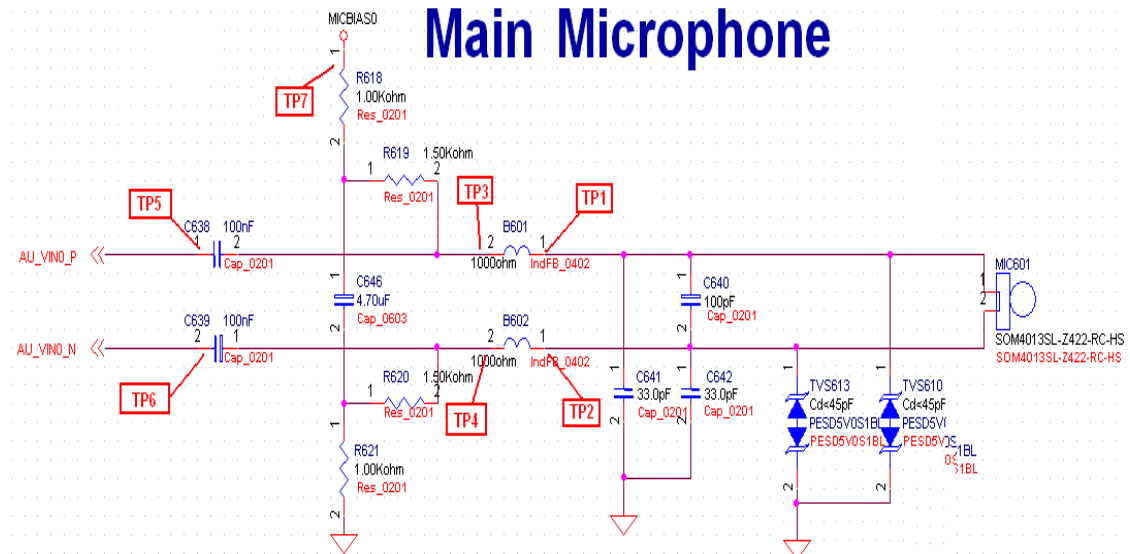


Image



Circuit Diagram

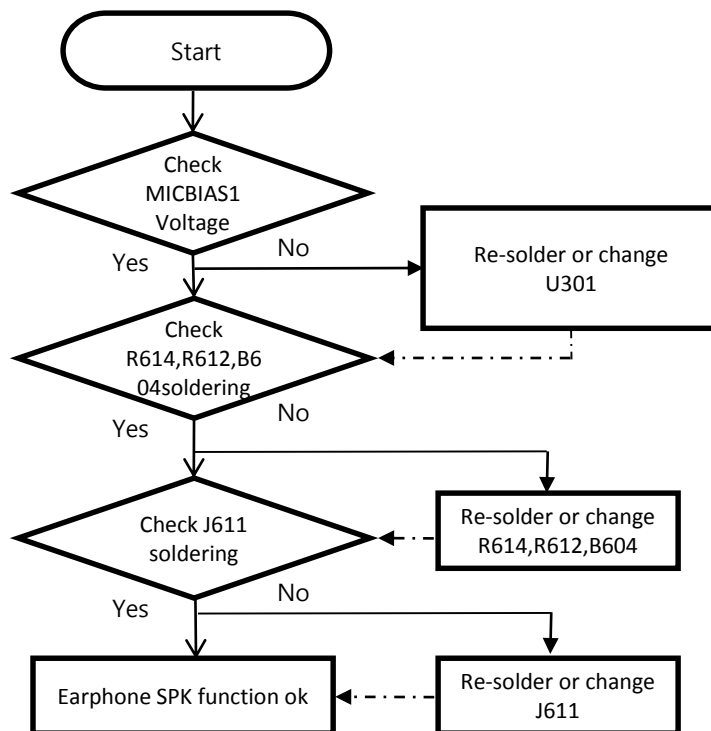
Main Microphone



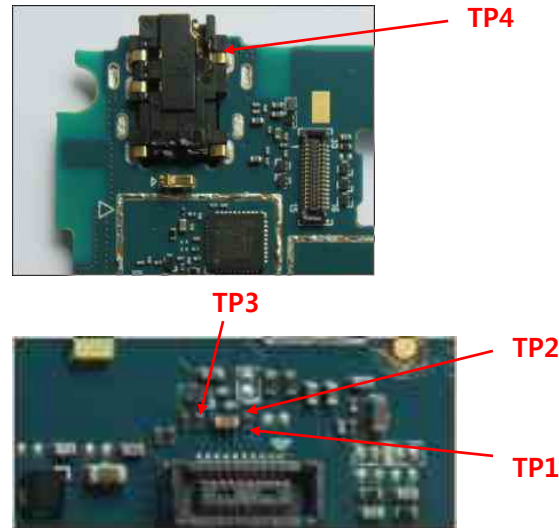
3.9.5 Audio_Ear MIC

Disable detecting headset insert or No sound from Earphone, Check the Ear Mic and PMU(U301).

Checking Flow



Image

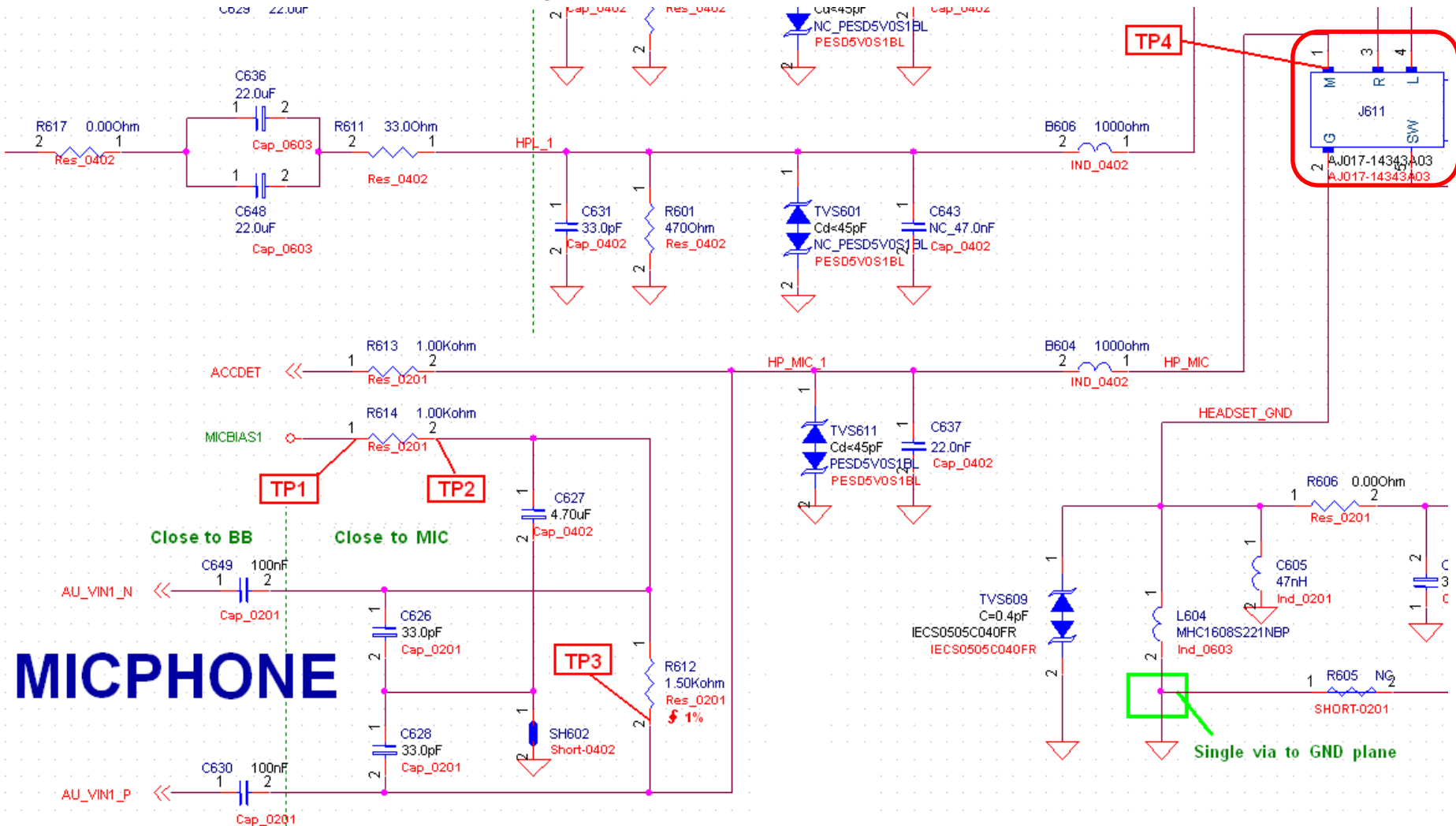


**Circuit Diagram
: Next Page**

3.9.5 Audio_Ear MIC

Disable detecting headset insert or No sound from Earphone, Check the Ear Mic and PMU(U301).

Circuit Diagram

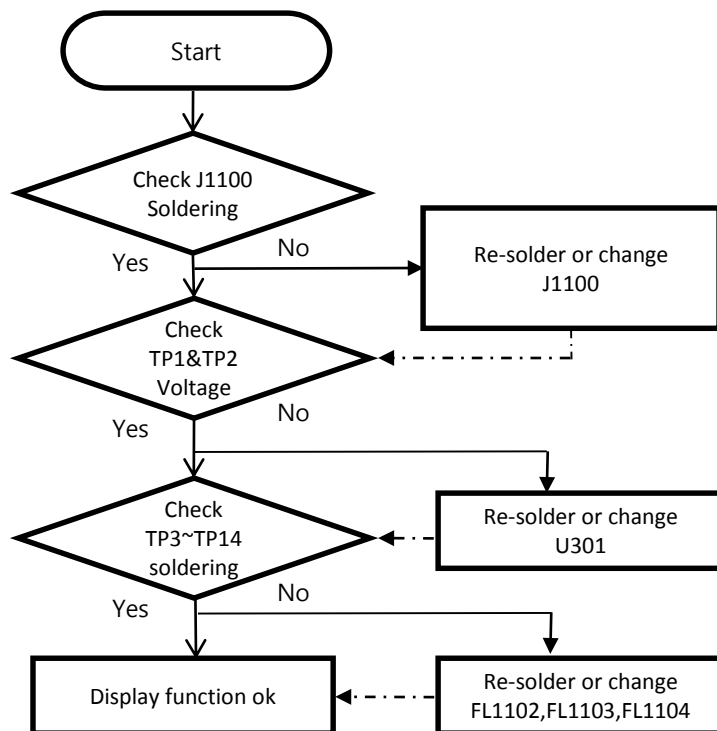


MICPHONE

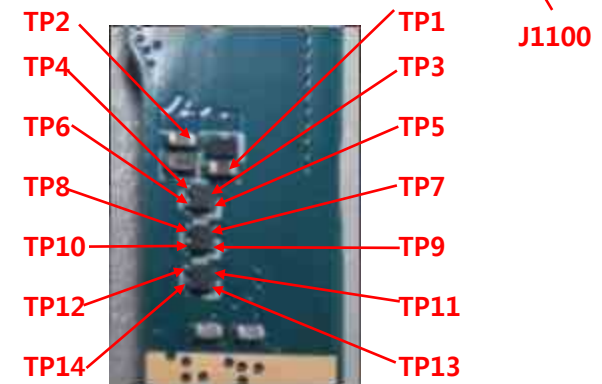
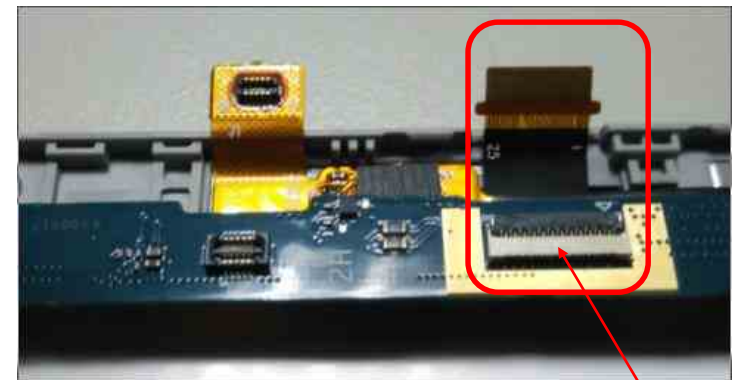
3.10 Checking LCD Block

The LCD control signals are generated by MT6582. Its interface is MIPI having four data lanes and one clock lane.

Checking Flow



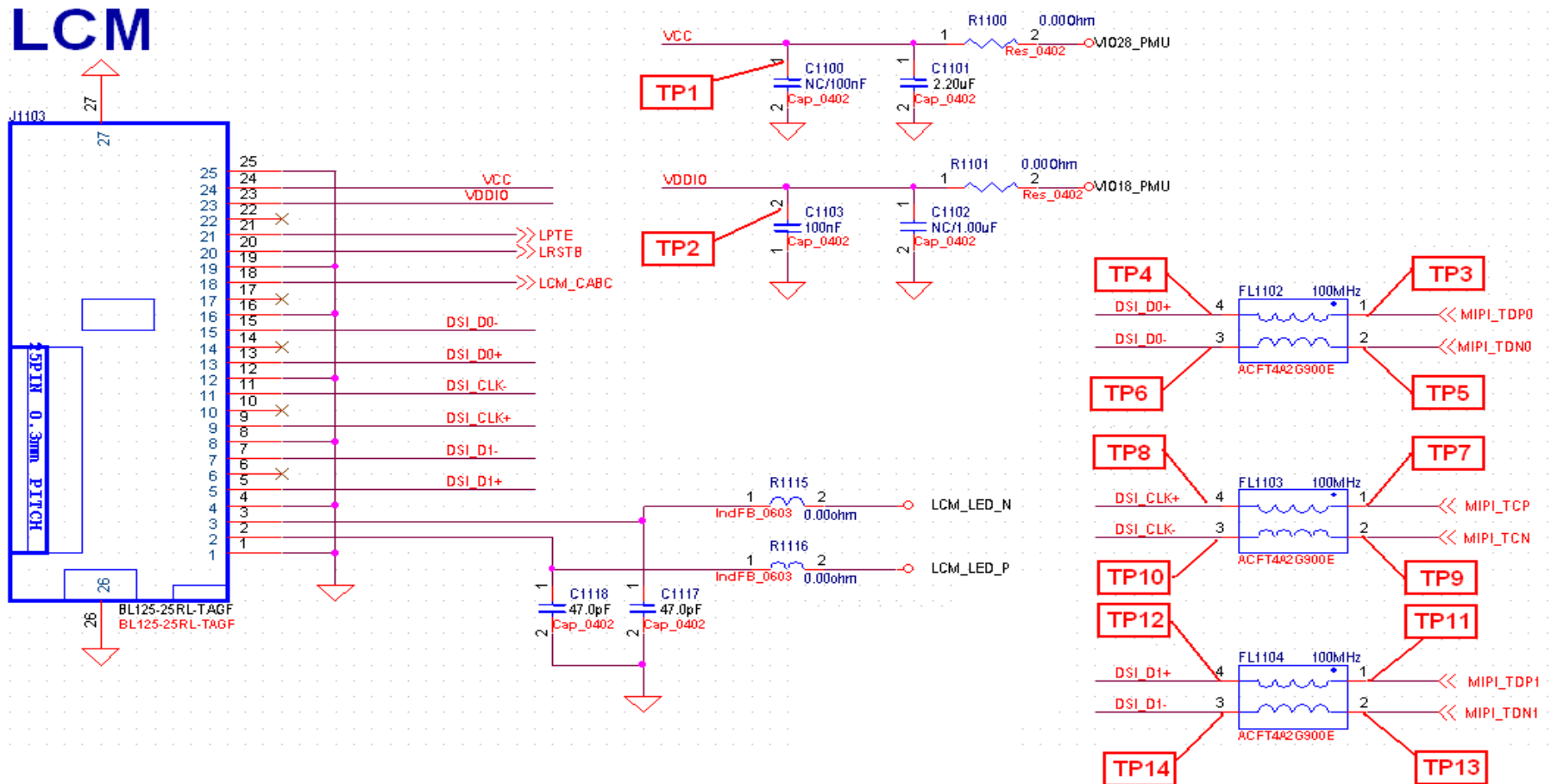
Image



3.10 Checking LCD Block

The LCD control signals are generated by MT6582. Its interface is MIPI having four data lanes and one clock lane.

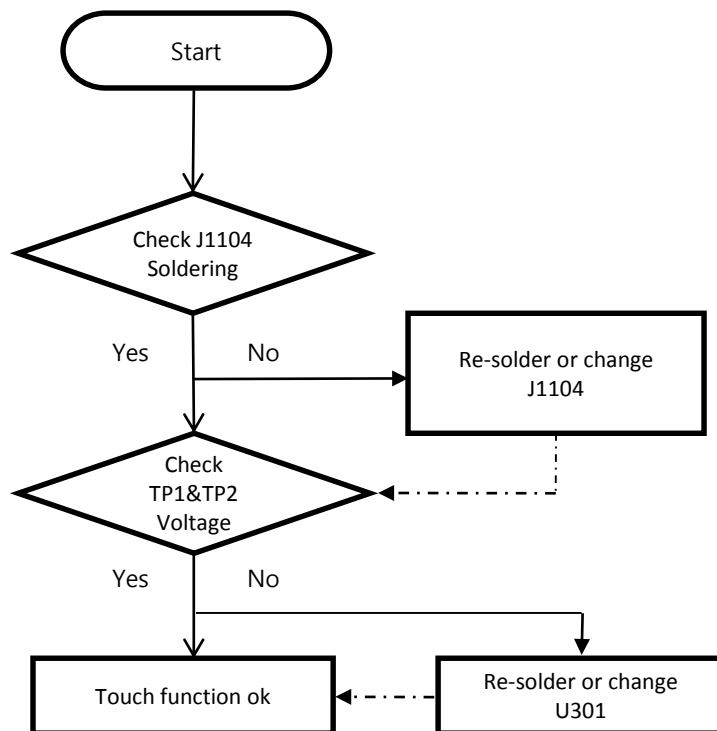
Circuit Diagram



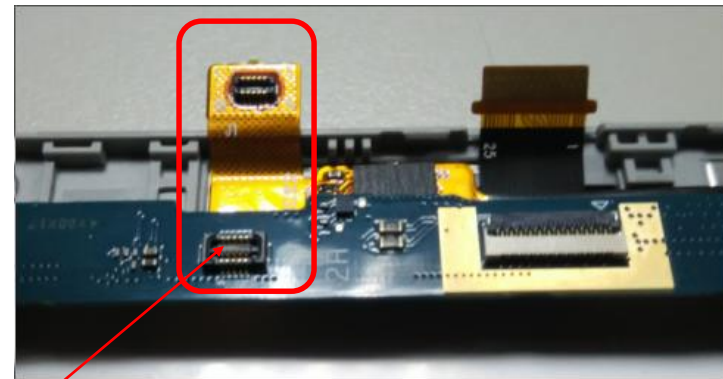
3.11 Checking Touch Block

The Touch control signals are generated by MT6582 and MT6323. It is assembled with LCD.

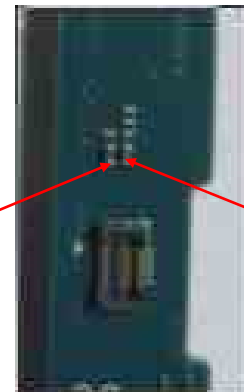
Checking Flow



Image



J1104



TP1

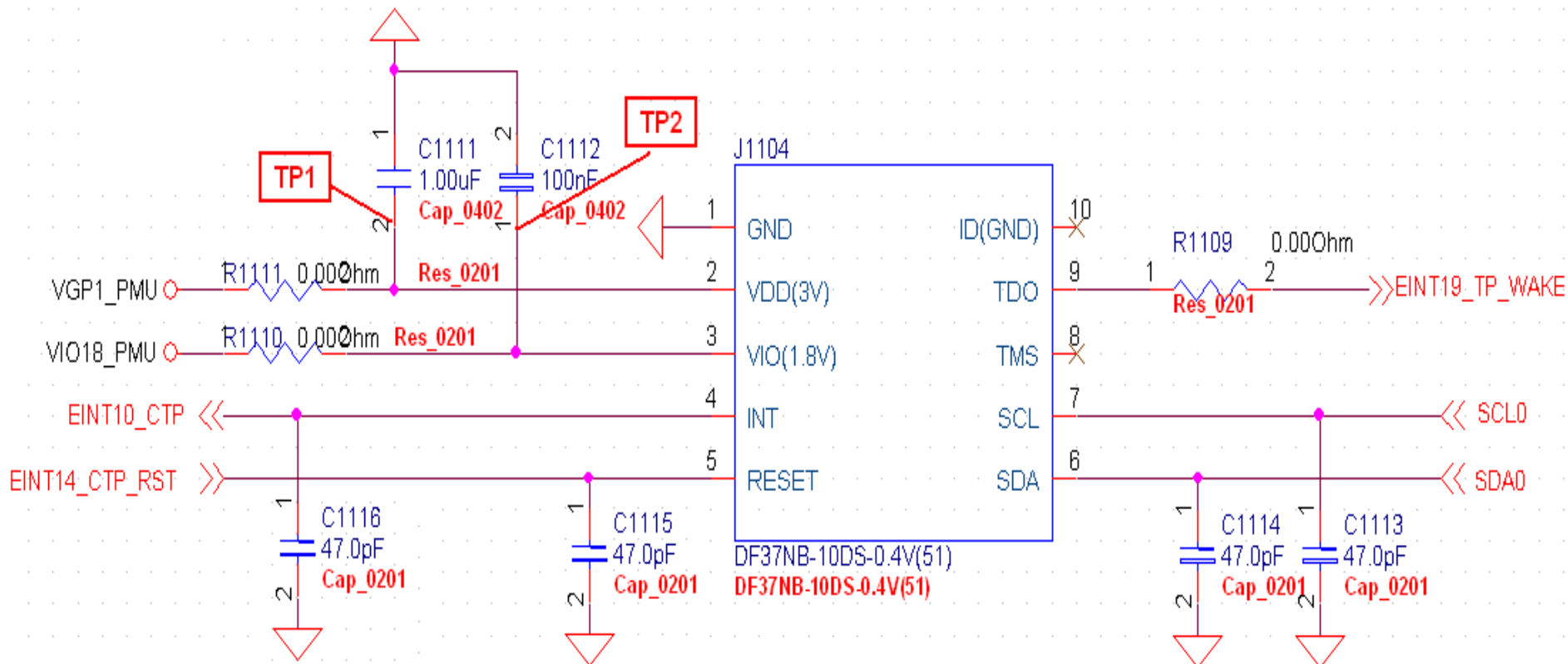
TP2

3.11 Touch

The Touch control signals are generated by MT6582 and MT6323. It is assembled with LCD.

Circuit Diagram

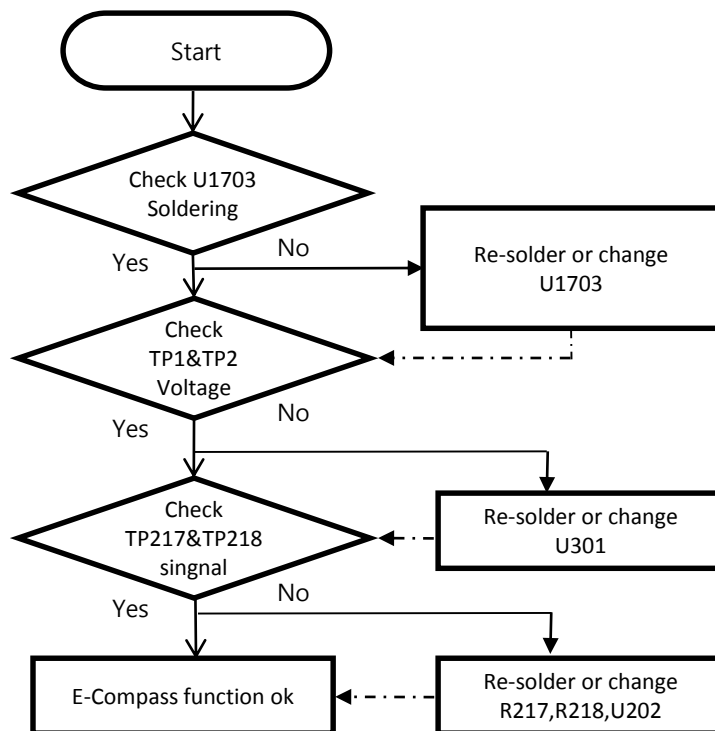
Capacitive Touch Panel



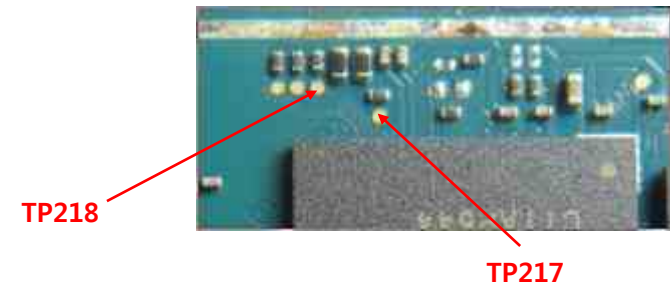
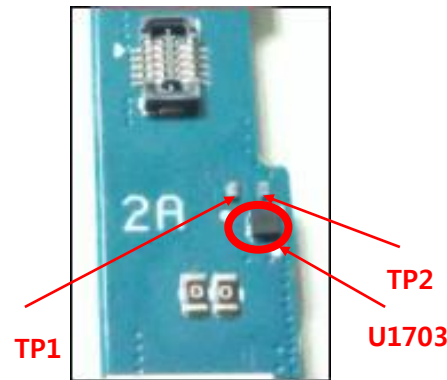
3.12 Checking compass Block

The compass is calibrated by using SW algorithm.

Checking Flow



Image

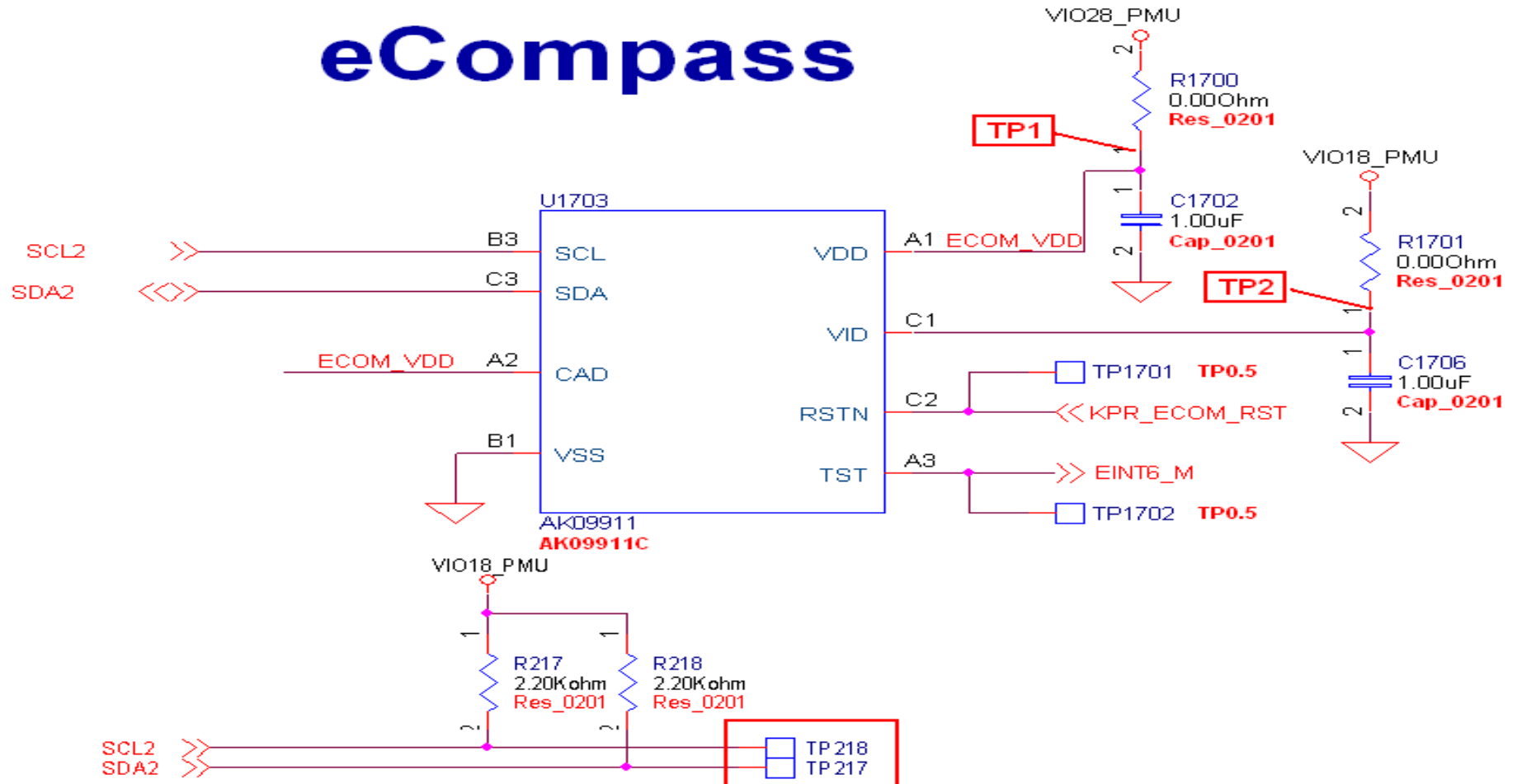


**Circuit Diagram
: Next Page**

3.12 Checking compass Block

The compass is calibrated by using SW algorithm.

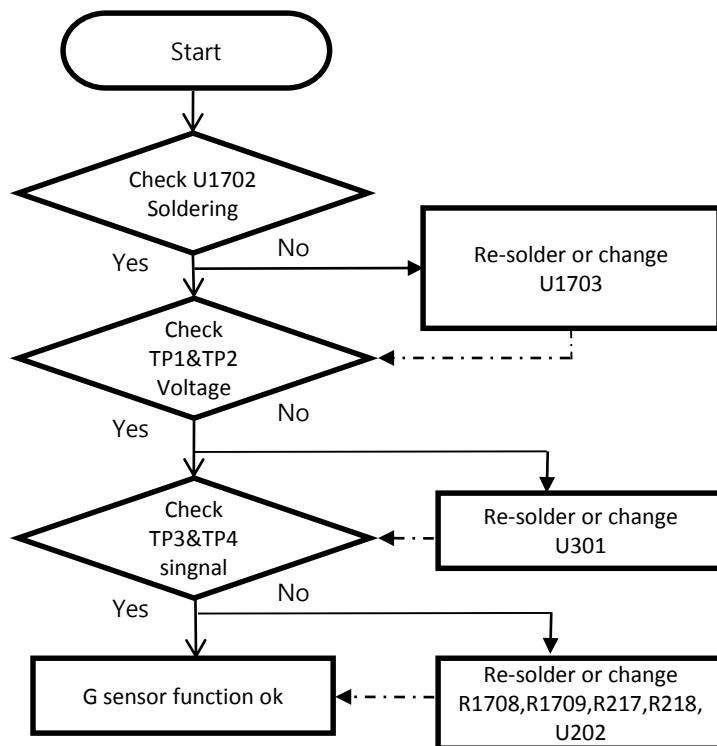
Circuit Diagram



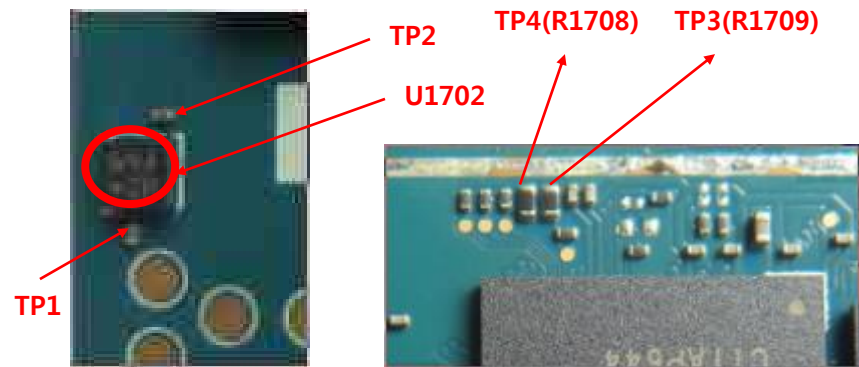
3.12 Checking G sensor Block

The G sensor is calibrated by using SW algorithm.

Checking Flow



Image

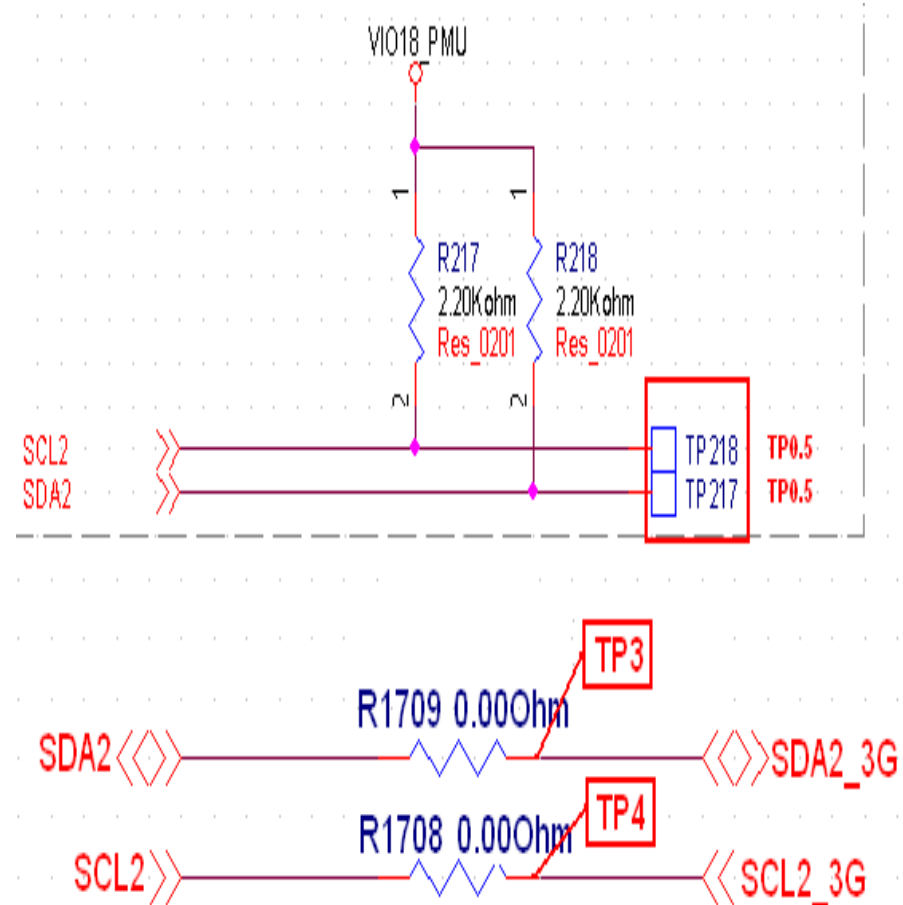
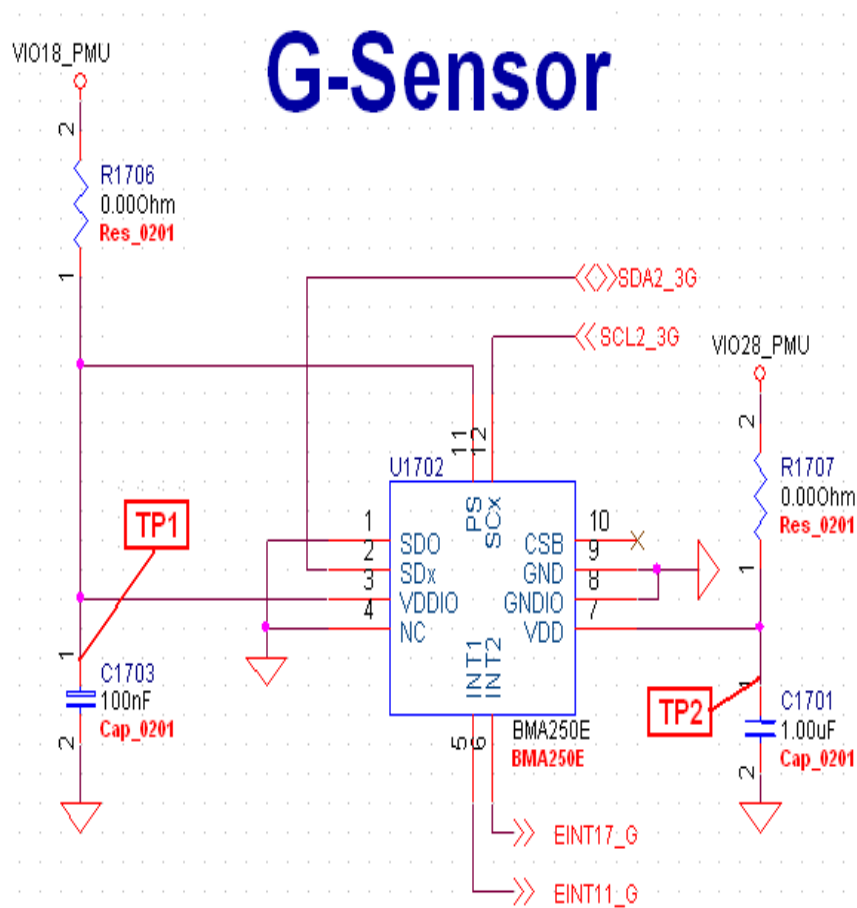


**Circuit Diagram
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3.12 Checking G sensor Block

The G sensor is calibrated by using SW algorithm.

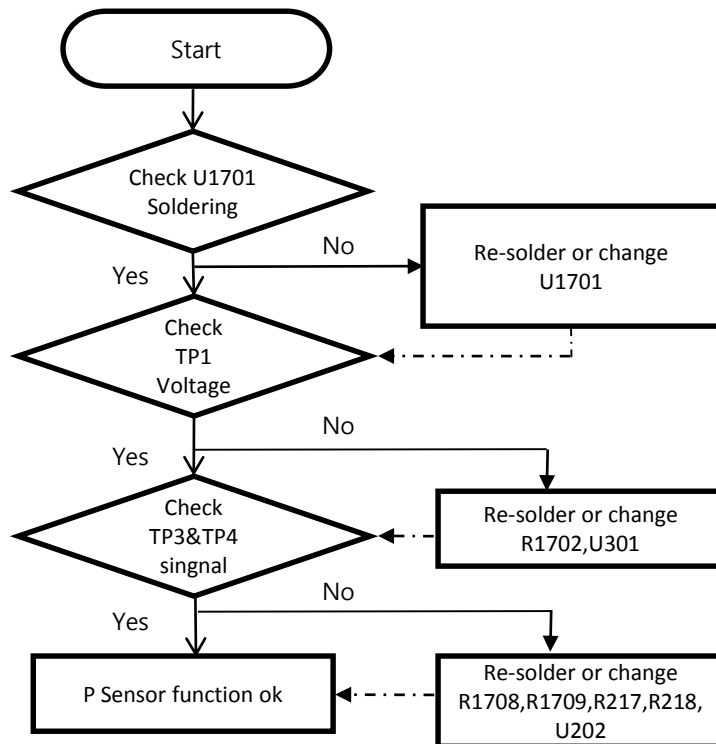
Circuit Diagram



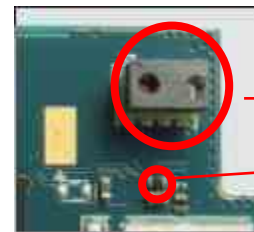
3.12 Checking proximity sensor Block

Proximity Sensor is worked as below: Send Key click → Phone number click → Call connected
→ Object moved at the sensor → Control the screen's on/off operation automatically

Checking Flow



Image

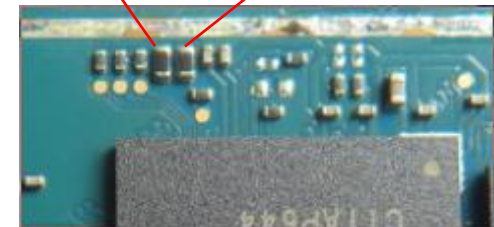


sensor

TP1(R1702)

TP4(R1708)

TP3(R1709)



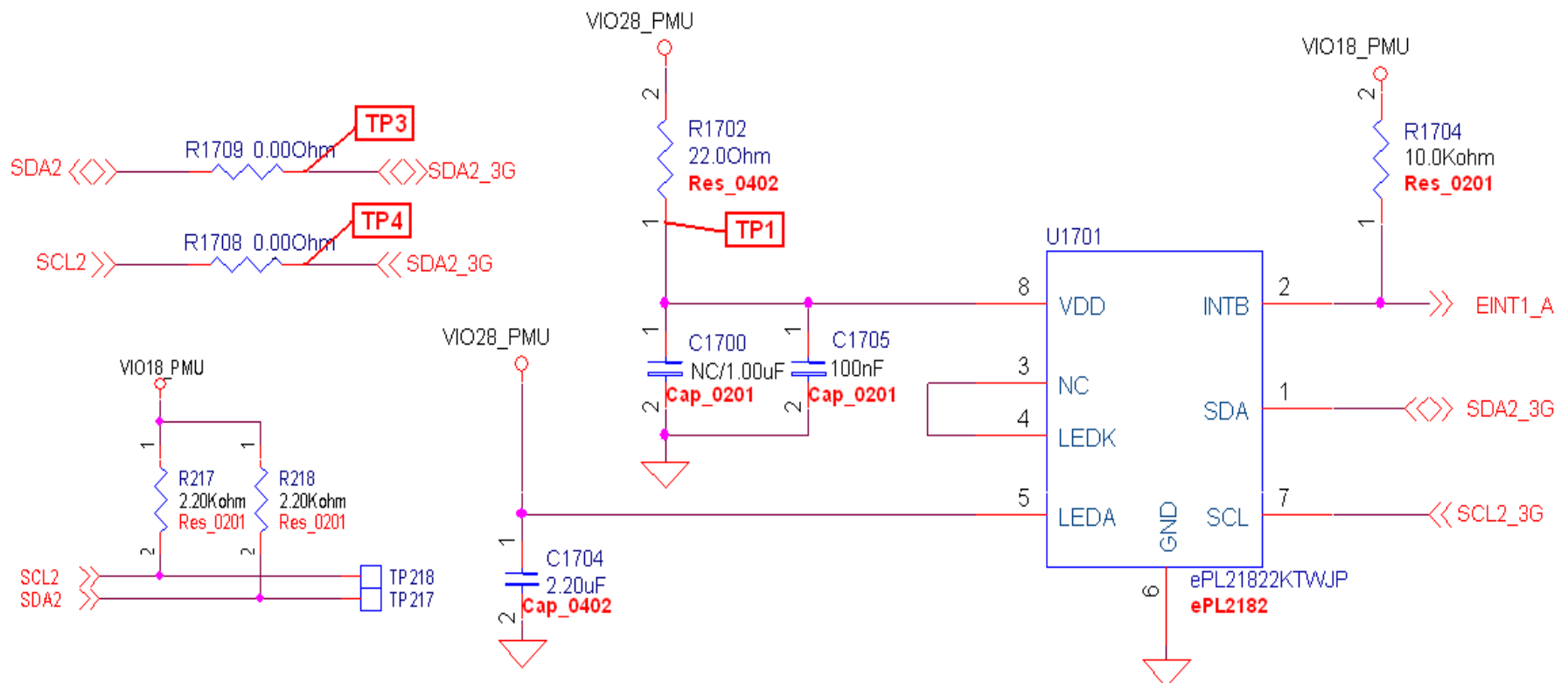
Circuit Diagram
: Next Page

3.12 Checking proximity sensor Block

Proximity Sensor is worked as below: Send Key click → Phone number click → Call connected
 → Object moved at the sensor → Control the screen's on/off operation automatically

Circuit Diagram

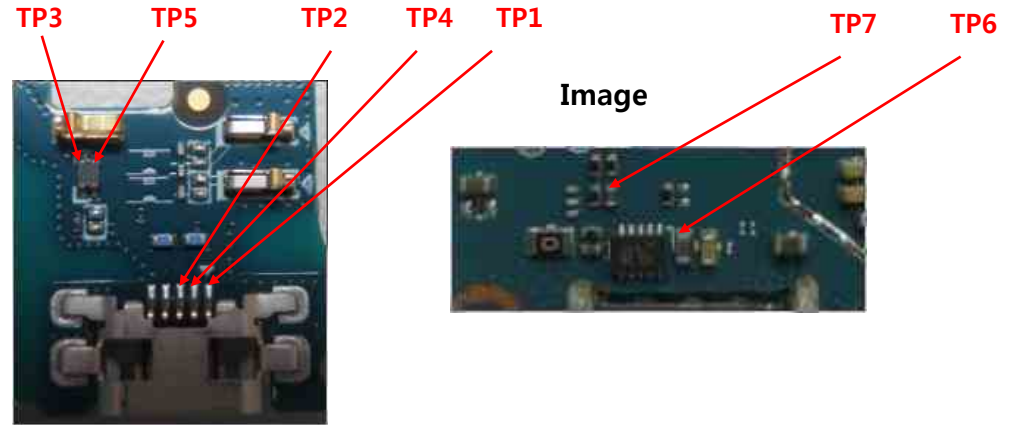
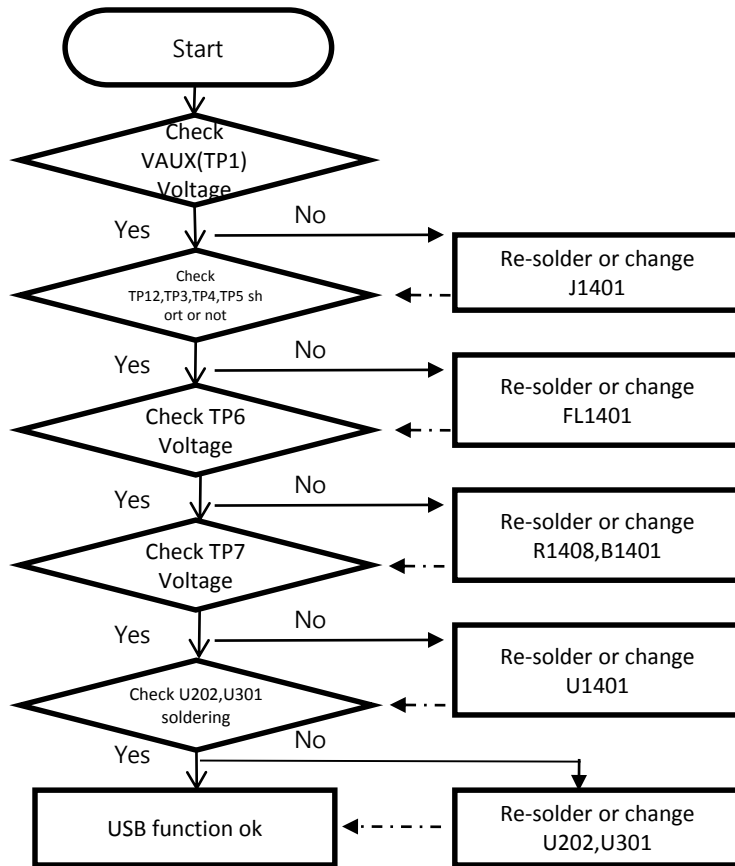
3 in 1 ALS+PS+IR



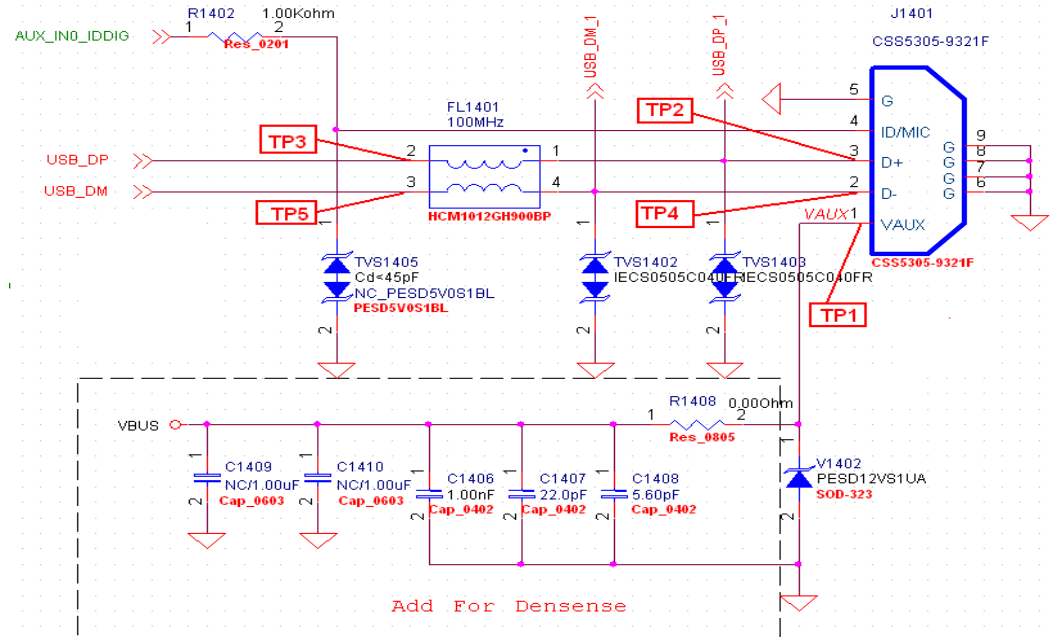
3.13 Checking USB Block

I/O connector is used as the USB port.

Checking Flow



Circuit Diagram

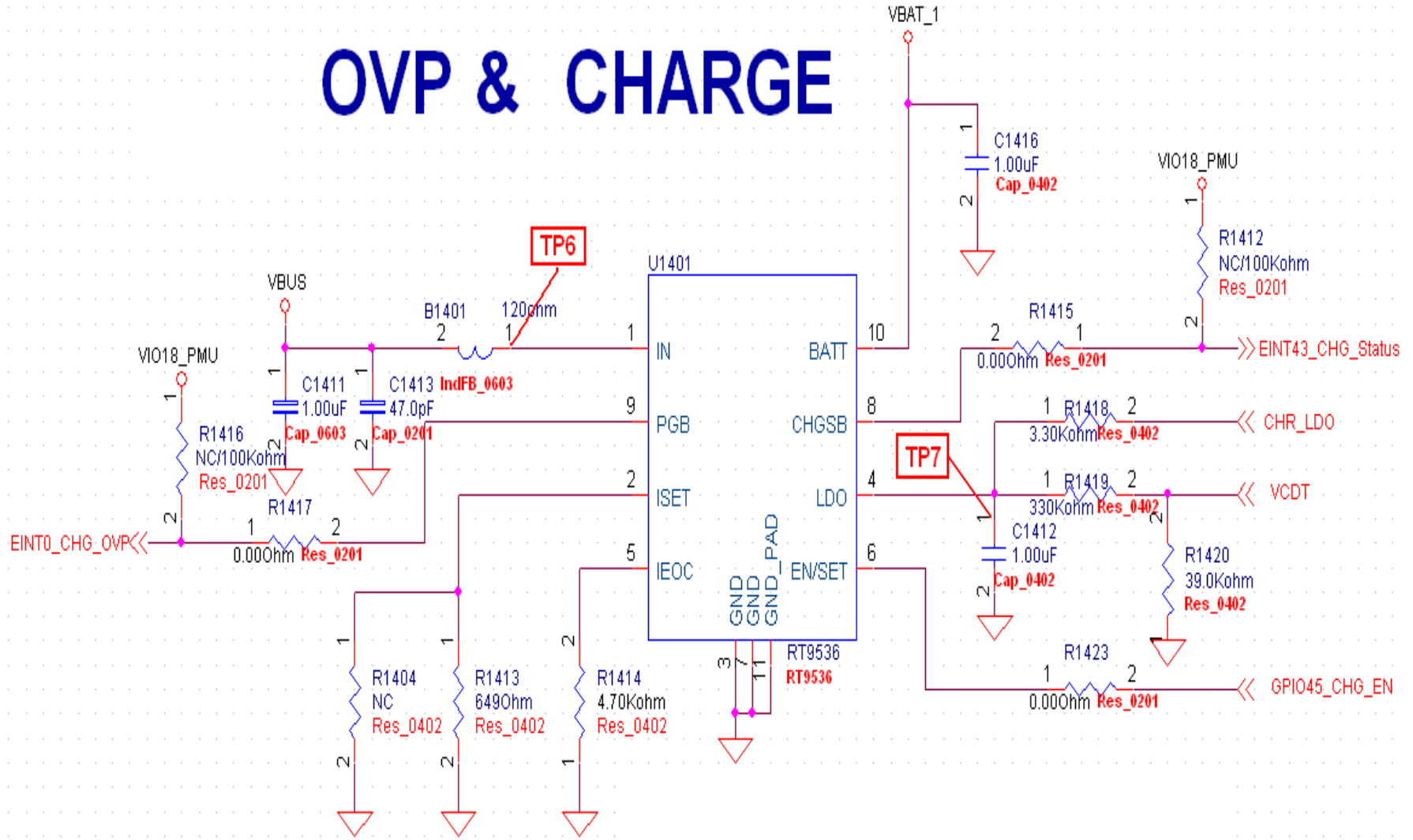


3.13 Checking USB Block

I/O connector is used as the USB port.

Circuit Diagram

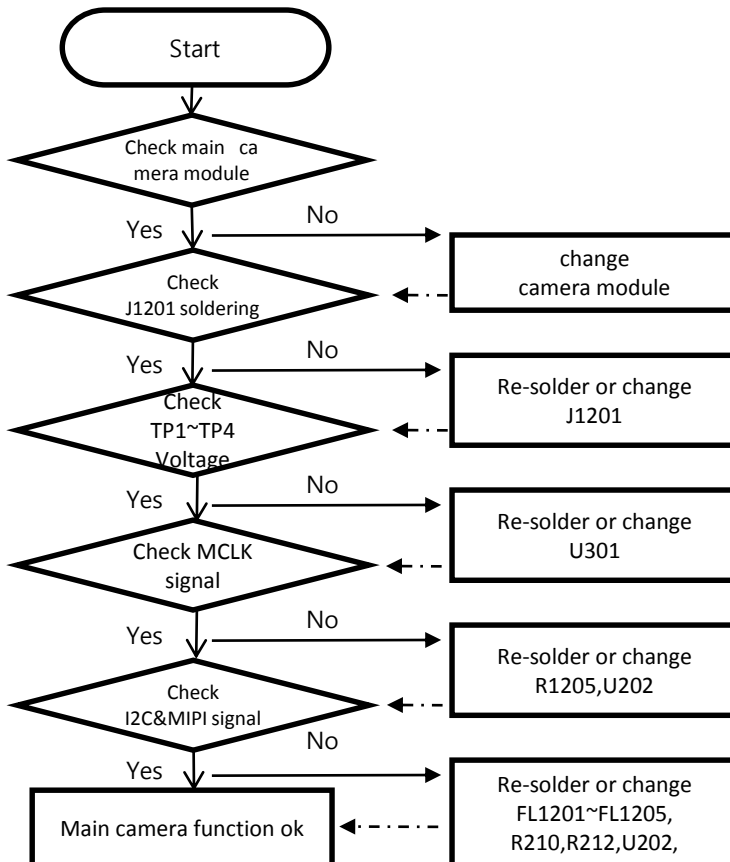
OVP & CHARGE



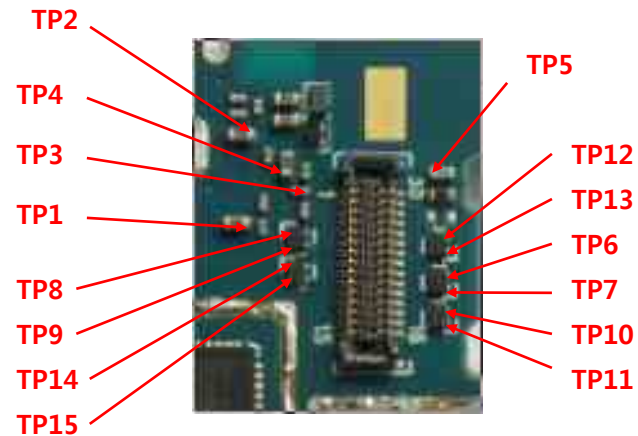
3.14 Checking Main camera Block

8M camera control signals are generated by MT6582(U202). And powered by PMU(U301).

Checking Flow



Image



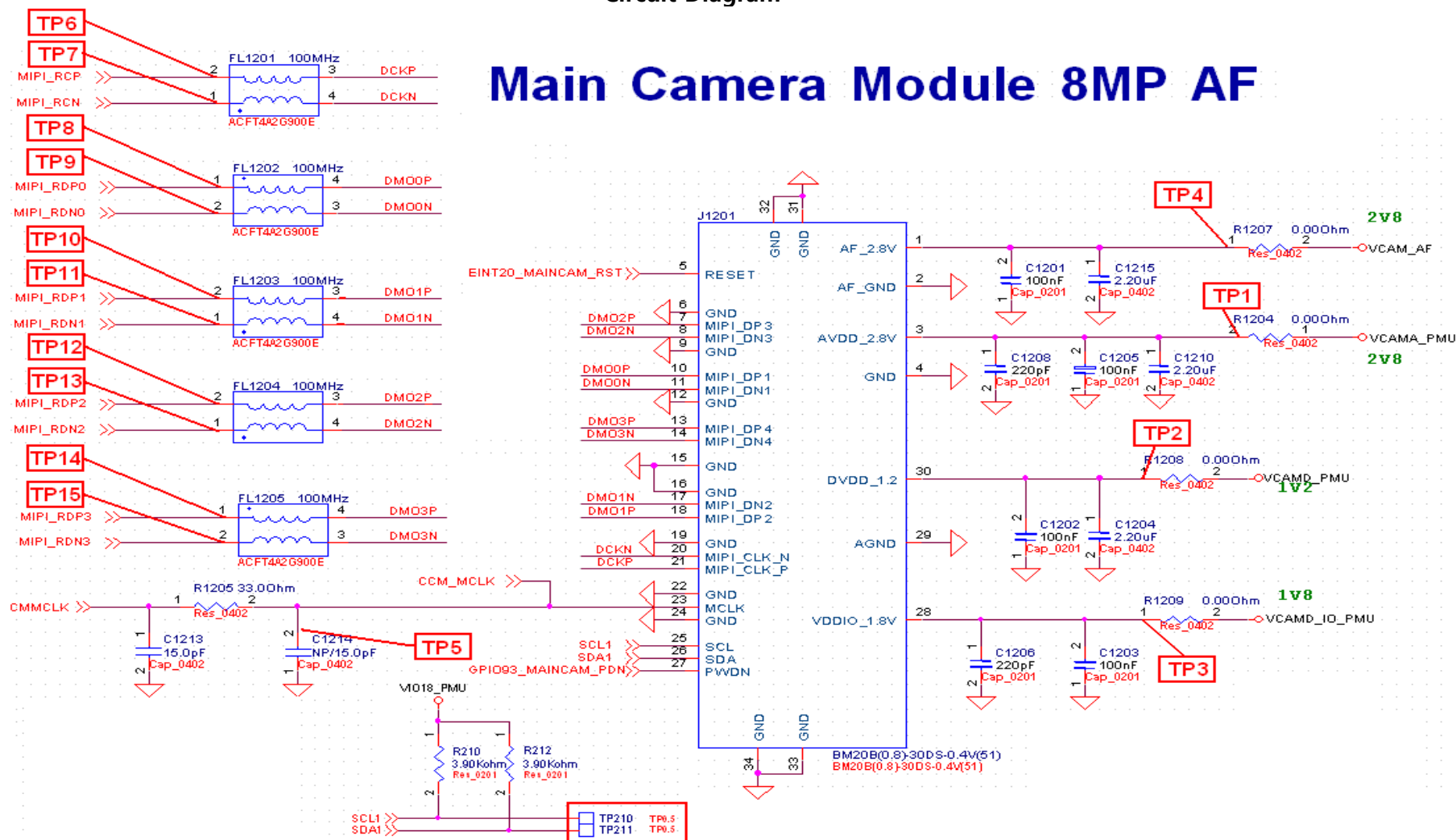
**Circuit Diagram
: Next Page**

3.14 Checking Main camera Block

8M camera control signals are generated by MT6582(U202). And powered by PMU(U301).

Circuit Diagram

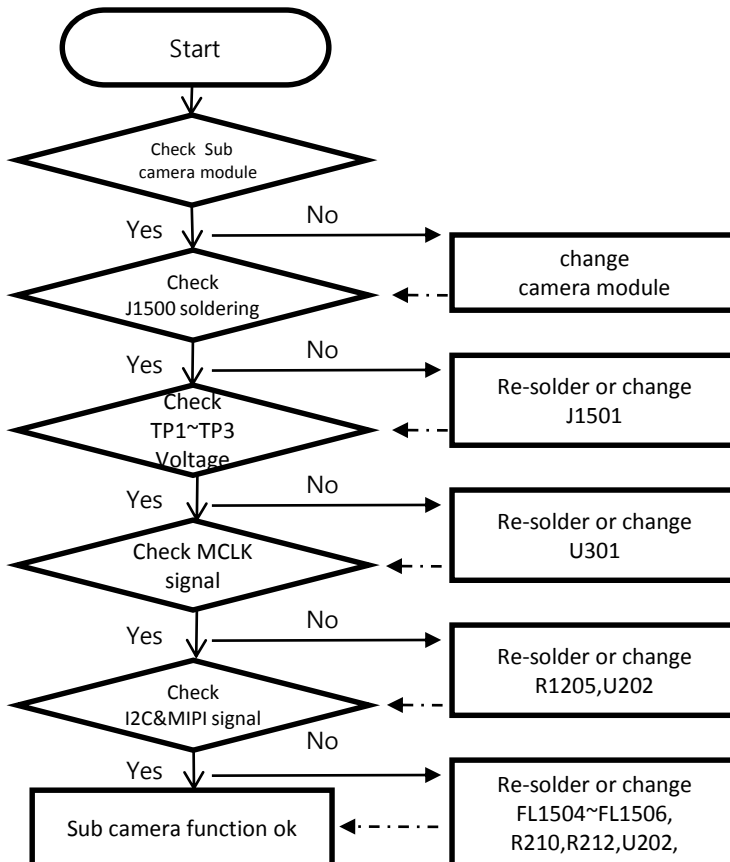
Main Camera Module 8MP AF



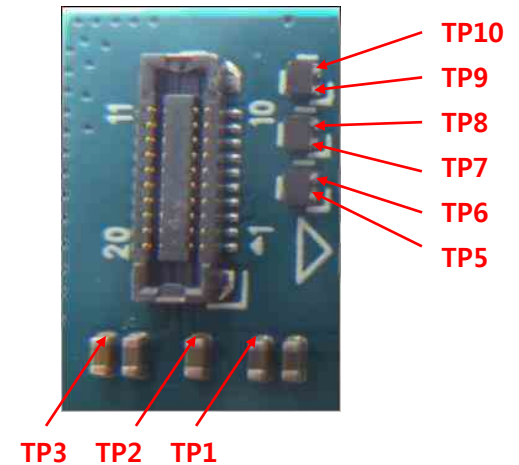
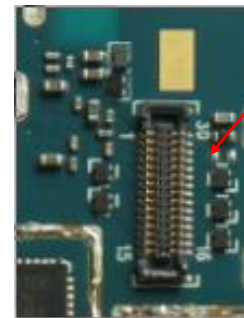
3.14 Checking sub camera Block

Sub camera control signals are generated by MT6582(U202). And powered by PMU(U301).

Checking Flow



Image



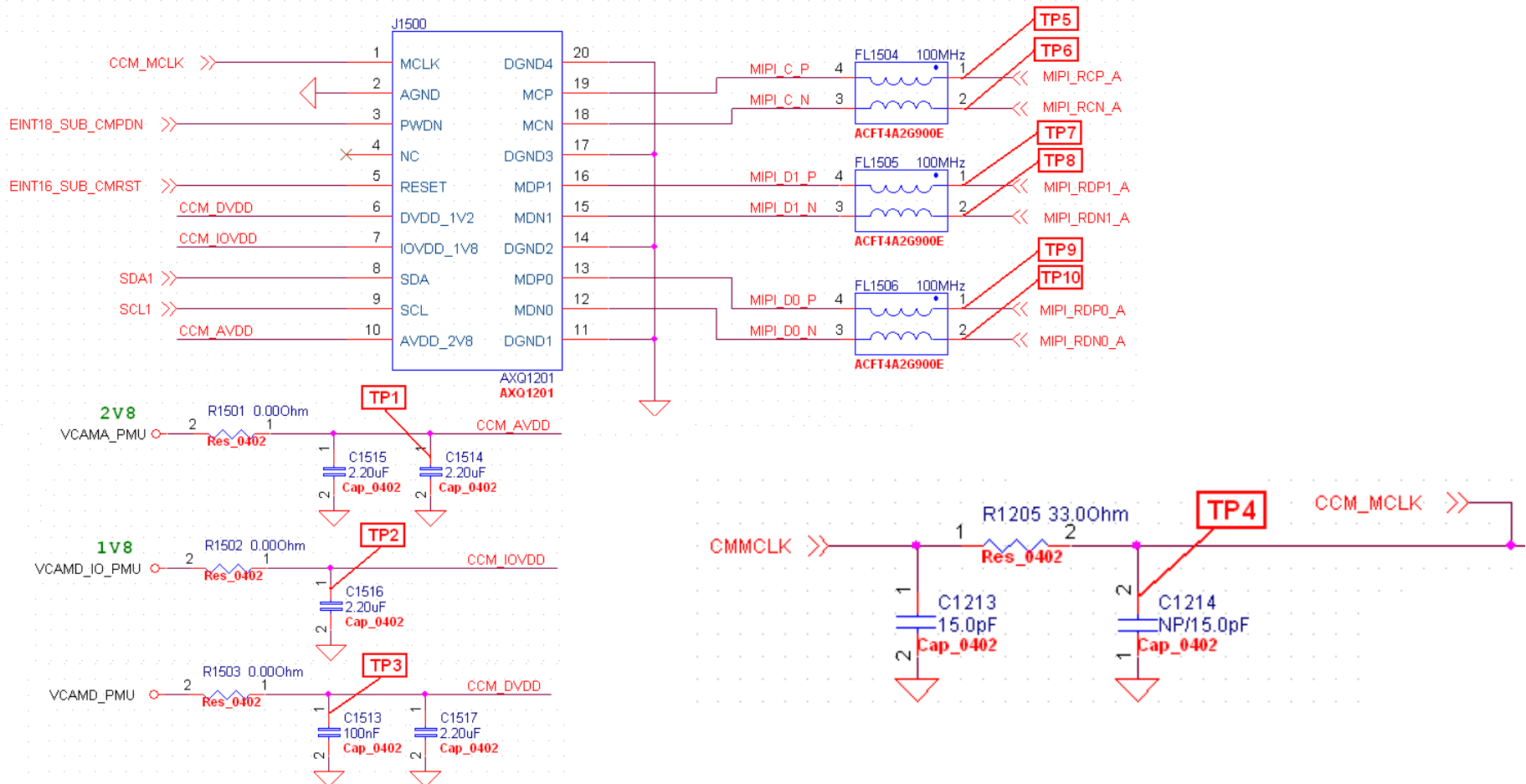
**Circuit Diagram
: Next Page**

3.14 Checking sub camera Block

Sub camera control signals are generated by MT6582(U202). And powered by PMU(U301).

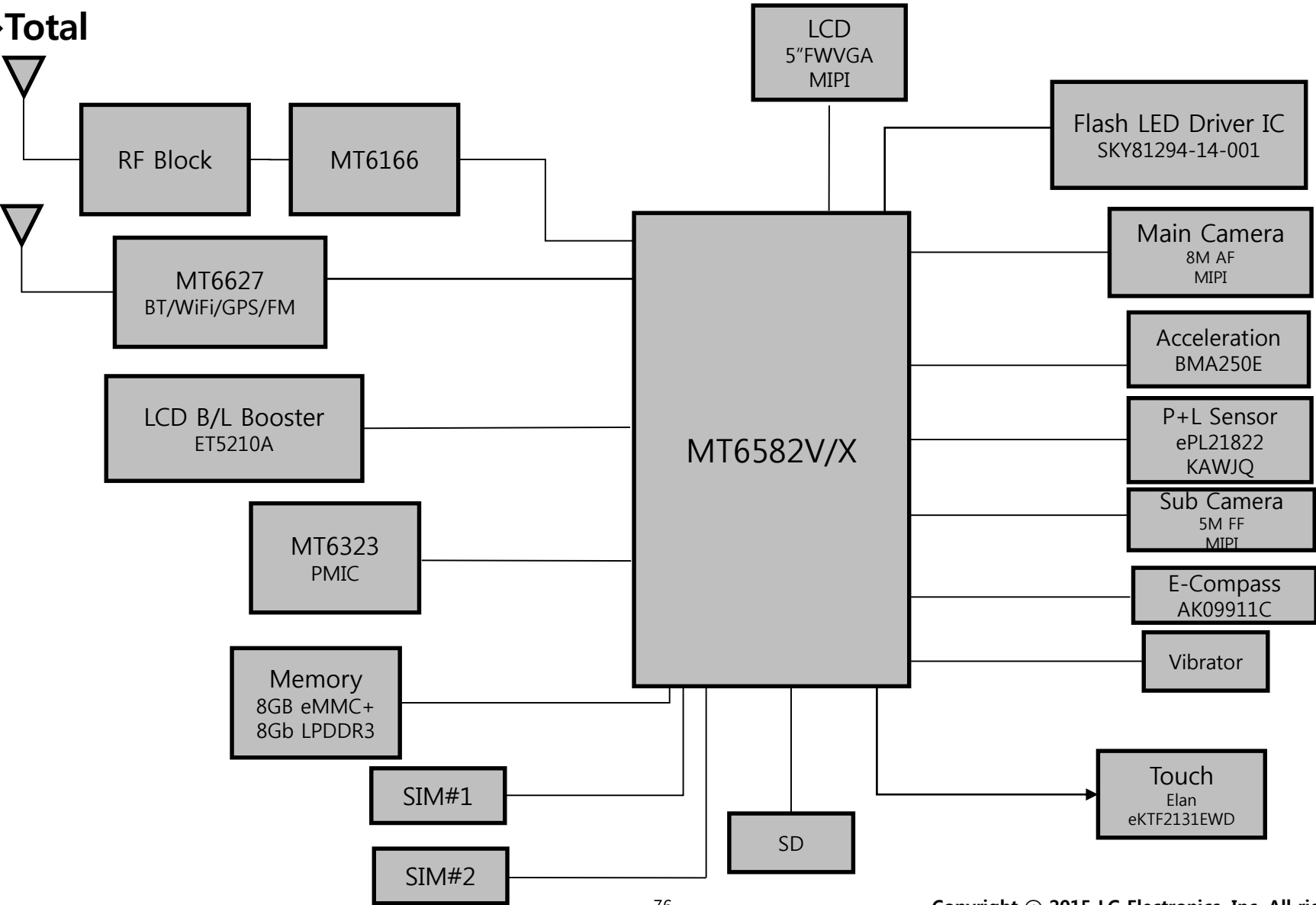
Circuit Diagram

SUB Camera Module 2M/5M Compatible

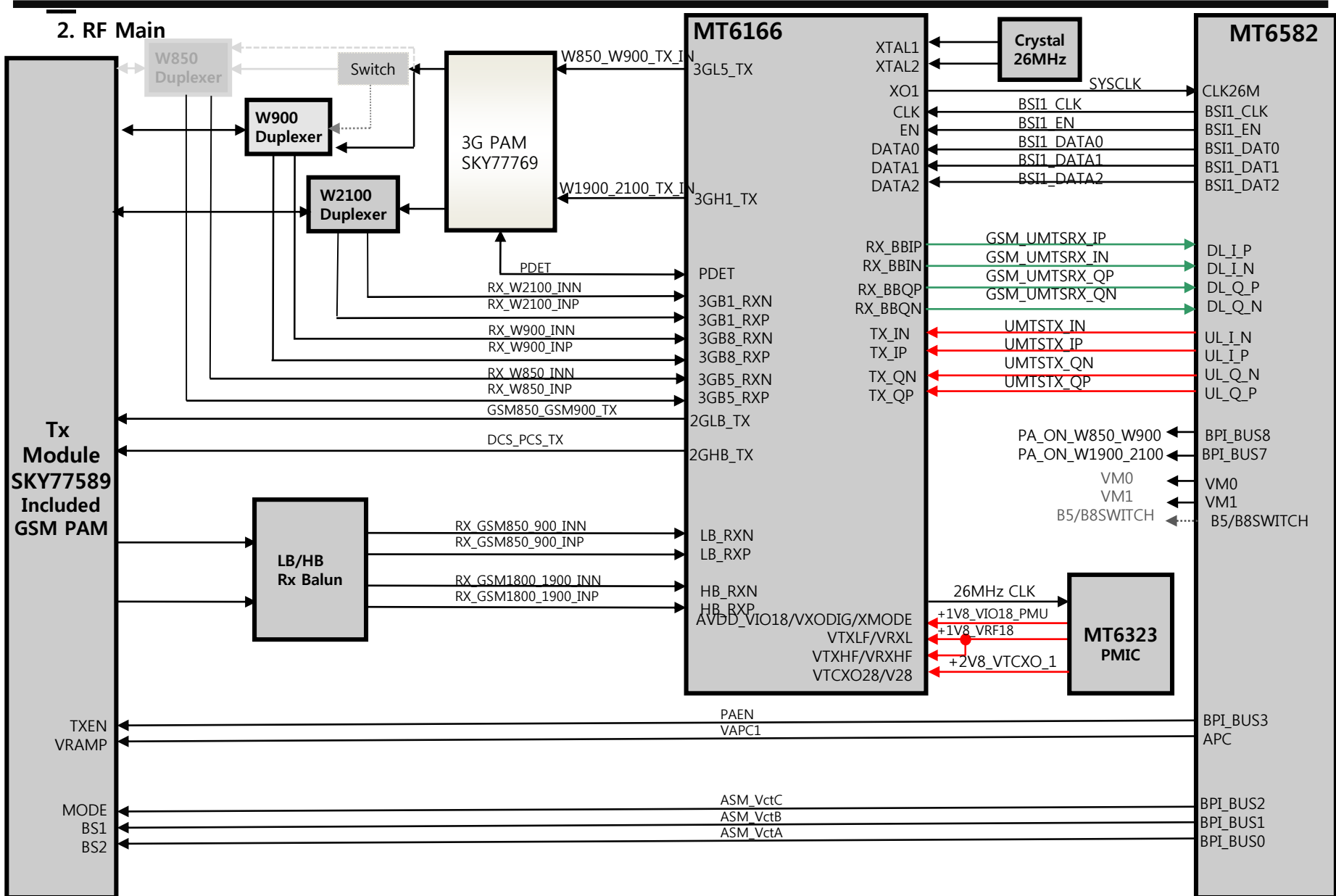


1. Total block diagram

◆ Total

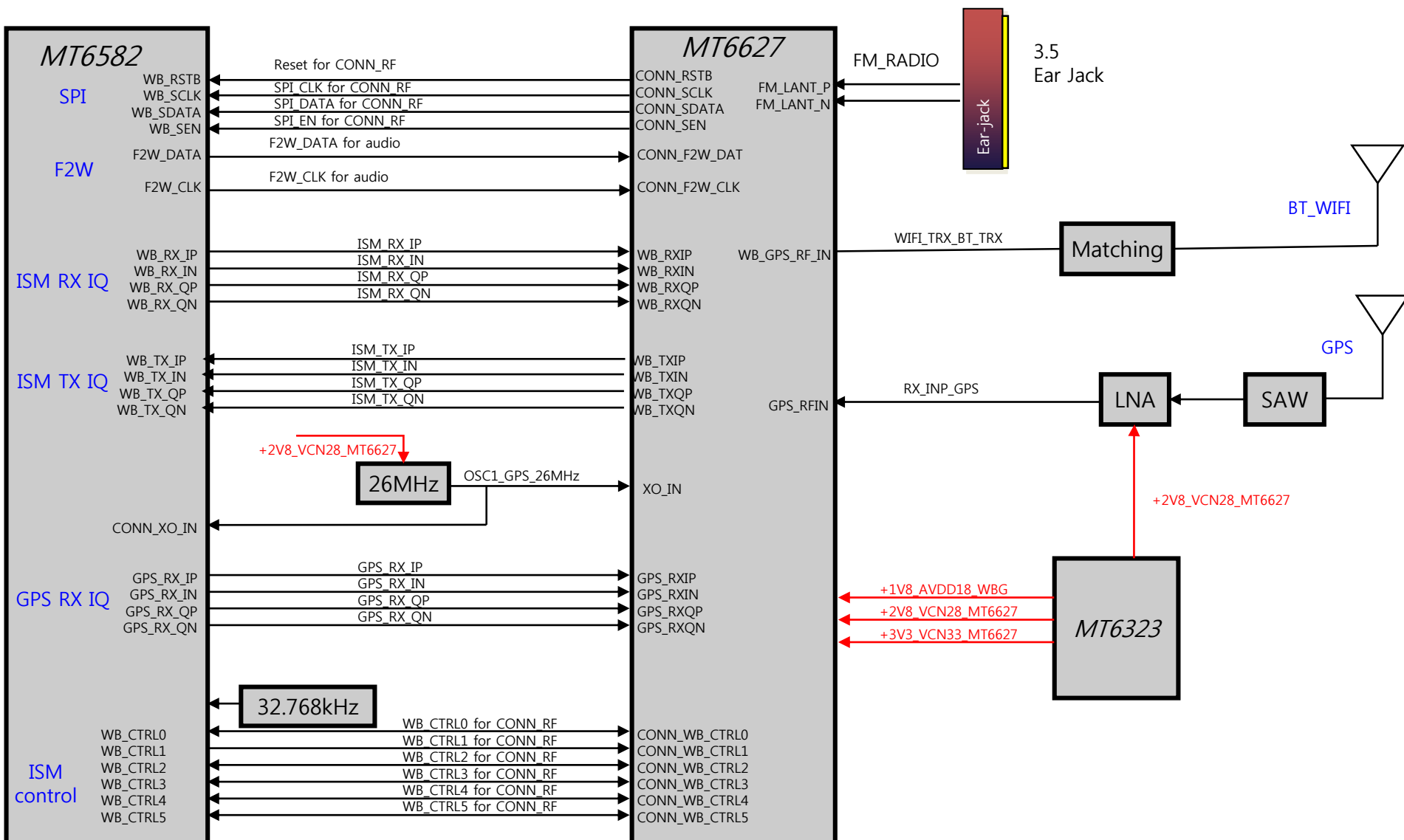


4. BLOCK DIAGRAM



4. BLOCK DIAGRAM

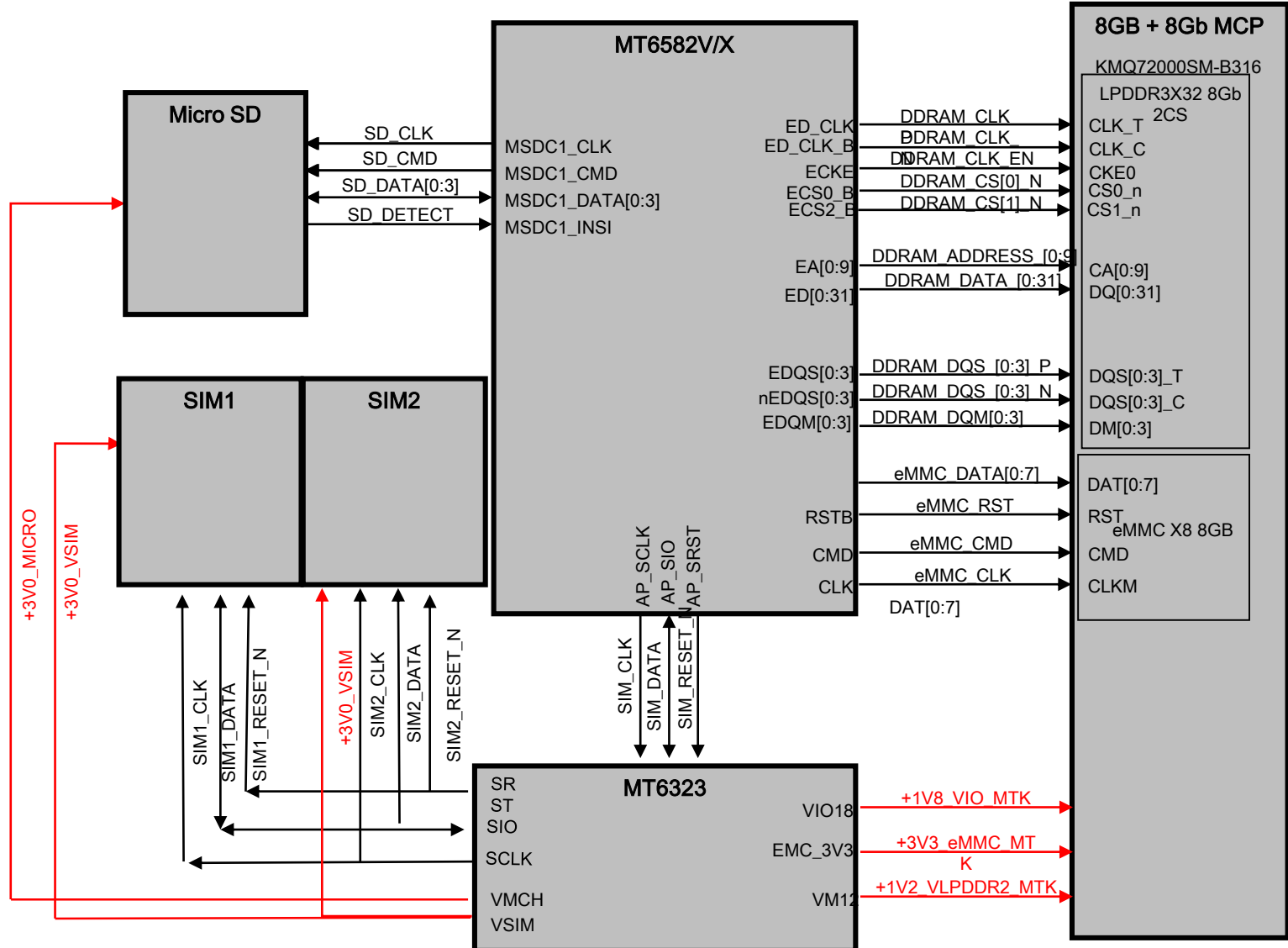
3. Connectivity (BT / WIFI / FM RADIO)



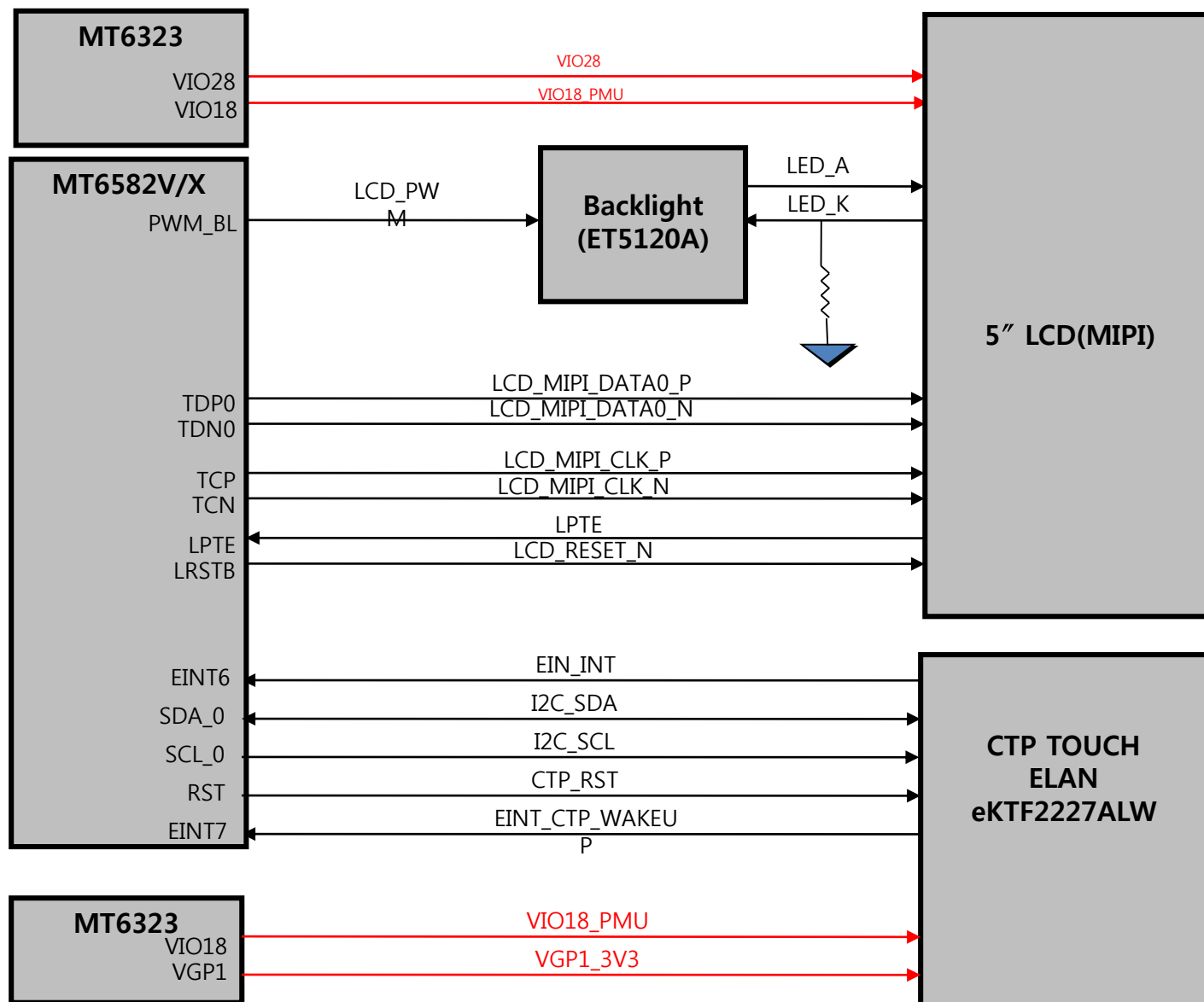
4. Power block diagram



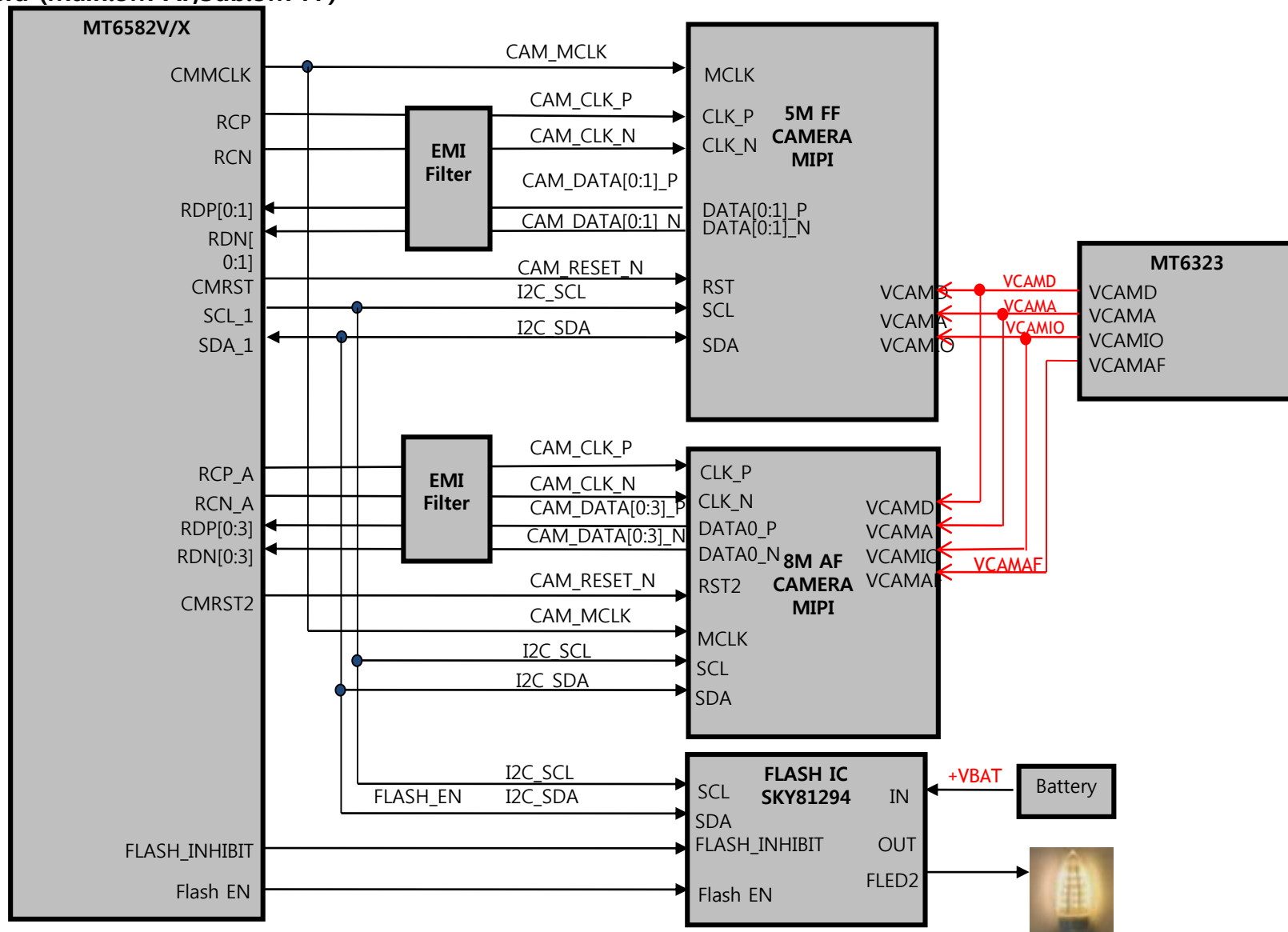
5.Memory, MICRO-SD, USIM



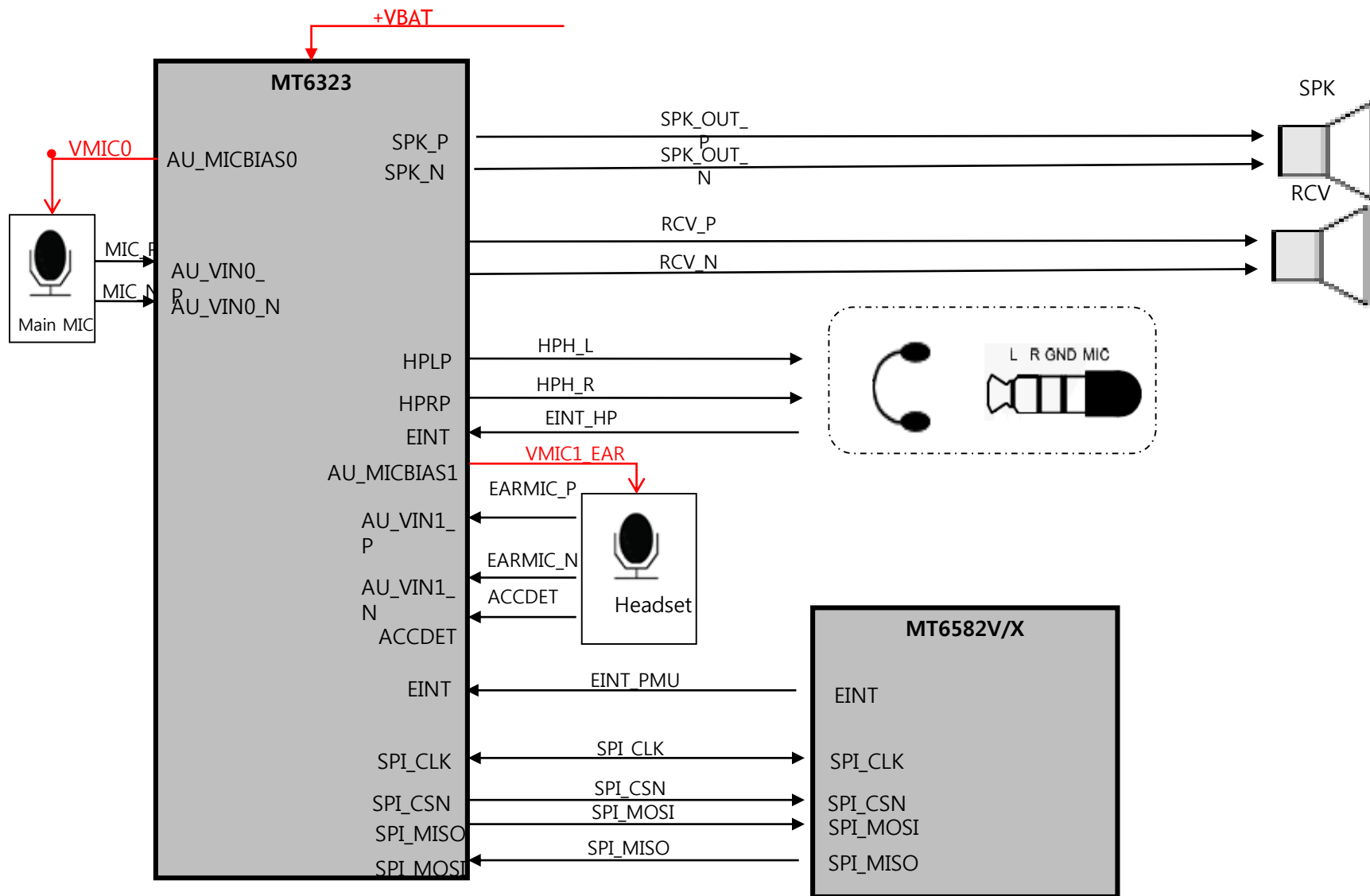
6. LCD(INCELL TOUCH)



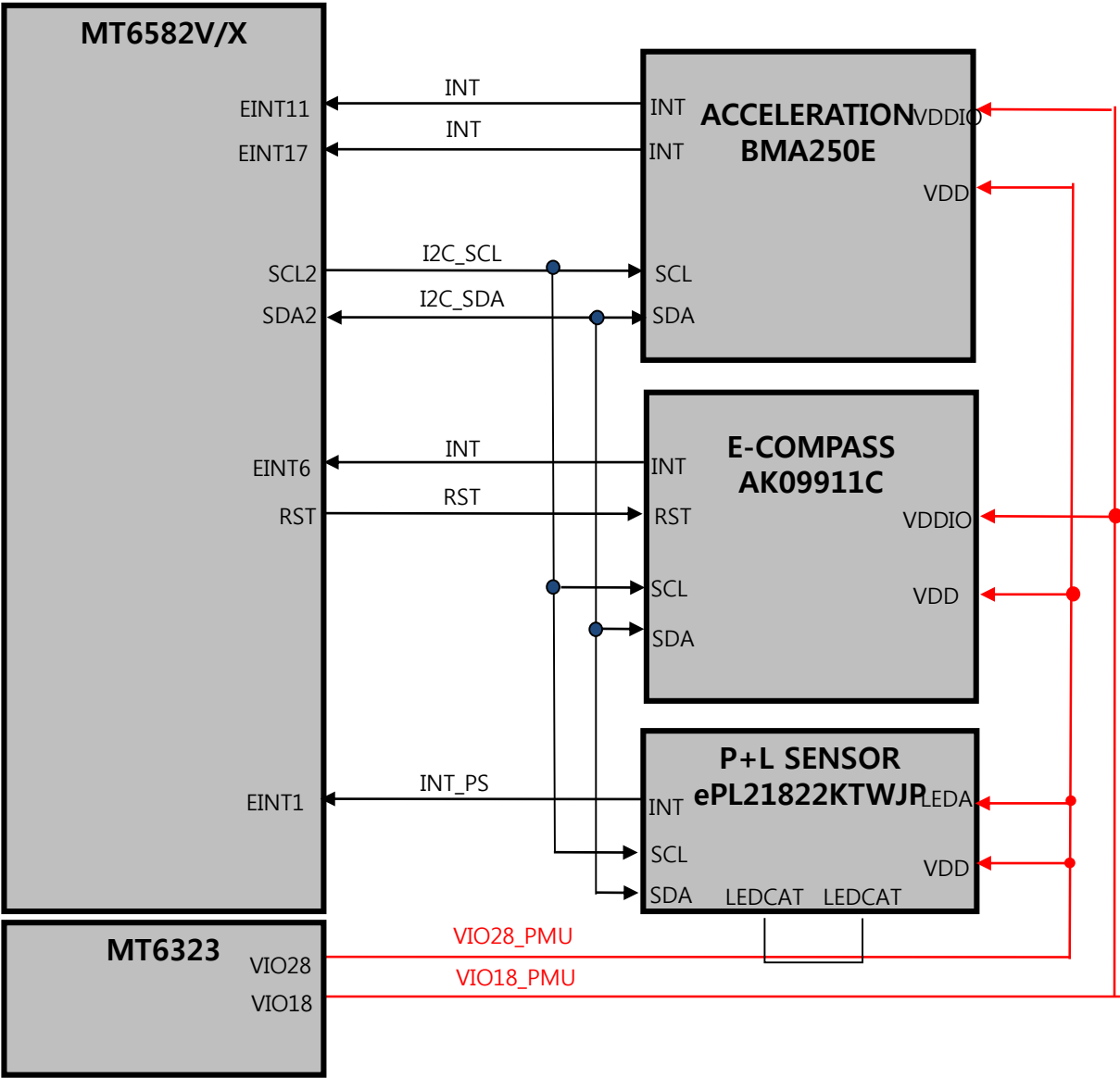
7. Camera (Main:5M AF,Sub:5M FF)



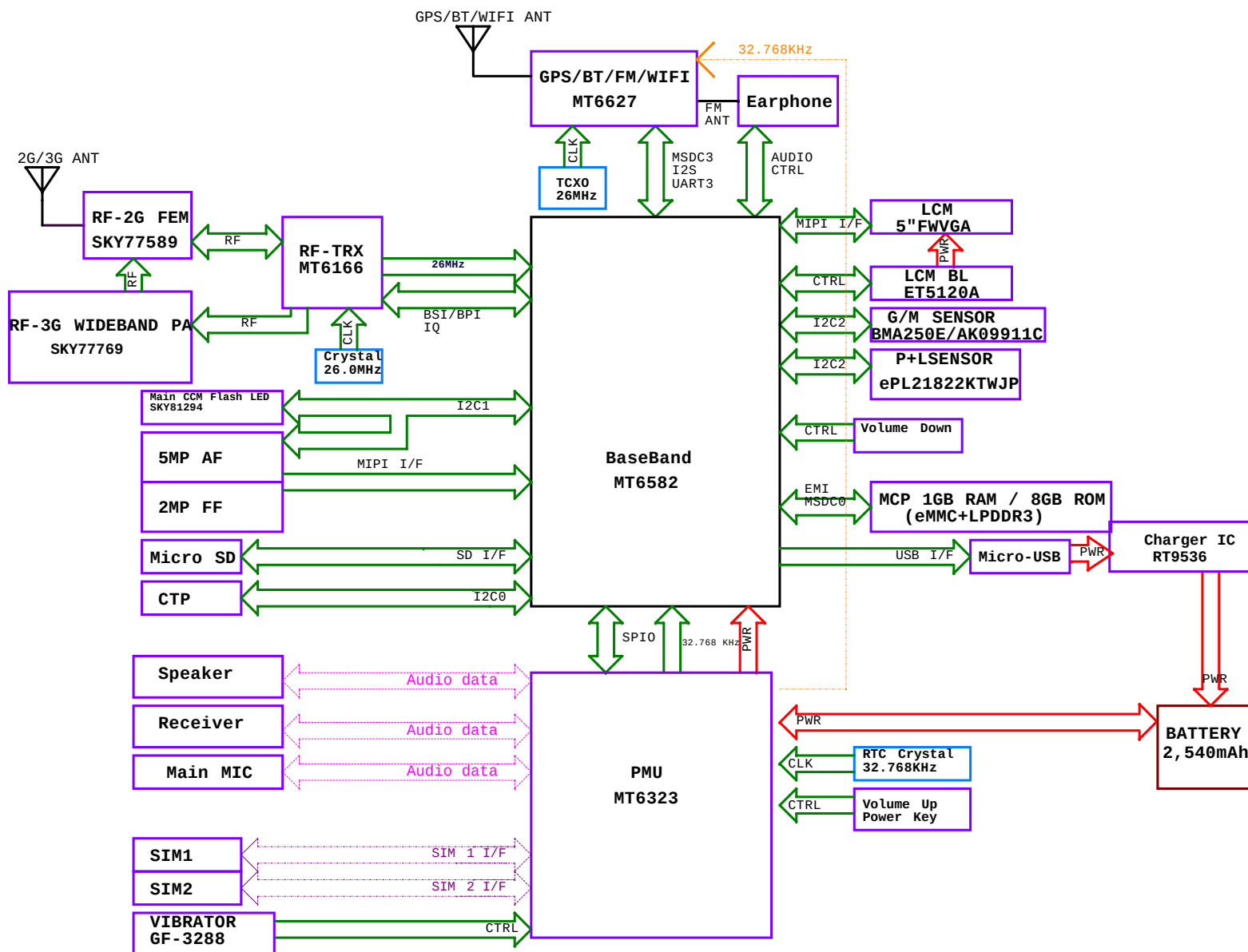
8. AUDIO



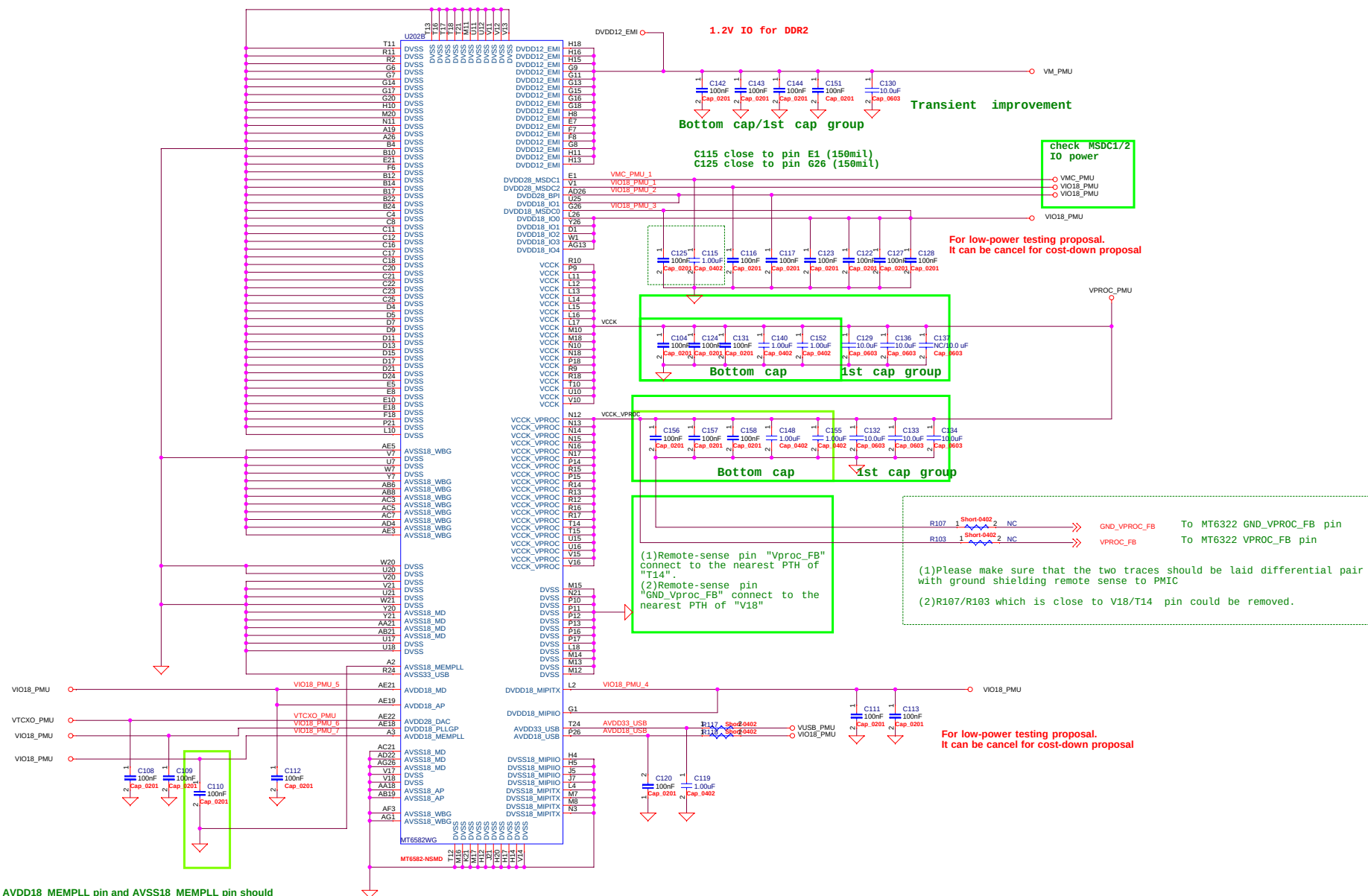
9. SENSORS



CIRCUIT DIAGRAM



MT6582_Power



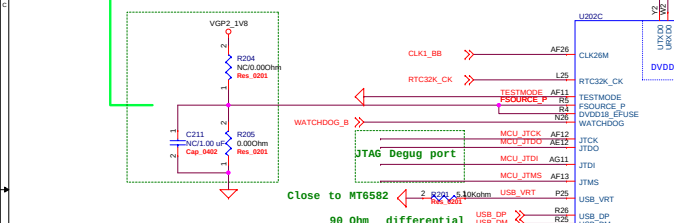
AVDD18_MEMPLL pin and AVSS18_MEMPLL pin should be connected with PCB CAP first, and then connected with PMU and PCB ground. CAP should be near MT6582 as possible as we can.

MT6582_Baseband

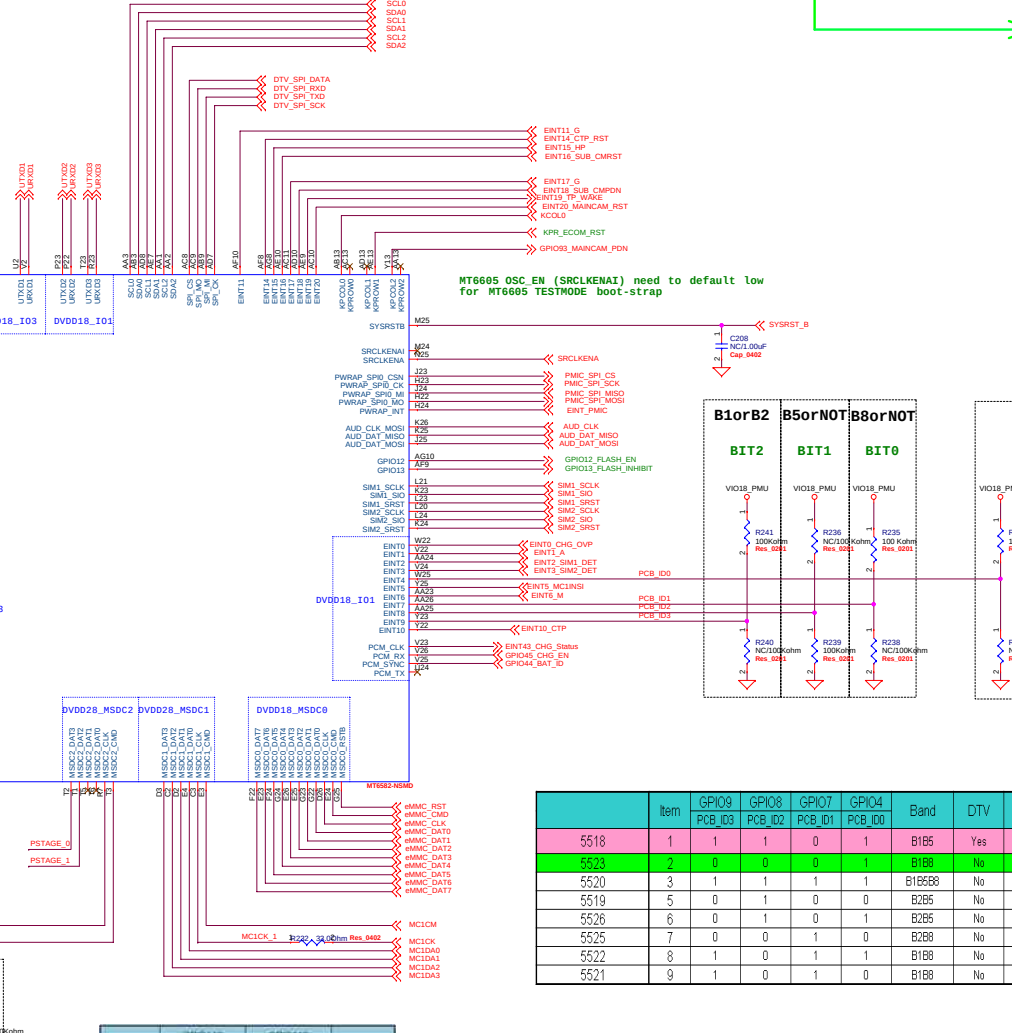
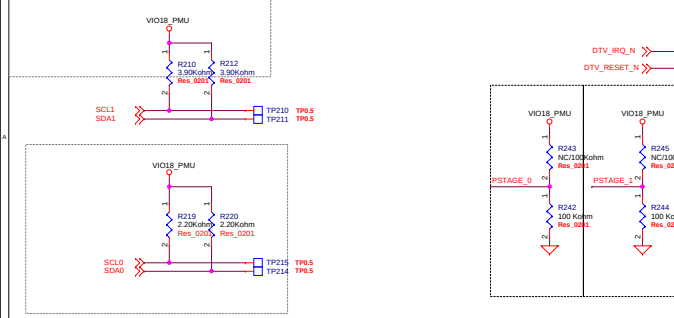
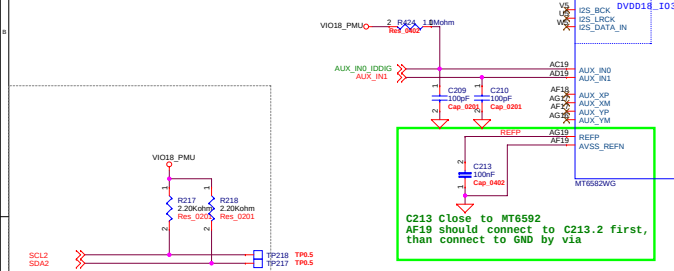
EFUSE Notice :
 (1)EFUSE power (VGPx) range = 1.71V~1.98V
 (2)EFUSE power (VGPx) should be only for EFUSE usage
 (not share with other application)
 (3)VGPx need 1uF bypass cap (pls refer to page 3 "LDO output voltage/current table")

Pin	DVDD18_EFUSE	FSOURCE_P	Description
Voltage	VGPx (1V8)		w/ EFUSE program
	GND		w/o EFUSE program

EFUSE Notice

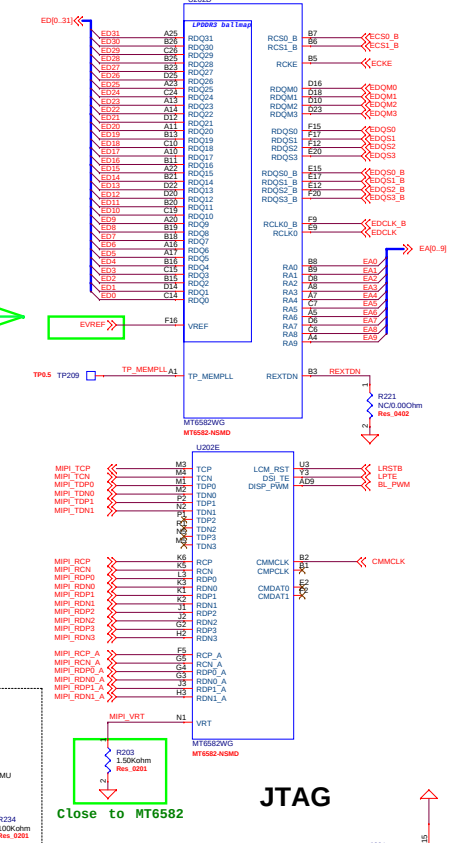


Please reserve bypass cap for AUX_ADC decoupling (from Antenna) proposal. It can be cancel for cost-down proposal



PCB Stage	GPI0118	GPI0119	Remark
002	1	1	PP
003	1	0	PP2
004	0	1	DV
005	0	0	PV

Item	GPI09	GPI08	GPI07	GPI04	Band	DTV	SIM	Camera	Antenna Model Name	LG Model Name
5518	1	1	1	0	1	B1B5	Yes	Dual	8M AF/5M FF	5518
5523	2	0	0	0	1	B1B6	No	Dual	5M AF/5M FF	5523
5520	3	1	1	1	1	B1B5B8	No	Dual	5M AF/5M FF	5520
5519	5	0	1	0	0	B2B5	No	single	8M AF/5M FF	5519
5526	6	0	1	0	1	B2B5	No	Dual	8M AF/5M FF	5526
5525	7	0	0	1	0	B2B8	No	single	8M AF/5M FF	5525
5522	8	1	0	1	1	B1B6	No	Dual	5M AF/2M FF	5522
5521	9	1	0	1	0	B1B6	No	single	5M AF/2M FF	5521

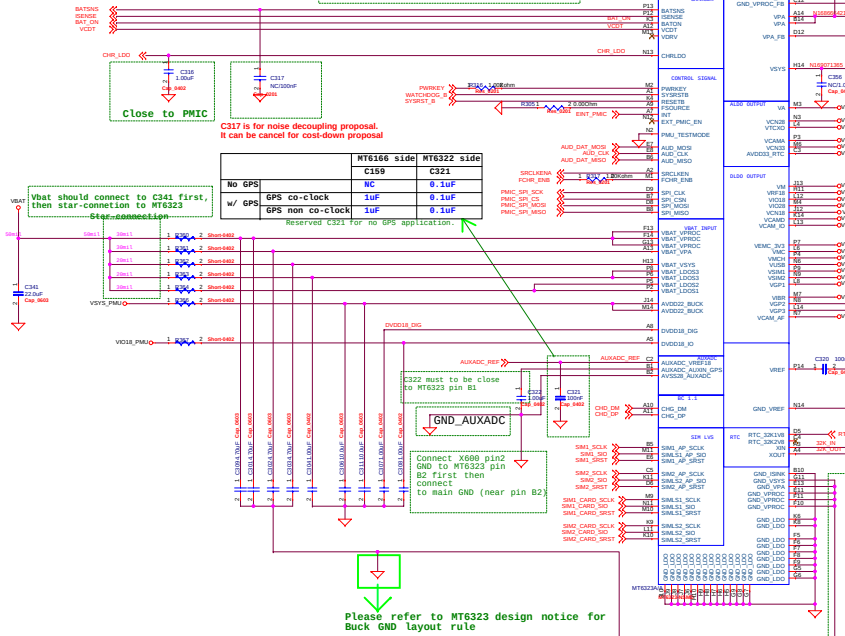


JTAG

PMIC_MT6323

Symbol	LP00R2/1.2V	PC00R3/1.35V	PC00R3/1.5V	LP00R1/1.8V	Default
SPI_CS#	N	L (20K)	N	L (20K)	PU
AUX_R0S1	L	N (20K)	N (20K)	L	PU

MT6323 switching charger function de-feature (Need to add external SWCHR)
Change notice:
Page 3
1. Add R335, MT6323 pin R13 (CHRLDO) connect to R335, then connect to VBUS_1
2. R330.1 connect to VBUS_1



Please refer to "MT6323_6322_for_MT6582_MT6592_Design_Notice" for R334/R336 value selection
Please reserve R331/NTC301 for thermal protection option(close to PA)

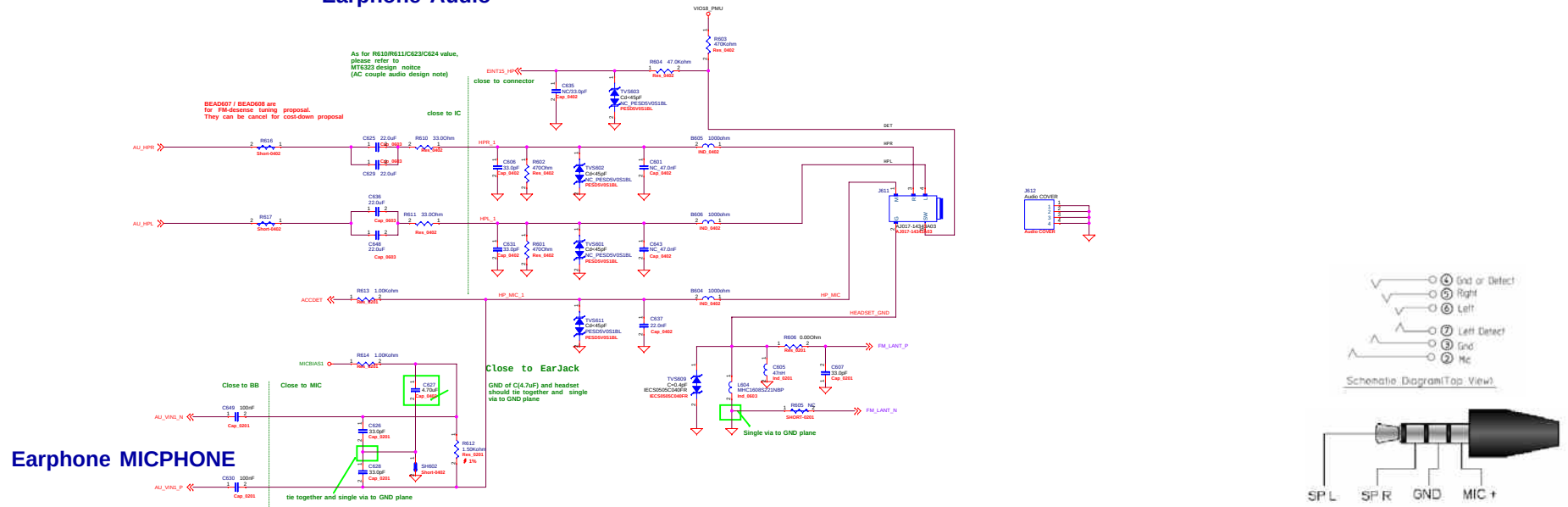
Need to check!!

Regulator	Output Voltage(V)	Output Current(mA)	Input Decoupling	Output Decoupling	Notes
VPROC	0.7~1.4	2800	>10uF	L=0.68uH, C=10uF*4	Total output cap>40uF
VSYS	2.2	1200	>10uF	L=0.68uH, C=10uF*2	Total output cap>20uF
VPA	0.5~3.4	600	>4.7uF	L=2.2uH, C=2.2uF+2.2uF	Output cap range 4.4uF +/-20%
LDO	Output Voltage(V)	Output Current(mA)	Input Decoupling	Output Decoupling	Notes
VM	1.24/1.39/1.54/1.84	700	10uF	-20%~+20%	Far-end bypass cap
VRF18	1.825	200	1uF	-20%~+200%	Far-end bypass cap
VT018	1.8	300	4.7uF	-20%~+200%	Far-end bypass cap
VCN18	1.8	120	1uF	-20%~+20%	Far-end bypass cap
VCAMD	1.2 / 1.3/1.5/1.8	150	1uF	-20%~+20%	Far-end bypass cap
VCAM IO	1.8	100	1uF	-20%~+20%	Far-end bypass cap
VGP3	1.2 / 1.3/1.5/1.8	200	1uF	-20%~+20%	Far-end bypass cap
VA	2.8	150	1uF	-20%~+20%	Far-end bypass cap
VTCX0	2.8	40	1uF	-20%~+20%	Far-end bypass cap
VCN28	2.8	30	1uF	-20%~+20%	Far-end bypass cap
VCAMA	2.8	150	3.2uF	-20%~+20%	1uF near-end, 2.2uF Far-end bypass cap
VCN33	3.3 / 3.4/3.5/3.6	240(MT6323) 350(MT6322)	4.7uF	-20%~+20%	Far-end bypass cap
VT028	2.8	200	2.2uF	-20%~+200%	Far-end bypass cap
VUSB	3.3	20	1uF	-20%~+20%	Far-end bypass cap
VMC	1.8 / 3.3	100	1uF	-20%~+20%	Far-end bypass cap
VMCH	3.0 / 3.3	400	2.2uF	-20%~+20%	Far-end bypass cap
VEM3_3V3	3.0 / 3.3	400	4.7uF	-20%~+20%	Far-end bypass cap
VCAM_AF	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VSIM1	1.8 / 3.0	50	1uF	-20%~+20%	Far-end bypass cap
VSIM2	1.8 / 3.0	50	1uF	-20%~+20%	Far-end bypass cap
VGP1	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VGP2	1.2/1.3/1.5/1.8/2.0 2.5/2.8/3.0	100	1uF	-20%~+20%	Far-end bypass cap
VTBR	1.2/1.3/1.5/1.8/2.0 2.8/3.0/3.3	100	1uF	-20%~+20%	Far-end bypass cap
VDI618	1.8	20	1uF	-20%~+20%	Far-end bypass cap
VRTC	2.8	2	0.1uF to 1000uF	-20%~+20%	Far-end bypass cap

VIO18_PMU 1 R418 2 DVDD18_EMI

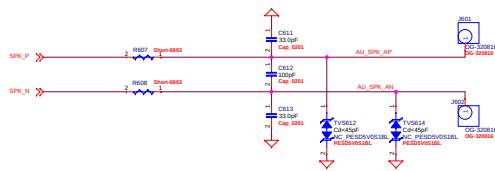


Earphone Audio

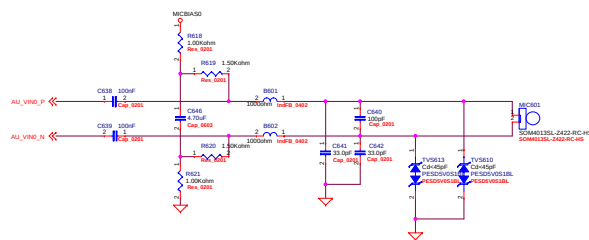


Ohm	Voltage	Function
0	0	Call/Play KEY
200	0.091503	Volume Up (+)
500	0.218069	Volume Down (-)
1180	0.465352	APR

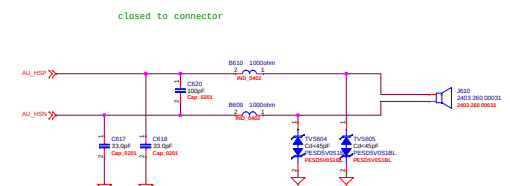
SPEAKER



Main Microphone



RECEIVER



D

C

B

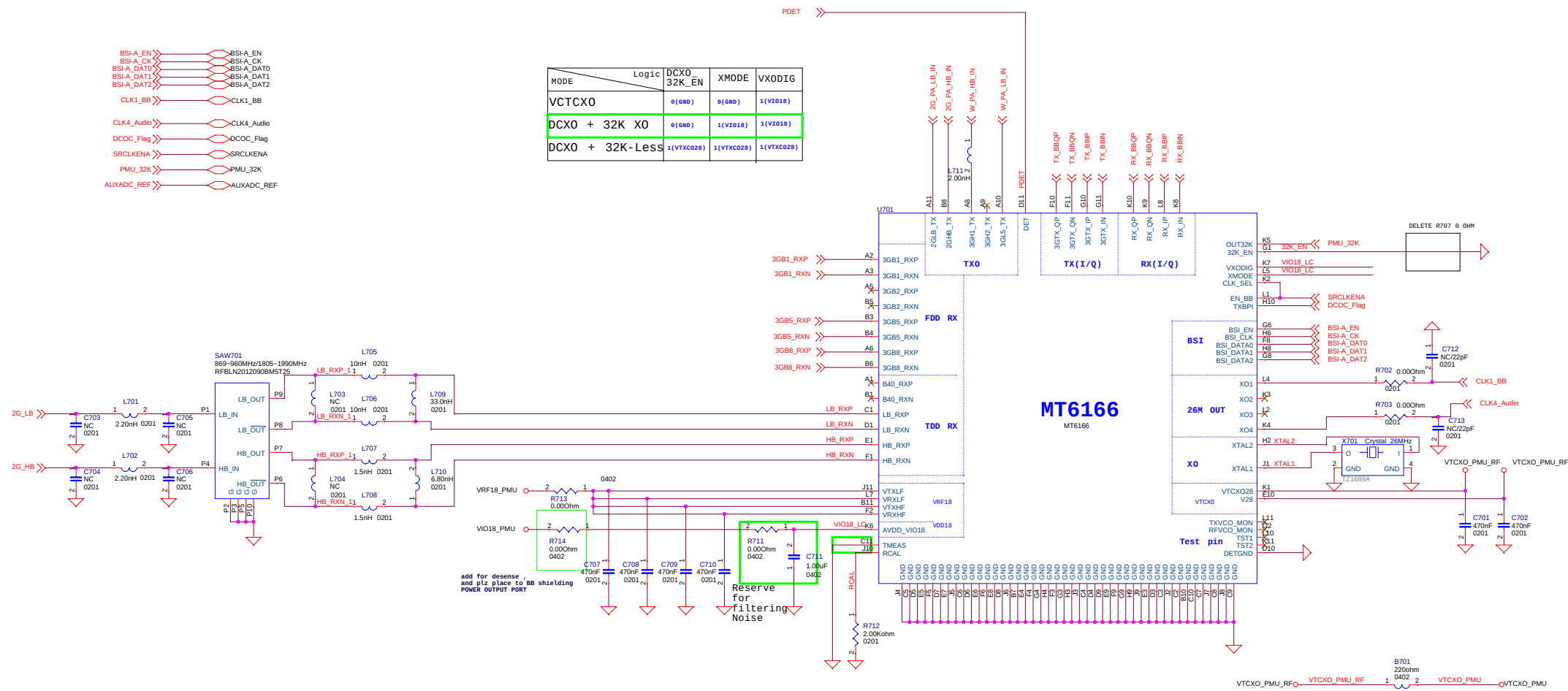
A

D

C

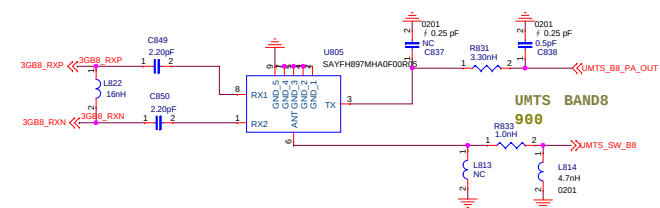
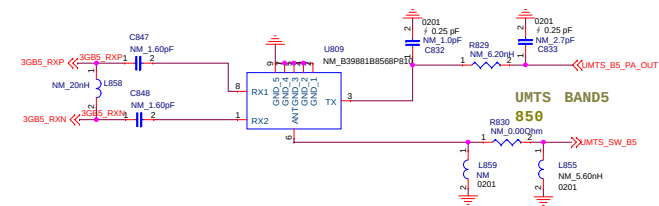
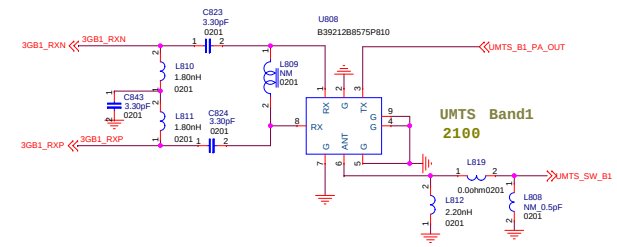
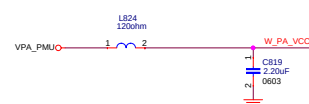
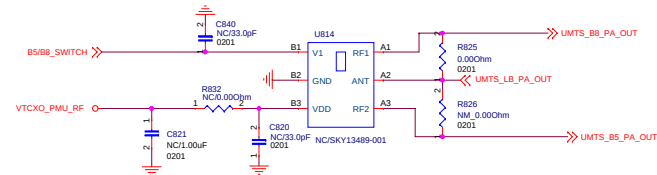
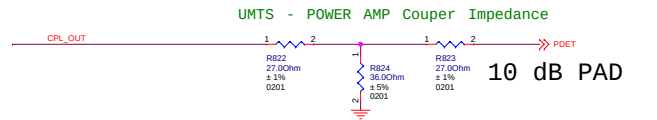
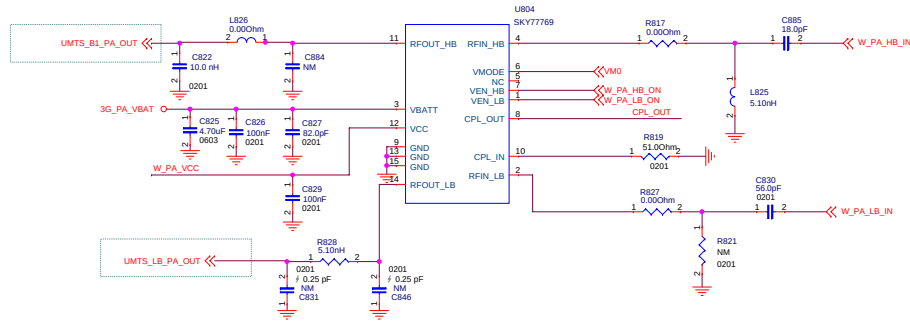
B

A



Net name	Pin name	Power domain	Max Current consumption (mA)
RXHF	F2	VRF18	100
VTCX028	K1	VTCX0	6
VRXLF	L7	VRF18	35
VTXLF	J11	VRF18	35
V28	E10	VTCX0	1
VTXHF	B11	VRF18	125
VIO18	K6	VIO18	1

UMTS -WINDBAND POWER AMP



FEM

Ant. Spring

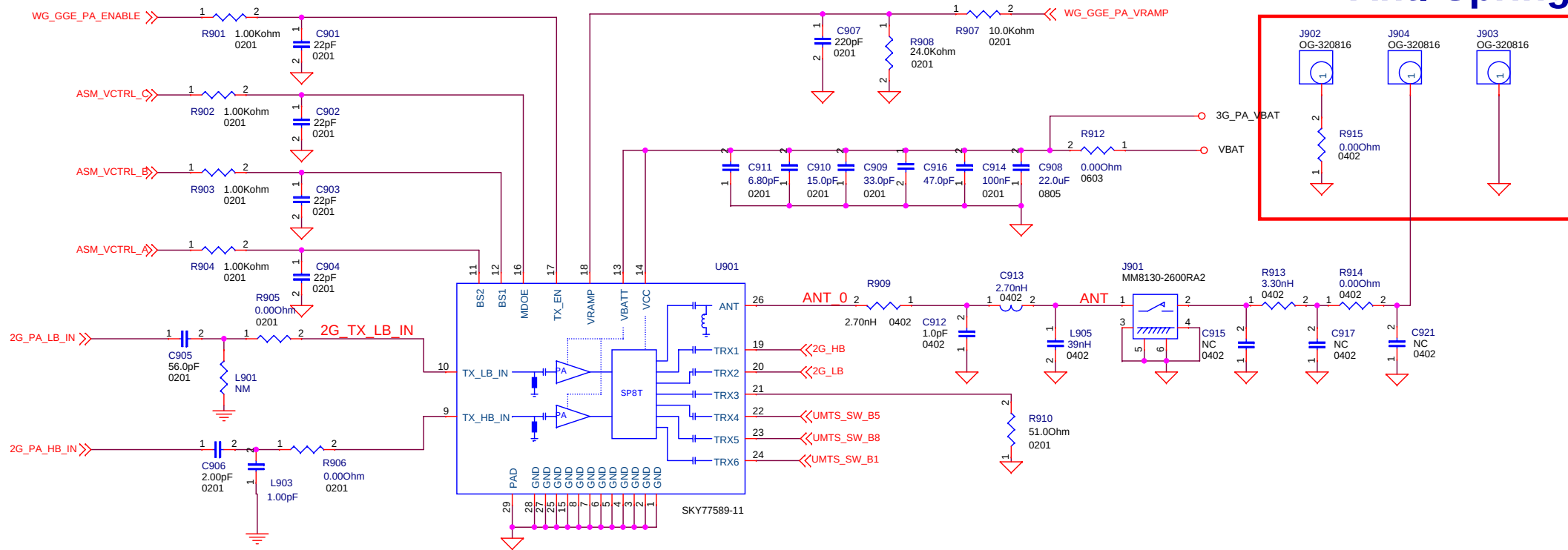


Table 3. SKY77589-11 Mode Control Logic

Mode	Input Control Bits			
	TdN	MODE	BS1	BS2
Standby	0	0	0	0
LB_GMSK_Tx	1	0	0	1
HB_GMSK_Tx	1	0	1	1
Trx1	0	1	0	0
Trx2	0	1	1	0
Trx3	0	1	0	1
Trx4	0	1	1	1
Trx5	0	0	1	0
Trx6	0	0	0	1

GPS xLNA

Close to ANT

matching value depends on LNA selected

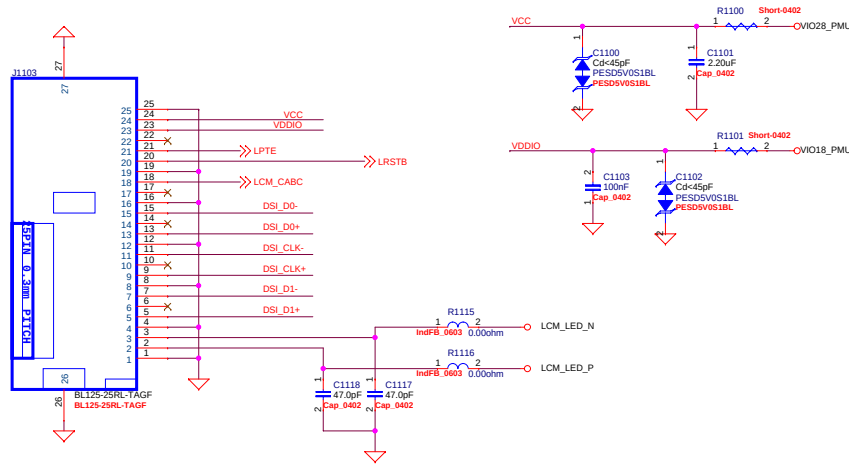
Bead1001 are for FM desense-free proposal.
They can be cancel for cost-down proposal

Close to MT6627

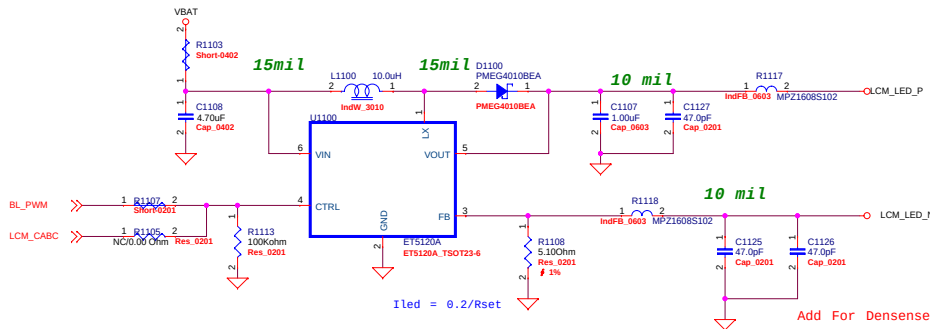
For MMD TP

MODE	Component	X1001	R1015	R1016	R1014
Xtal Mode		QVL TCXO	0ohm	NC	
GPS co-clock		NC	NC	0 ohm	

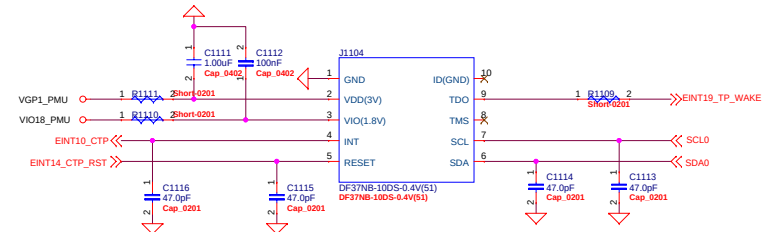
LCM



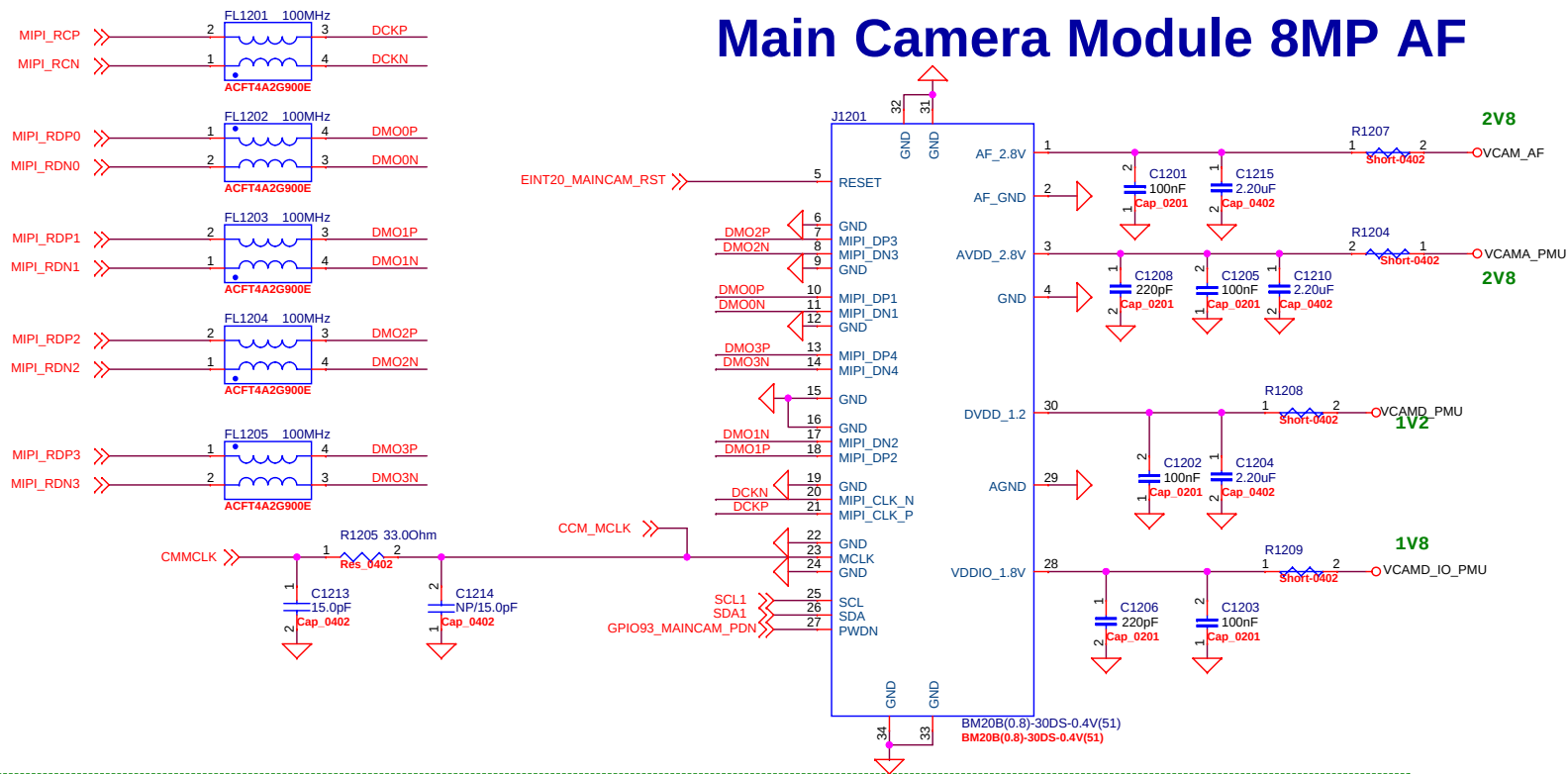
Backlight LED DRV



Capacitive Touch Panel

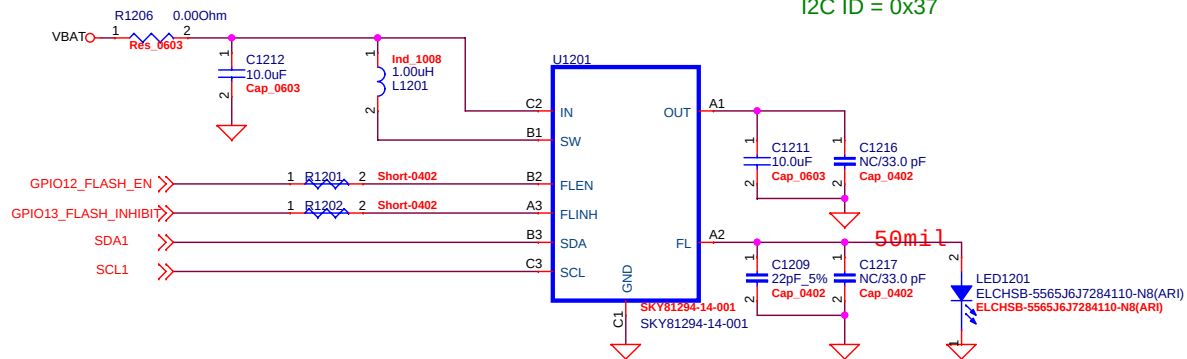


Main Camera Module 8MP AF

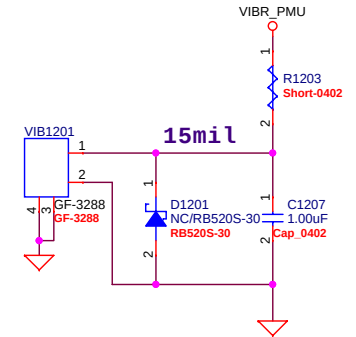


MAIN CCM FLASH LED Driver

I2C ID = 0x37

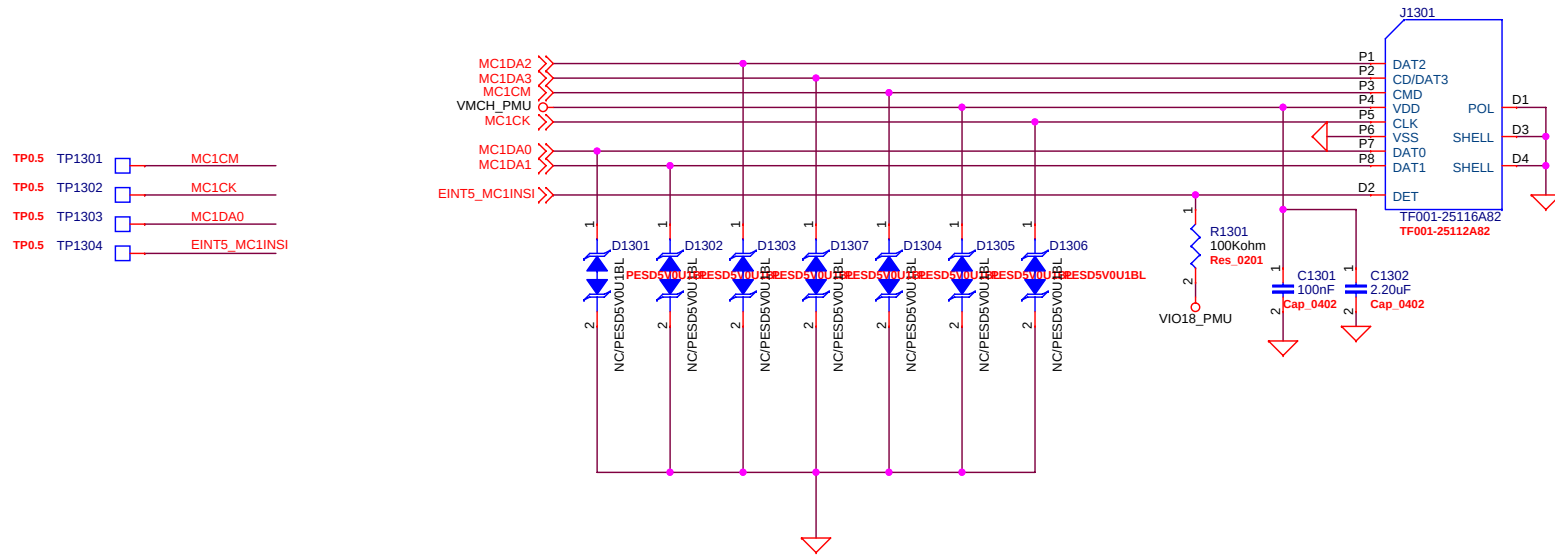


Vibrator

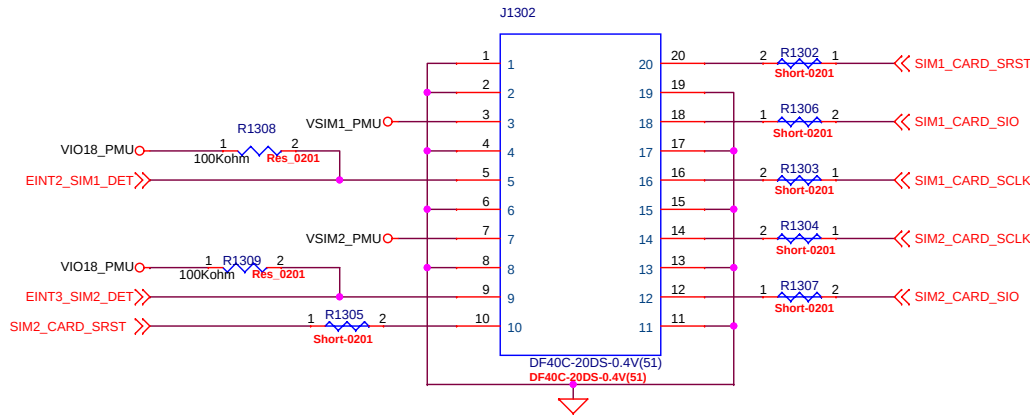


Arima Communications Corp.	
Title	
X155 5522 MainBoard PV	
Size	Document ID
12_Main Camera_FlashLED_Vib	
Date:	Thursday, June 18, 2015
Sheet	11 of 16
Rev	PA1

mSD CARD



SIM Board I/F



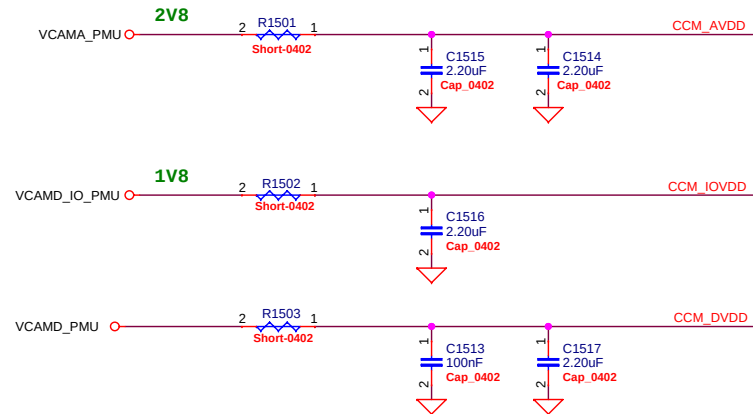
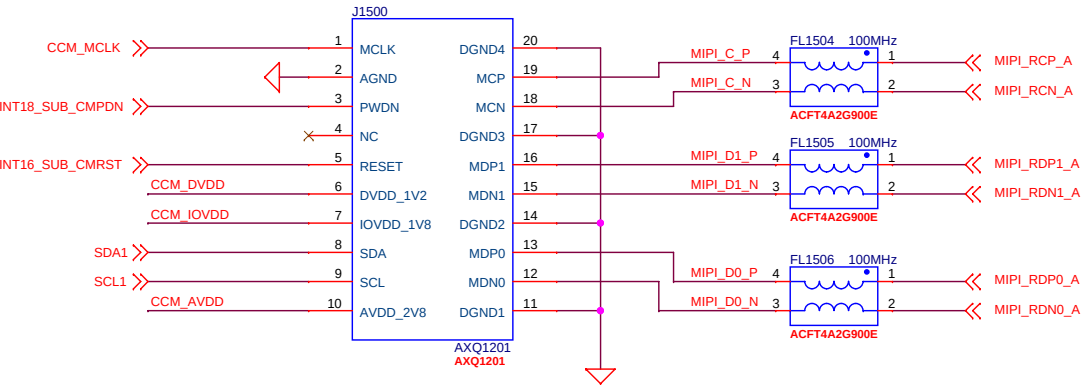
		Arima Communications Corp.	
Title			
X155 5522 MainBoard PV			
Size	Document ID	Rev	
B	13_Memory_CARD_DSIM_BT0B	PA1	
Date:	Thursday, June 18, 2015	Sheet	12 of 16

Close to MT6323 AUXADC_VREF18 PIN,
and the path need shielding with GND

[illegible][illegible]

 Arima Communications Corp.			
Title			
X155 5522 MainBoard PV			
Size	Document ID	Rev	
B	14_USB_OVP_CHG_BAT	PA1	
Date:	Thursday, June 15, 2015	Sheet	13 of 16

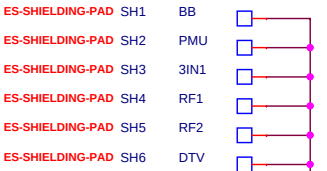
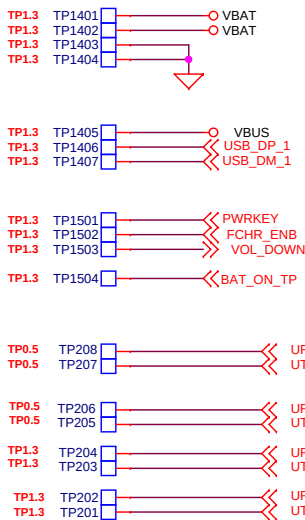
SUB Camera Module 2M/5M Compatible



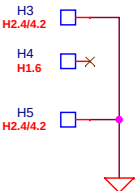
Test point

Shelding case

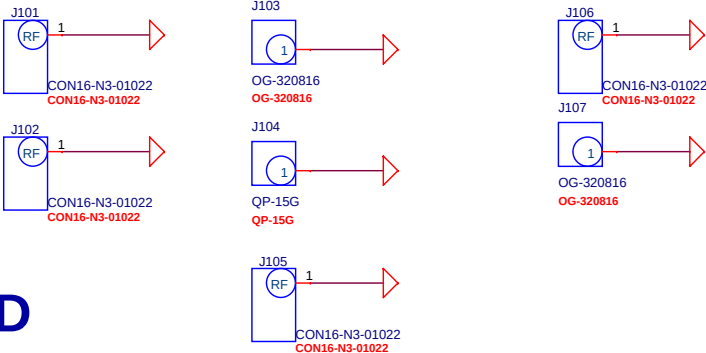
Spring Connector



Screw Hole

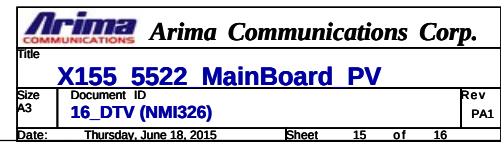


FID

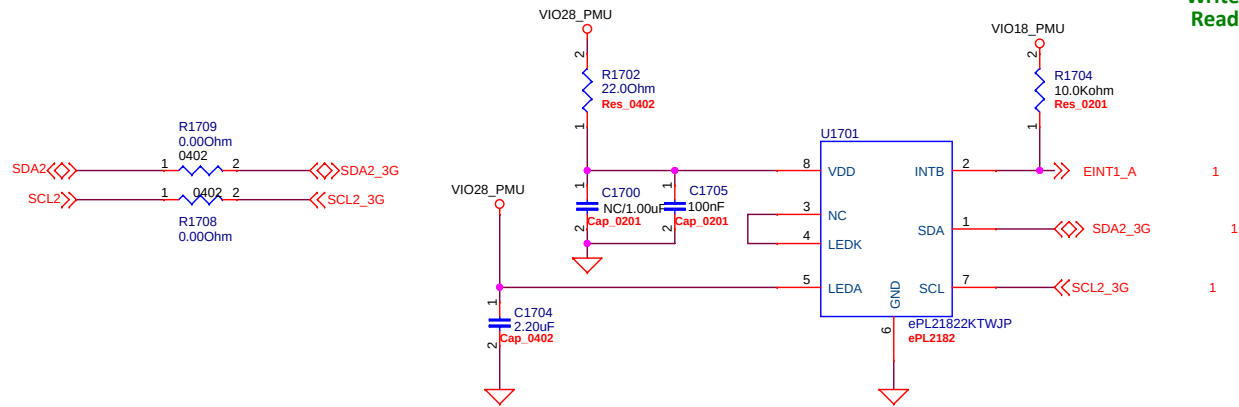


Arima Communications Corp.		
Title		
X155 5522 MainBoard PV		
Size	Document ID	Rev
B	15_Sub Camera_Test point	PA1
Date:	Thursday, June 18, 2015	Sheet 14 of 16

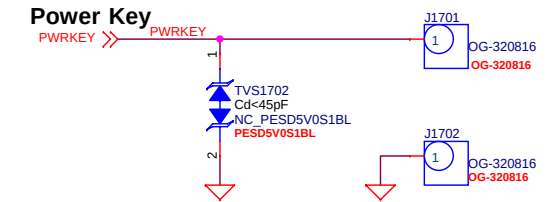
ANT



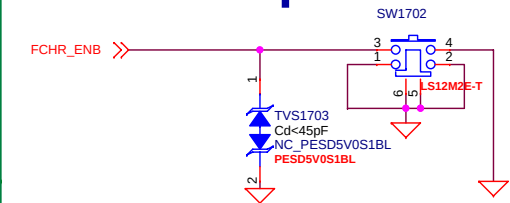
3 in 1 ALS+PS+IR



Power Key

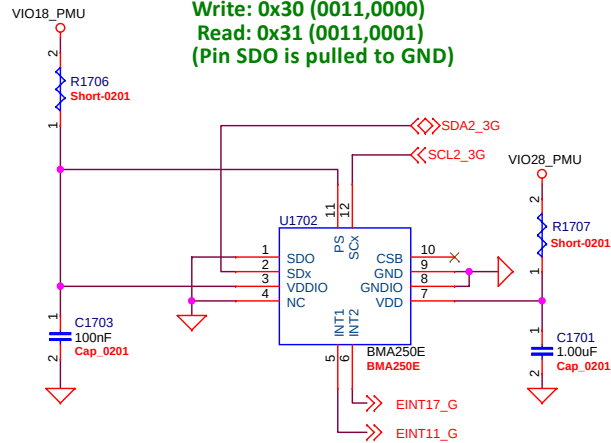


Volume Up



G-Sensor

Write: 0x30 (0011,0000)
Read: 0x31 (0011,0001)
(Pin SDO is pulled to GND)

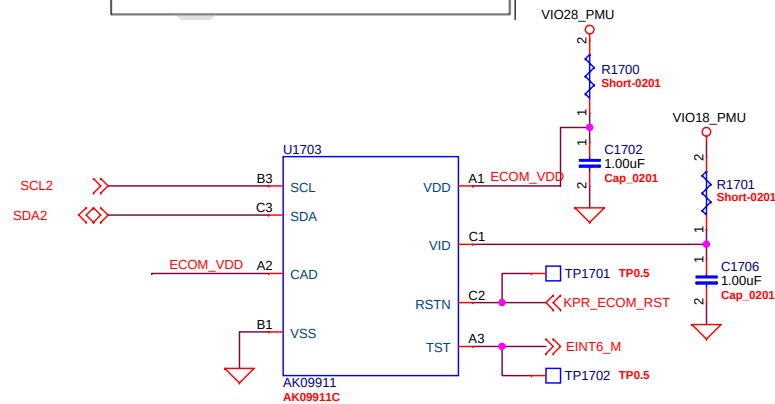


I2C ID = 0x18

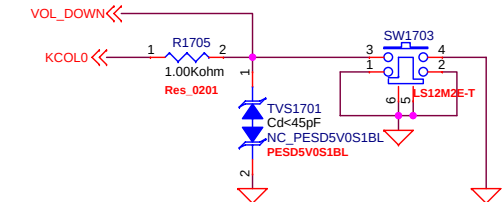
eCompass

Slave address select

CAD	address
VSS	0 0 0 1 1 0 0 R/W
VDD	0 0 0 1 1 0 1 R/W

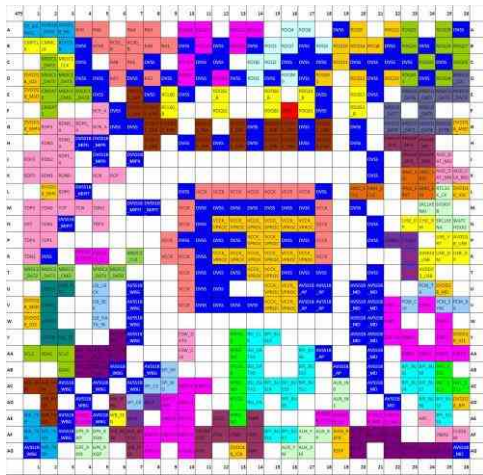


Volume Down



Arima Communications Corp.	
Title	
X155 5522 MainBoard PV	
Size	Document ID
B	17_Sensors_KEY
Date:	Thursday, June 18, 2015
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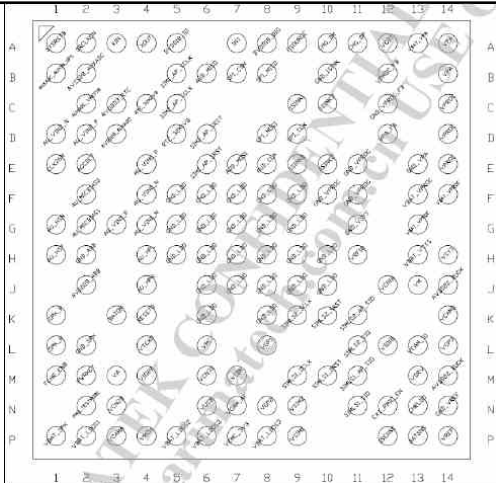
U202_MT6582_BB chipIC,Digital Baseband Processor(Top)



U501_KMQ7X000SA-B315,IC,MCP. eMMC+LPDDR3(Top)



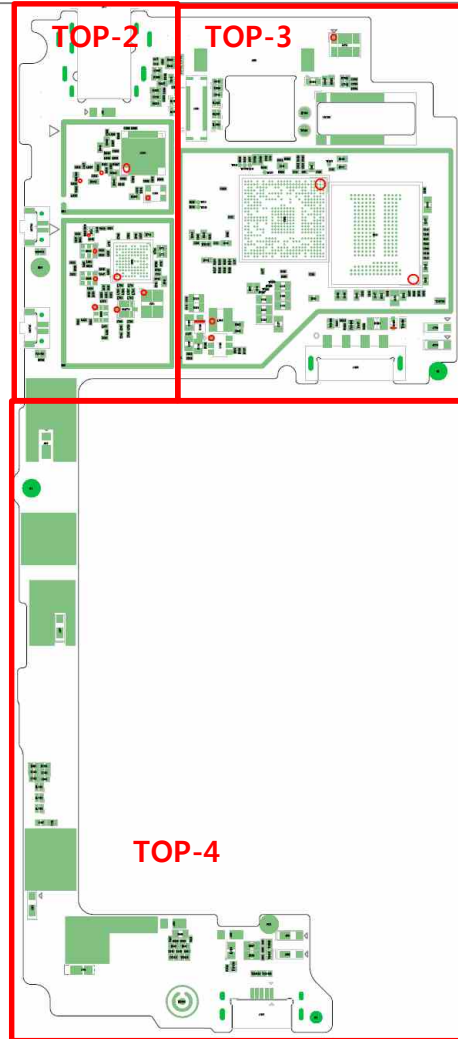
U301_MT6323_IC,PMIC (Top)



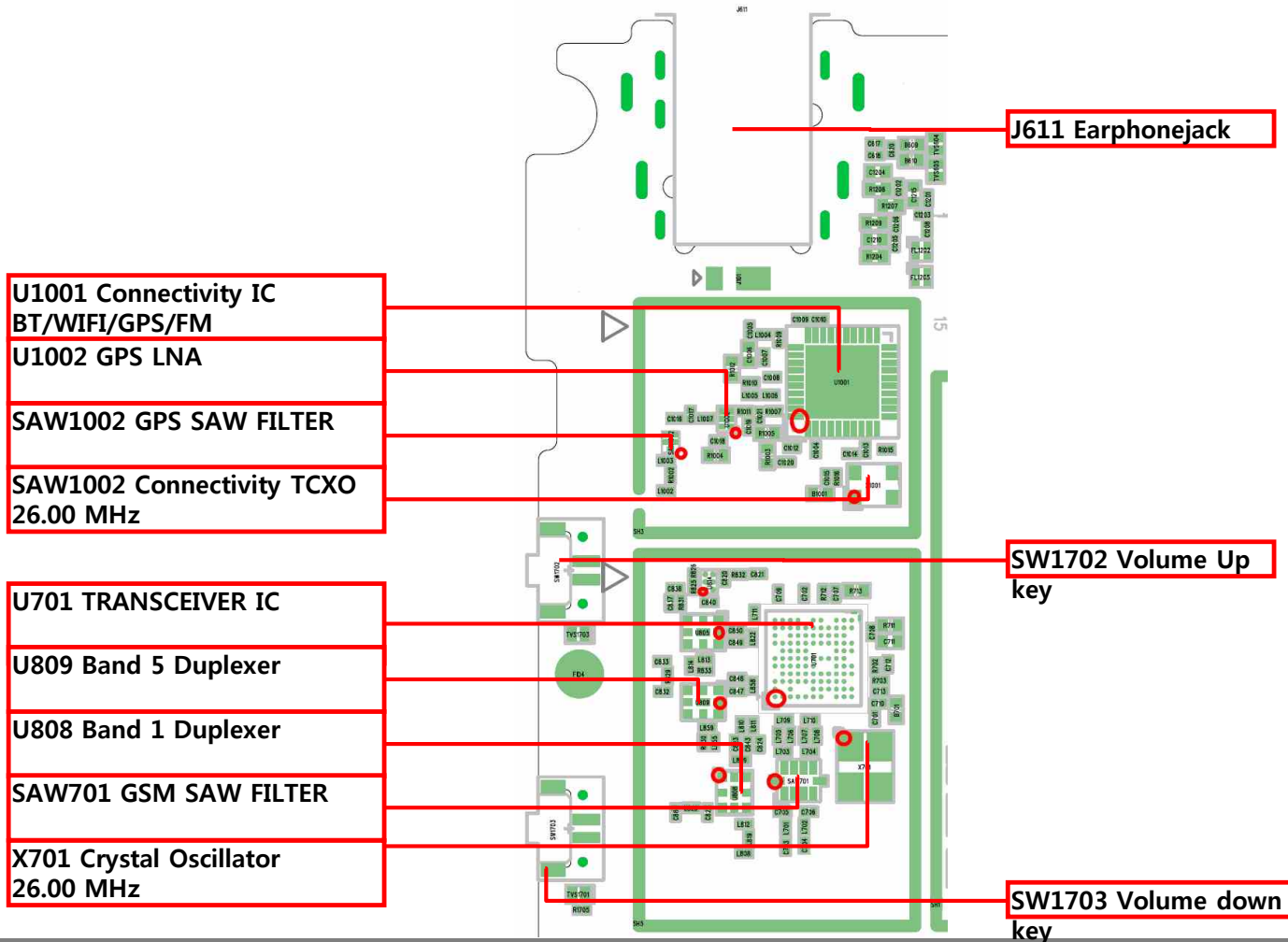
U701_MT6166V_IC ,Transceiver (Top)



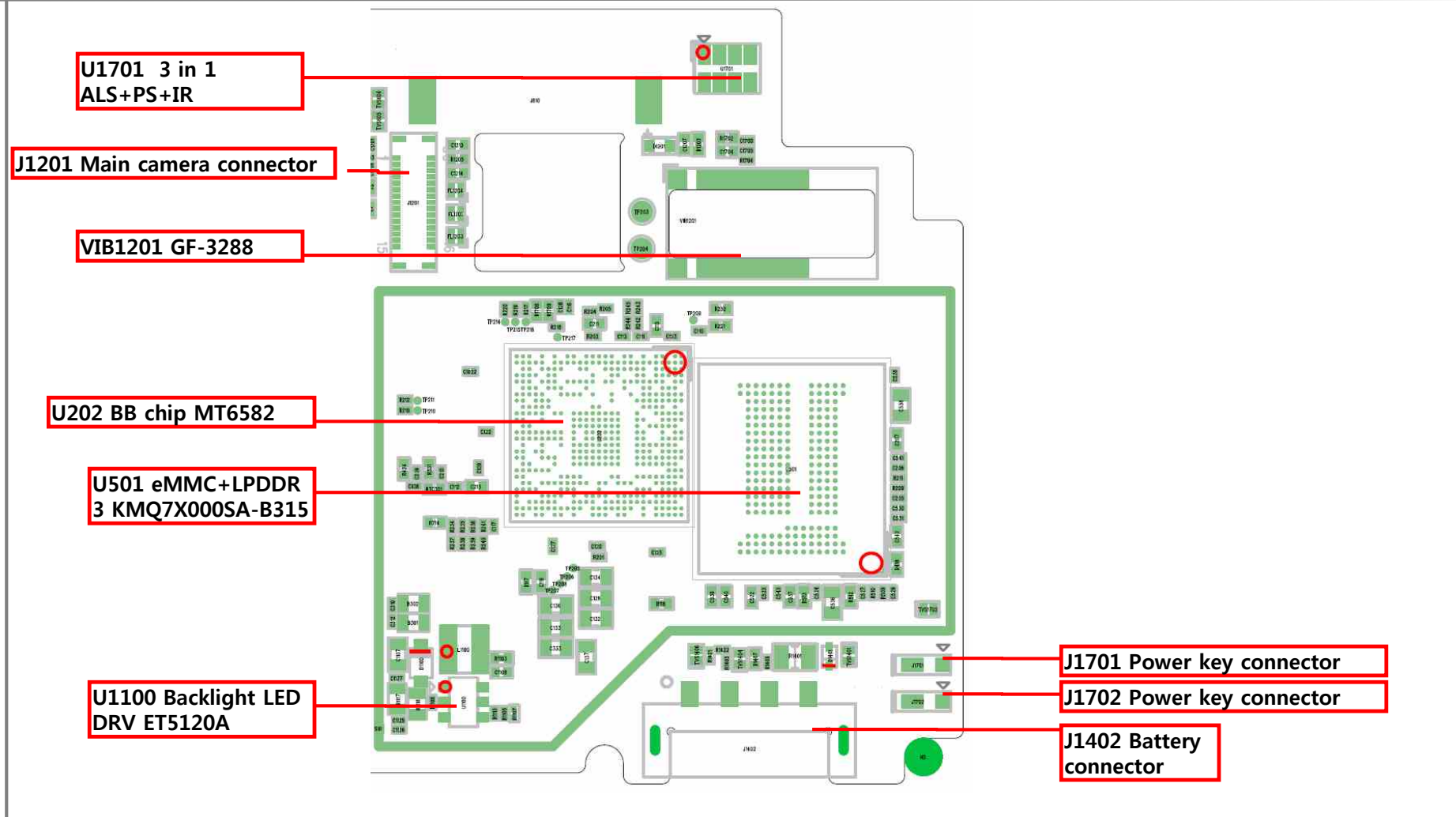
LG-X155-MAIN_TOP-1



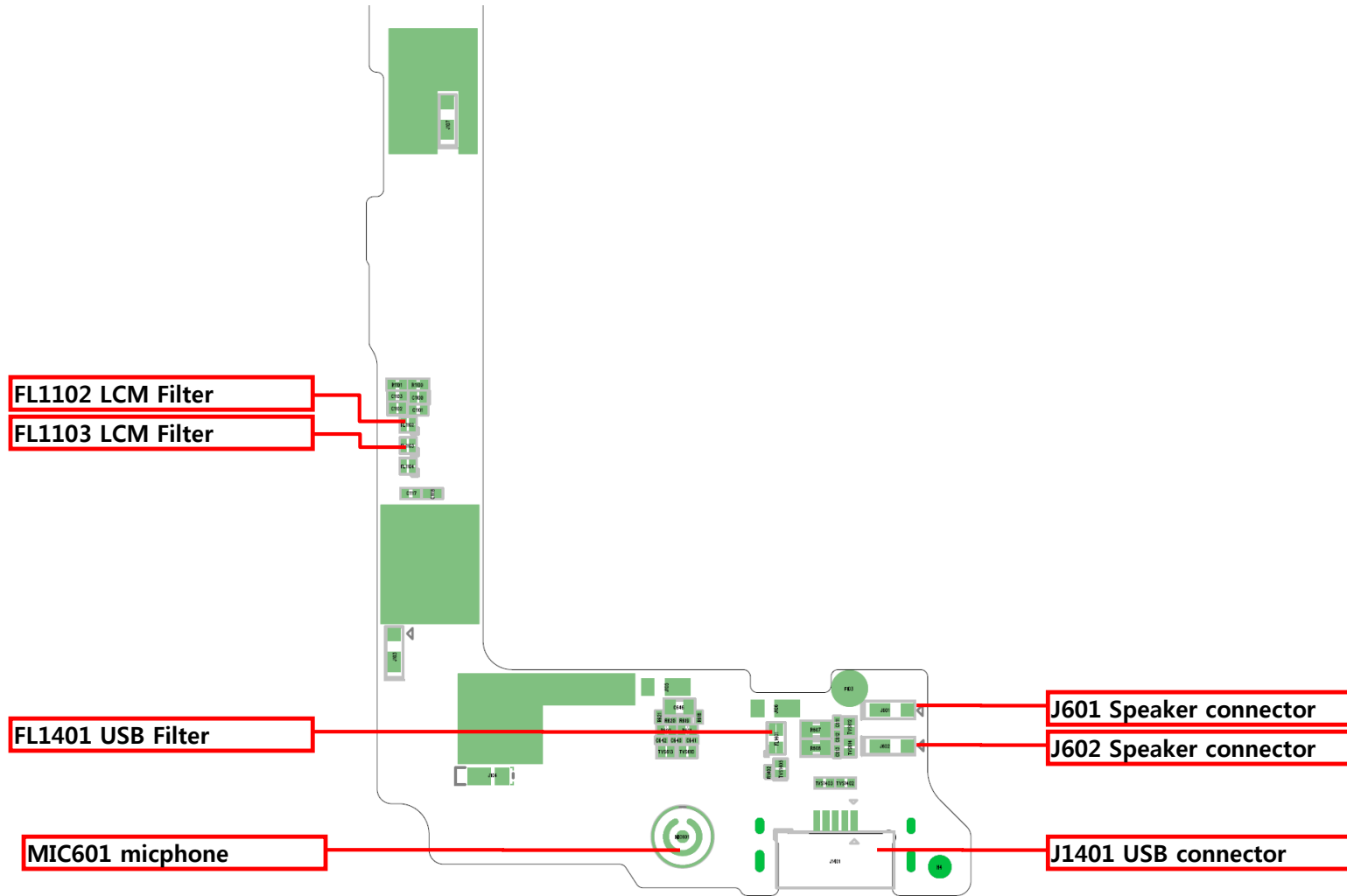
LG-X155-MAIN_TOP-2

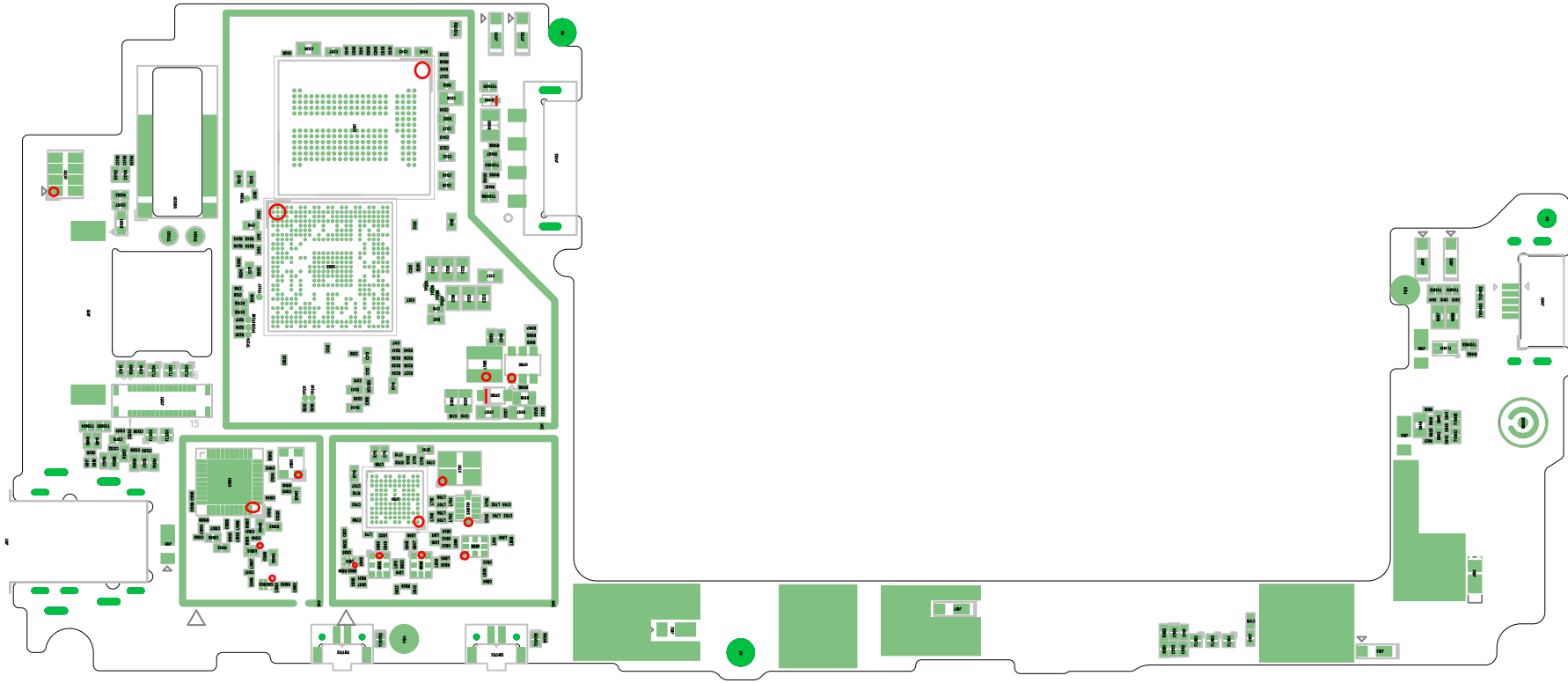


LG-X155-MAIN_TOP-3

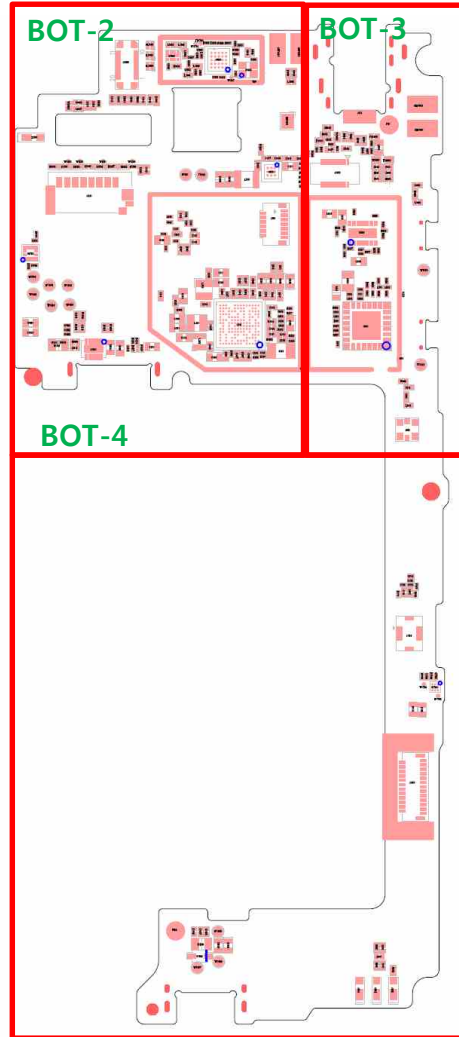


LG-X155-MAIN_TOP-4

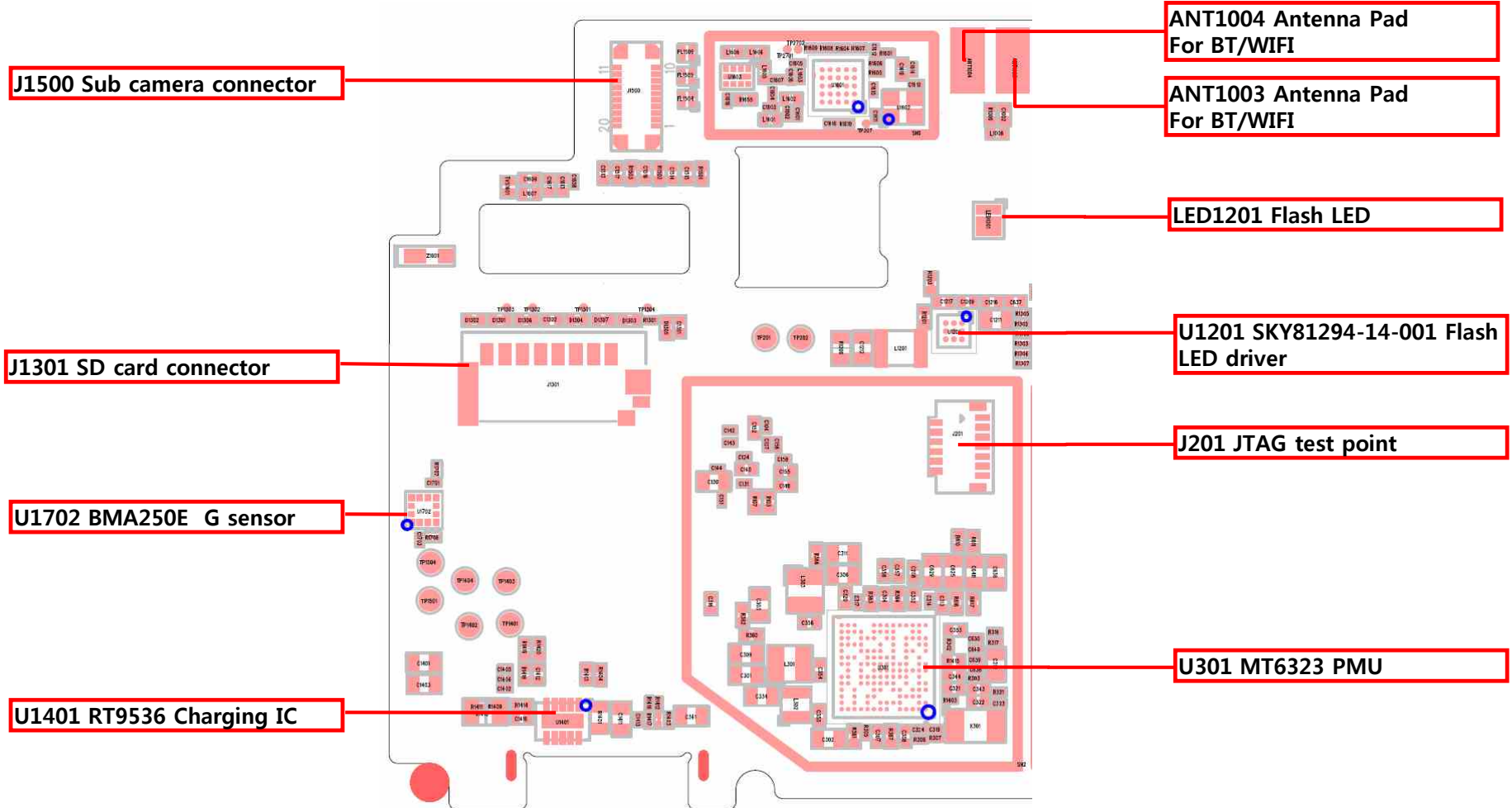




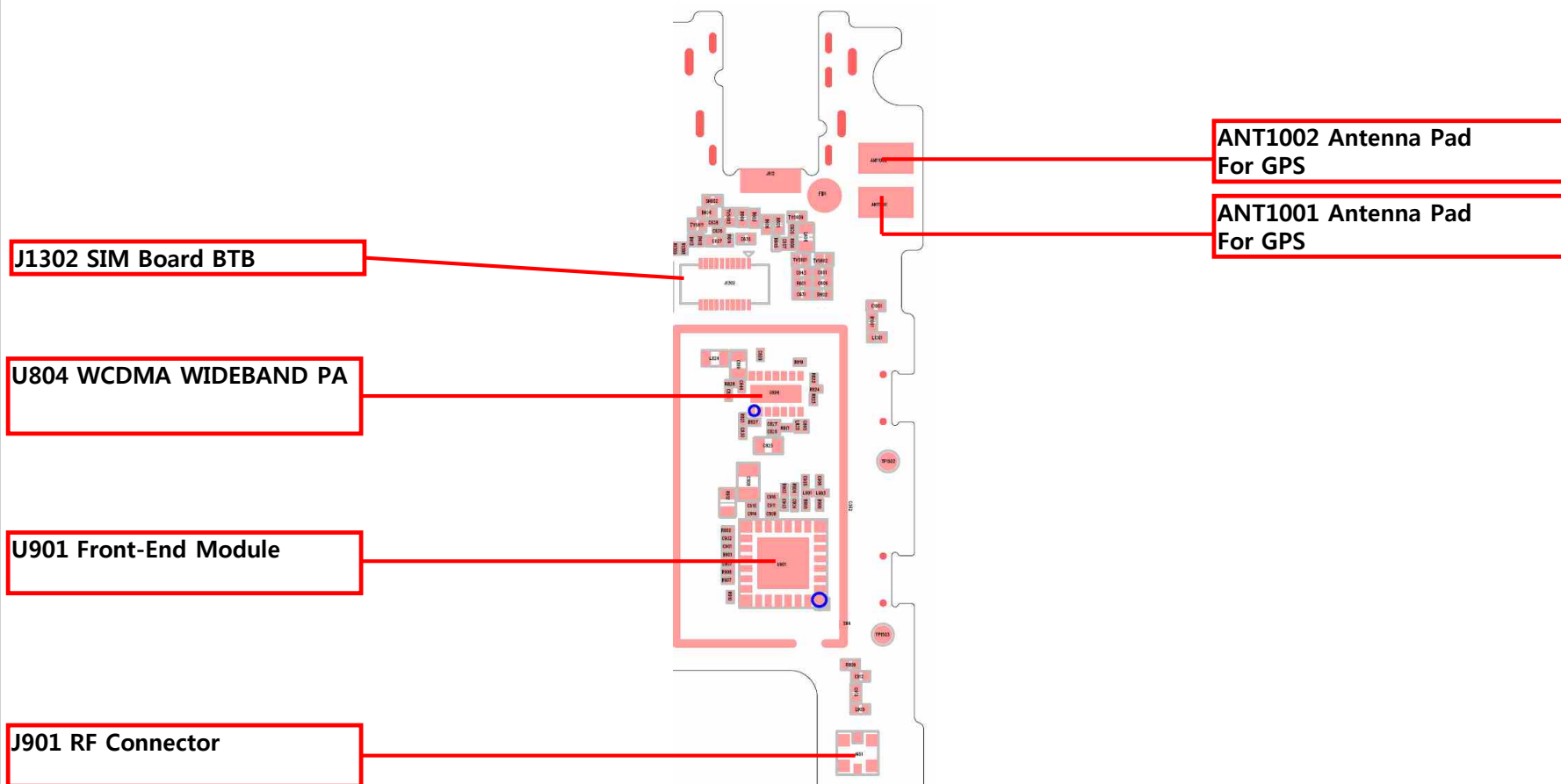
LG-X155-MAIN_BOT-1



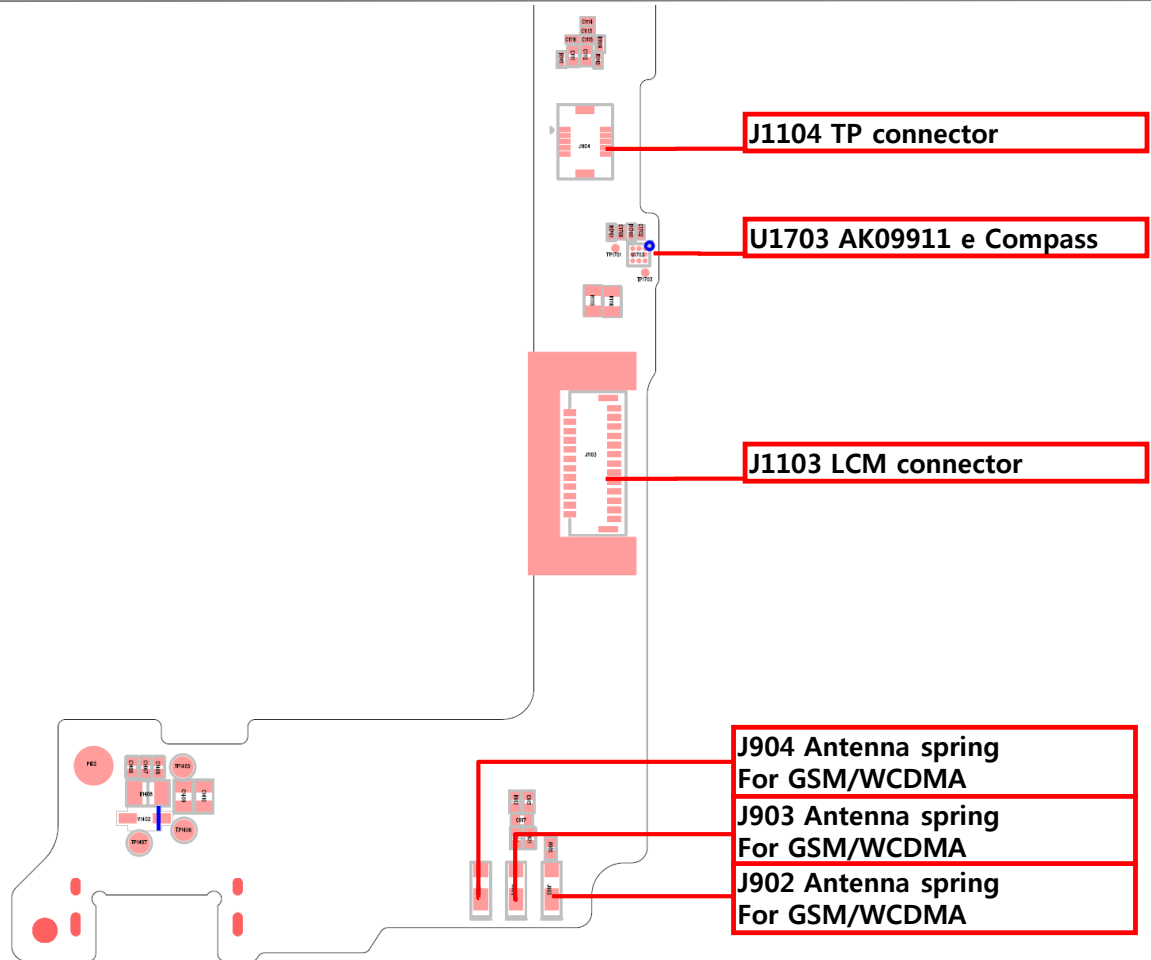
LG-X155-MAIN_BOT-2



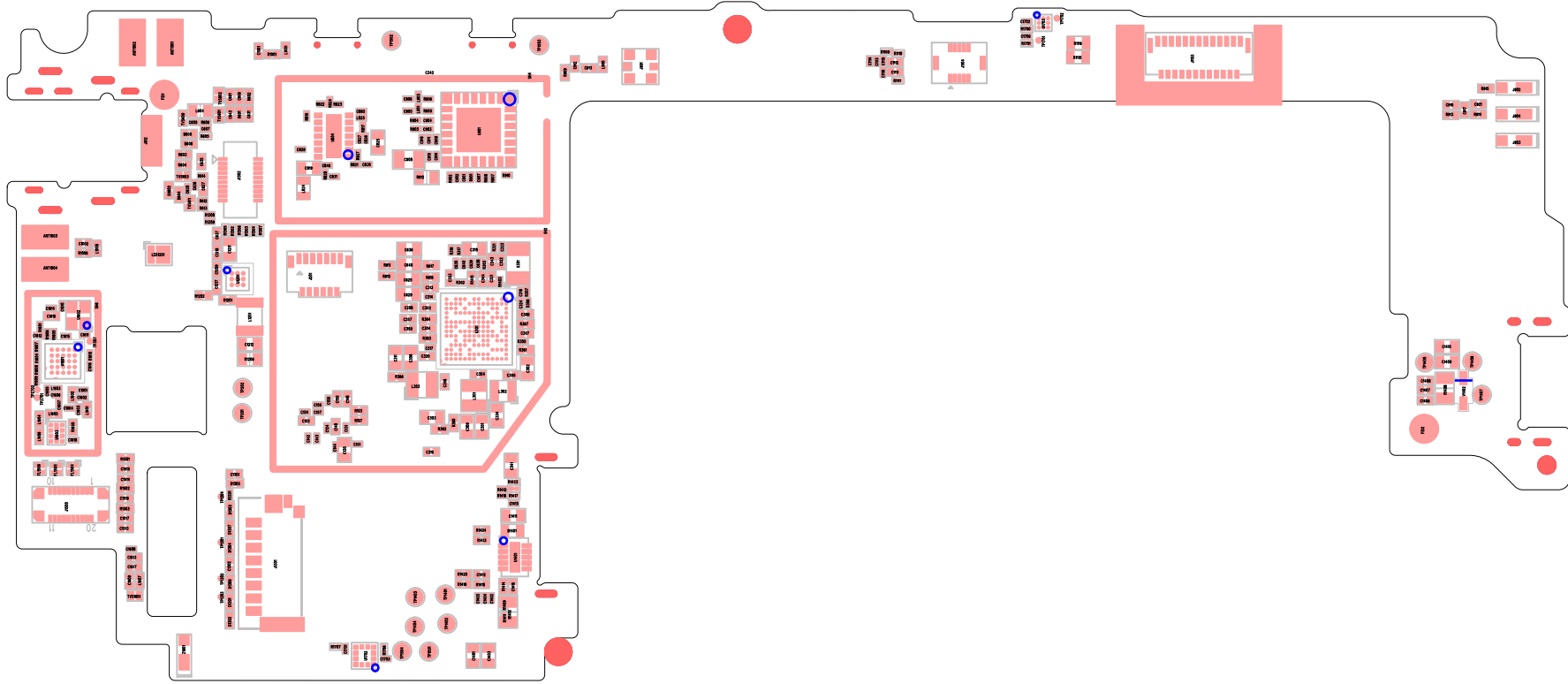
LG-X155-MAIN_BOT-3



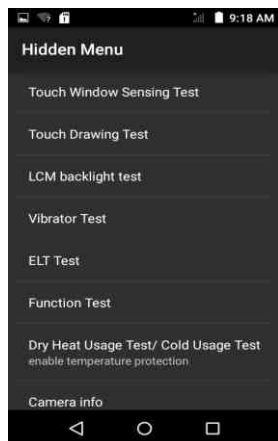
LG-X155-MAIN_BOT-4



X155_5522_MB_005_Location_Bottom

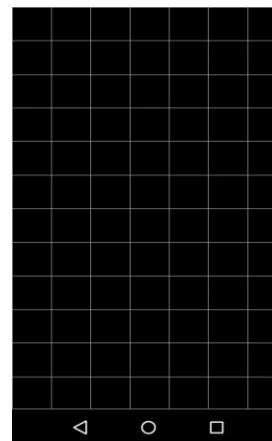


1. Hidden Menu Start



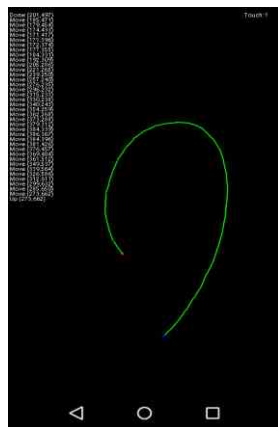
- Start shortcut key: ***#546368##155#**
- Hidden Menu List
: Start the desired menu, click

2. Touch window sensing test



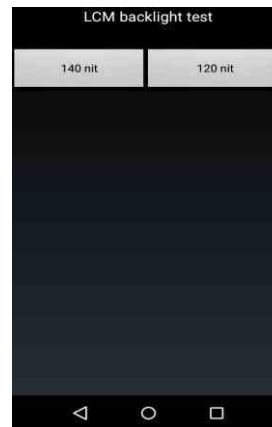
- Recording touch count
- Recording touch point
- Recording touch key count
- press volume up to back

3. Touch draw test



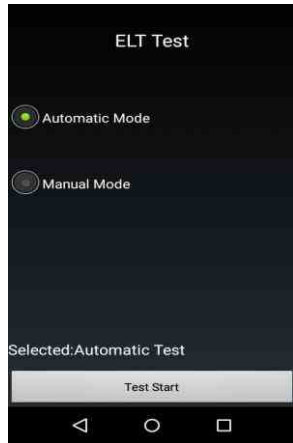
- Recording touch trace when slide on screen
- Recording touch count when slide on screen

4. LCM backlight test



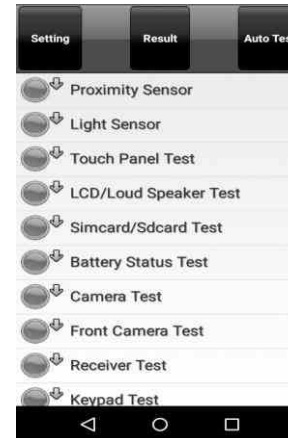
- Test back light (two levels: 140 nit ,120 nit)

5. ELT Test



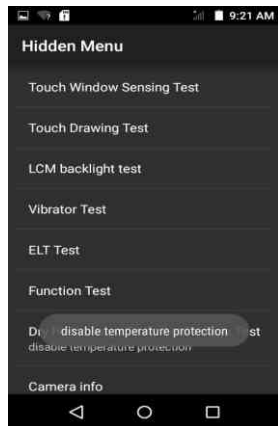
- ELT auto mode test
- ELT manual mode test
- press test start button to start ELT test

6. Function Test



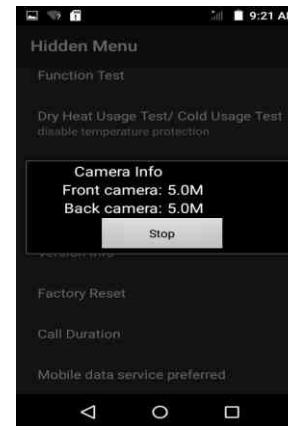
- test function one by one

7. Dry Heat Usage Test / Cold Usage Test

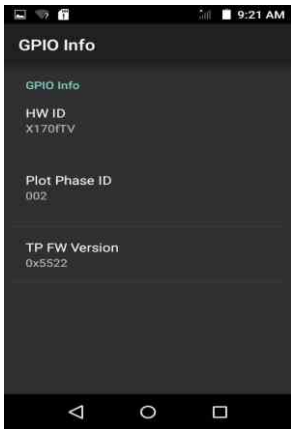
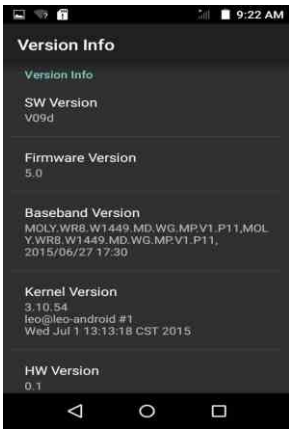
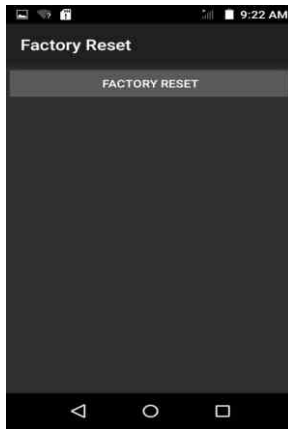
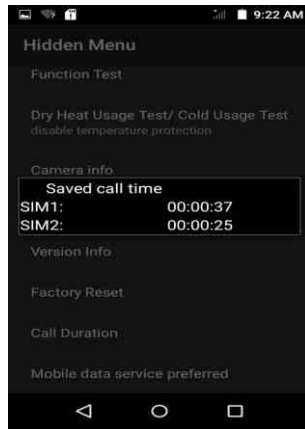
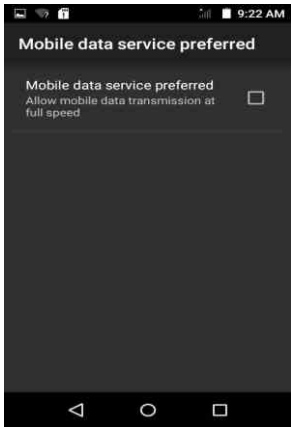


- Over temperature protection switch

8. Camera info

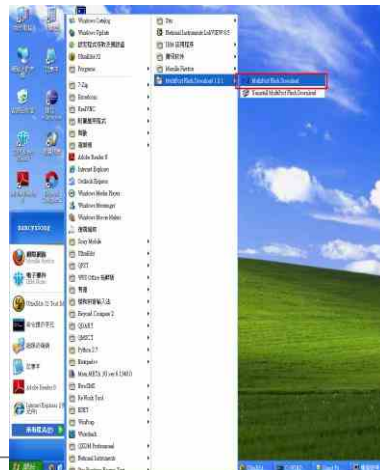


- Display Front camera info
- Display back camera info

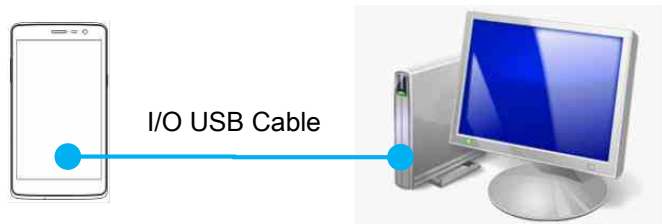
9. GPIO info	10. Version info	11. Factory Reset	12. Call Duration
 The screenshot shows the 'GPIO Info' screen with the following details: - GPIO Info - HW ID: X170FTV - Plot Phase ID: 002 - TP FW Version: 0x5522	 The screenshot shows the 'Version Info' screen with the following details: - Version Info - SW Version: V09d - Firmware Version: 5.0 - Baseband Version: MDLY.WR8.W1449.MD.WG.MPV1.P11.MOL.Y.WR8.W1449.MD.WG.MPV1.P11, 2015/06/27 17:30 - Kernel Version: 3.10.54, leo@leo-android #1, Wed Jul 1 13:13:18 CST 2015 - HW Version: 0.1	 The screenshot shows the 'Factory Reset' screen with a single button labeled 'FACTORY RESET'.	 The screenshot shows the 'Hidden Menu' screen with the following options: - Function Test - Dry Heat Usage Test/ Cold Usage Test (disable temperature protection) - Camera info - Saved call time - SIM1: 00:00:37 - SIM2: 00:00:25 - Version Info - Factory Reset - Call Duration - Mobile data service preferred
13. Mobile data service preferred			
 The screenshot shows the 'Mobile data service preferred' screen with the following details: - Mobile data service preferred - Allow mobile data transmission at full speed: ☐			

1. Summary

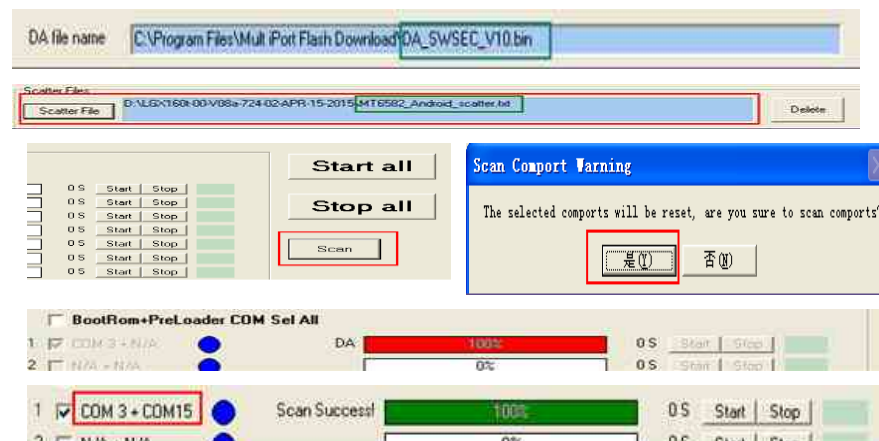
Tool Version	DLL name	USB Driver	
MultiPort Flash Download Setup v1.0.1	FlashToolLib.dll (version 1448)	LGUnitedMobileDriver_S51MAN312AP22_ML_WHQL_Ver_3.12.3	
H/W			
	Name	Part No	SW
D/L Cable	USB Cable 2.0	N/A	N/A



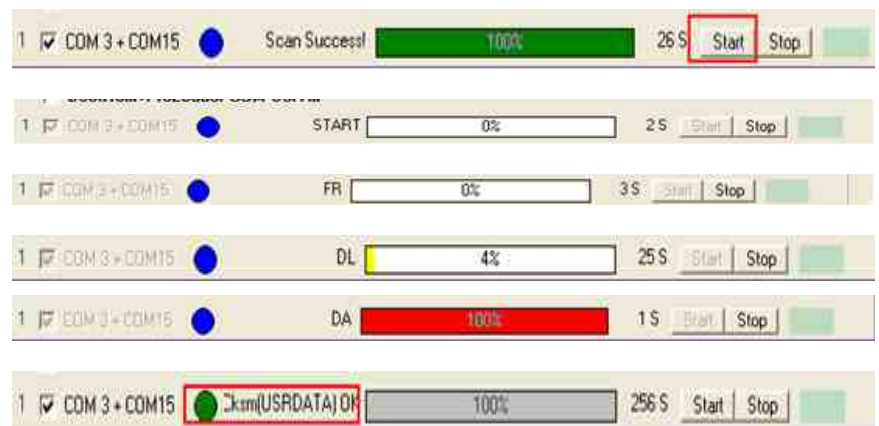
3. USB D/L Cable setting



2. . USB COM port Setting



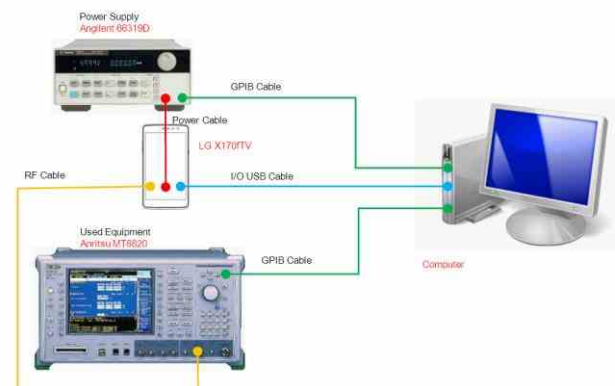
4. Flash tool D/L setting



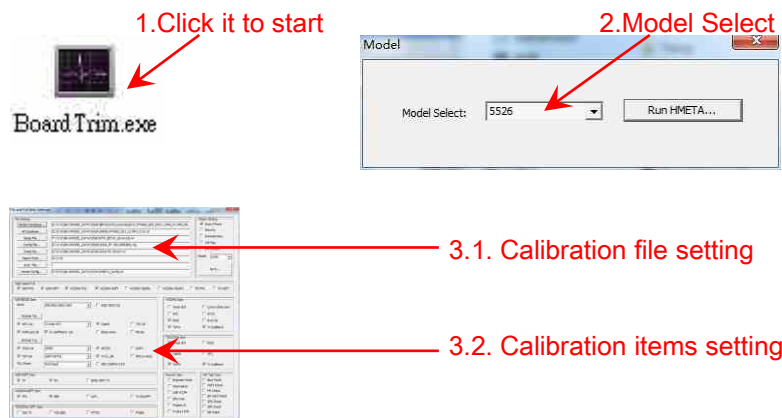
1. Summary

CAL INFORMATION	
Summary	
H/W	Name
USB Cable	USB Cable 2.0
Power Cable	DC Power Cable
RF Cable	MXHQ87WJ3000
Power Supply	Angilent 66319D
RF Test Equipment	Anritsu MT8820
USBCOM Port Driver	LGUnitedMobileDriver_S51MAN312AP22_ML_WHQL_Ver_3.12.3

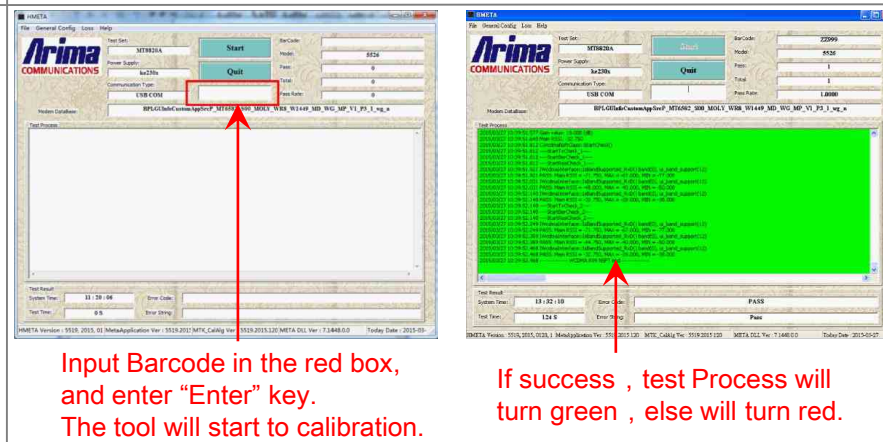
2. Calibration Cable setting



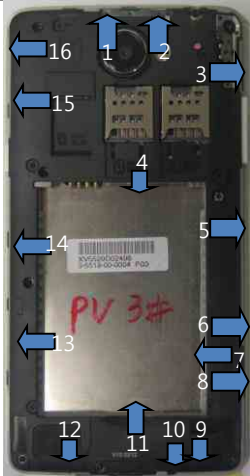
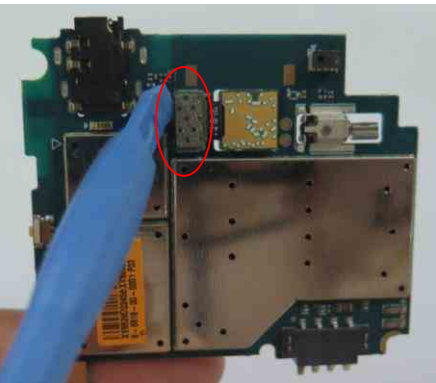
3. Calibration setting 1



4. Calibration setting 2



11. DISASSEMBLE GUIDE

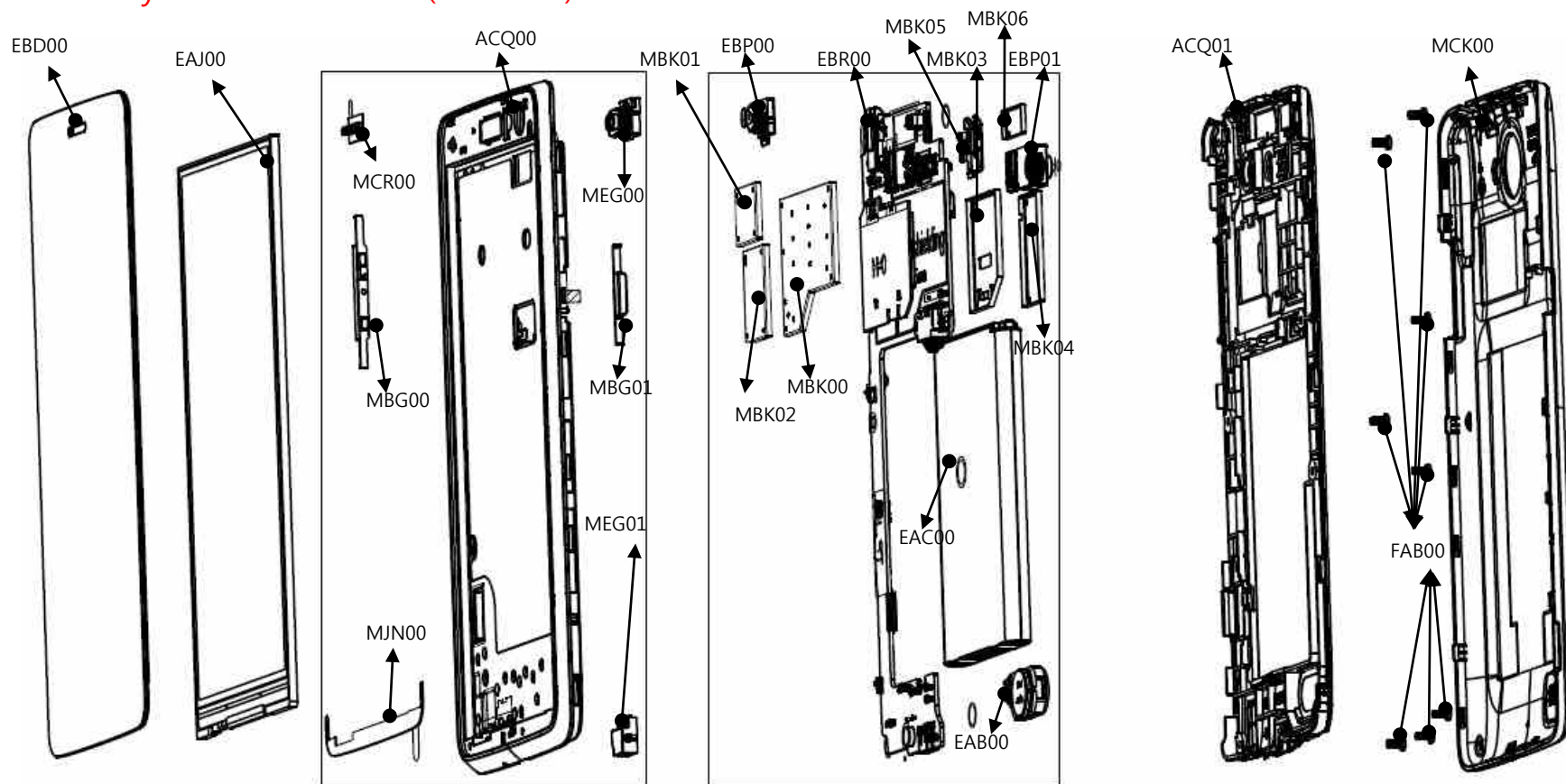
1. Remove screws (8point) 	2. Disassemble Rear cover 	3. Disassemble Connector 	4. Disassemble Main PCB 
5. Disassemble HW parts(2ea) 	6. Disassemble HW parts(1ea) 	7. Disassemble SIM PCB 	8. Disassemble SIM PCB 

9. Complete Disassembling



12. EXPLODED VIEW

※ only EXPLODED VIEW(blue cell) of SBOM file



Location no	Description	Location no	Description	Location no	Description	Location no	Description
EBD00	Touch Window Assembly	MEG00	Holder	EBP00	Camera Module	MBK04	Can,Shield
EAJ00	LCD,Module-TFT	MBG01	Button,Side	EBR00	PCB Assembly,Main	EAC00	Rechargeable Battery,Lithium Ion
ACQ00	Cover Assembly,Front	MEG01	Holder	MBK05	Can,Shield	EAB00	Speaker Module
MCR00	Decor	MBK01	Can,Shield	MBK03	Can,Shield	ACQ01	Cover Assembly,Rear
MBG00	Button,Side	MBK02	Can,Shield	MBK06	Can,Shield	FAB00	Screw,Machine
MJN00	Tape,Window	MBK00	Can,Shield	EBP01	Camera Module	MCK00	Cover,Battery

13. REPLACEMENT PART LIST

Seq	Qty	Manufacture	Manufacture P/N	Ref Des
10	1	TRIPOD	5518MB-005	NA
15	3	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-JP-PA,RM02JTN000,RTT01000JTH,RC0201JR-070RL	R1016,R205,R305
20	12	RALEC	RTT01000JTH	C1005,L819,R1002,R1015,R606,R702,R703,R817,R825,R827,R905,R906
20	0	TA-I	RM02JTN000	C1005,L819,R1002,R1015,R606,R702,R703,R817,R825,R827,R905,R906
20	0	KAMAYA	RMC1/20-JP-PA	C1005,L819,R1002,R1015,R606,R702,R703,R817,R825,R827,R905,R906
25	9	TA-I,RALEC,YAGEO	RM04JTN0,RTT02000JTH,RC0402JR-070RL	R1003,R1004,R1005,R1012,R1708,R1709,R711,R713,R714
30	4	RALEC	RTT02000JTH	R1001,R1006,R914,R915
30	0	TA-I	RM04JTN0	R1001,R1006,R914,R915
40	4	YAGEO,TA-I,RALEC,KAMAYA	RC0603JR-070RL,RM06JTN000,RTT03000JTP,RMC1/16-JP-TP	R1115,R1116,R1206,R912
50	3	YAGEO,TA-I,RALEC	RC0805JR-070RL,RM10JTN000,RTT05000JTP	R1401,R1408,R1410
60	1	YAGEO,KAMAYA,TA-I,RALEC	RC0201FR-075R1L,RMC1/20-5R1FPA,RM02FTN5R10,RTT015R10FTH	R1108
70	1	YAGEO,TA-I,RALEC	RC0402JR-0722RL,RM04JTN220,RTT02220JTH	R1702
80	2	RALEC	RTT0127R0FTH	R822,R823
80	0	TA-I	RM02FTN27R0	R822,R823
90	4	YAGEO,TA-I,RALEC	RC0402FR-0733RL,RM04FTN33R0,RTT0233R0FTH	R1205,R232,R610,R611
100	1	RALEC	RTT01360JTH	R824
100	0	TA-I	RM02JTN360	R824
110	2	RALEC	RTT01510JTH	R819,R910
110	0	TA-I	RM02JTN510	R819,R910
120	2	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-241-F-PA,RM02FTN2400,RTT012400FTH,RC0201FR-07240RL	R509,R510
130	2	YAGEO,RALEC,TA-I	RC0402FR-07470RL,RTT024700FTH,RM04FTN4700	R601,R602
140	1	RALEC	RTT026490FTH	R1413
150	13	YAGEO,TA-I,RALEC,KAMAYA	RC0201JR-071KL,RM02JTN102,RTT01102JTH,RMC1/20-102-J-PA	R1402,R1405,R1705,R316,R317,R613,R614,R618,R621,R901,R902,R903,R904
160	5	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-152-F-PA,RM02FTN1501,RTT011501FTH,RC0201FR-071K5L	R203,R321,R612,R619,R620
170	1	RALEC	RTT012001FTH	R712
170	0	TA-I	RM02FTN2001	R712
180	4	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-222-J-PA,RM02JTN222,RTT01222JTH,RC0201JR-072K2L	R217,R218,R219,R220
200	1	YAGEO,TA-I,RALEC	RC0402JR-073K3L,RM04JTN332,RTT02332JTH	R1418
210	2	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-392-J-PA,RM02JTN392,RTT01392JTH,RC0201JR-073K9L	R210,R212
215	1	YAGEO,TA-I,RALEC	RC0402FR-074K7L,RM04FTN4701,RTT024701FTH	R1414
220	1	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-512-F-PA,RM02FTN5101,RTT015101FTH,RC0201FR-075K1L	R201
230	2	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-822-F-PA,RM02FTN8201,RTT018201FTH,RC0201FR-078K2L	R209,R211
235	1	KAMAYA,TA-I,RALEC,YAGEO	RMC1/20-103-J-PA,RM02JTN103,RTT01103JTH,RC0201JR-0710KL	R1704
240	1	RALEC	RTT01103JTH	R907
240	0	TA-I	RM02JTN103	R907
250	1	RALEC	RTT01243JTH	R908
250	0	TA-I	RM02JTN243	R908
260	1	YAGEO,TA-I,RALEC	RC0402FR-0739KL,RM04FTN3902,RTT023902FTH	R1420
270	1	YAGEO,RALEC,TA-I	RC0402JR-0747KL,RTT02473JTH,RM04JTN473	R604

13. REPLACEMENT PART LIST

280	1	RALEC	RTT016192FTH	R1403
290	10	YAGEO,TA-I,RALEC,KAMAYA	RC0201JR-07100KL,RM02JTN104,RTT01104JTH,RMC1/20-104-J-PA	R1113,R1301,R1308,R1309,R234,R235,R239,R241,R242,R244
300	1	RALEC	RTT011023FTH	R1406
310	1	YAGEO,TA-I,RALEC	RC0402FR-07330KL,RM04FTN3303,RTT023303FTH	R1419
320	1	YAGEO,TA-I,RALEC	RC0402JR-07470KL,RM04JTN474,RTT02474JTH	R603
330	1	YAGEO,TA-I,RALEC	RC0402JR-071ML,RM04JTN105,RTT02105JTH	R424
340	2	MURATA	GRM0335C1HR50B	C838,C884
350	1	MURATA	GRM0335C1E1R0C	C1017
360	1	MURATA	GRM0335C1E1R0BA01D	L903
370	1	MURATA	GRM1555C1H1R0C	C912
380	1	MURATA	GRM0335C1E2R0C	C906
390	3	MURATA	GRM0335C1E2R2C	C1021,C849,C850
400	4	MURATA	GRM0335C1E3R3C	C823,C824,C843,R1009
410	1	DARFON	C1005NP0569DGT	C1408
420	1	DARFON	C0603NP0689DFT	C911
430	1	MURATA	GRM0335C1H100J	C1018
440	2	DARFON	C0603NP0150JFT	C1404,C910
450	1	DARFON	C1005NP0150JFT	C1213
460	2	MURATA	GRM0335C1H180J	C1016,C885
470	7	MURATA	GRM0335C1H220J	C1402,C319,C324,C901,C902,C903,C904
480	2	MURATA	GRM1555C1H220J	C1209,C1407
490	1	MURATA	GRM0335C1E330J	R1011
500	11	DARFON	C0603NP0330JFT	C1405,C607,C611,C613,C617,C618,C626,C628,C641,C642,C909
510	2	DARFON	C1005NP0330JET	C606,C631
520	9	MURATA	GRM0335C1H470J	C1113,C1114,C1115,C1116,C1125,C1126,C1127,C1413,C916
530	2	MURATA	GRM1555C1H470J	C1117,C1118
540	2	MURATA	GRM0335C1H560J	C830,C905
550	1	MURATA	GRM0335C1E820J	C827
560	8	MURATA	GRM0335C1E101J	C1007,C1010,C1014,C209,C210,C612,C620,C640
570	3	MURATA,TDK	GRM033R71E221K,C0603X7R1E221KT	C1206,C1208,C907
580	2	MURATA,AVX KYOCERA,PHYCOMP	GRM033R71C102K,CM03X7R102K16AH,CC0201KRX7R7BB102	C310,C312
590	1	MURATA	GRM155R71H222K	C1406
600	1	MURATA,PHYCOMP,AVX KYOCERA	GRM033R71A472K,CC0201KRX7R6BB472,CM03X7R472K10AH	C1012

13. REPLACEMENT PART LIST

610	1	MURATA,PHYCOMP,AVX KYOCERA	GRM033R71A103K,CC0201KRX7R6BB103,CM03X7R103K10AH	C1008
620	1	MURATA,AVX KYOCERA,TDK,PHYCOMP	GRM155R61C223K,CM05X5R223K16AH,C1005X5R1C223KT,CC0402KRX5R7BB223	C637
630	51	MURATA,TDK,PHYCOMP	GRM033R61A104K,C0603X5R1A104KT,CC0201KRX5R6BB104	C1009,C1022,C104,C108,C109,C110,C111,C112,C113,C116,C117,C120,C1201,C1202,C1203,C1205,C122,C123,C124,C125,C127,C128,C131,C142,C143,C144,C151,C156,C157,C158,C1703,C1705,C205,C206,C314,C323,C523,C527,C528,C529,C530,C531,C541,C543,C630,C638,C639,C649,C826,C829,C914
640	8	MURATA,TAIYOYUDEN,PHYCOMP,AVX KYOCERA,TDK	GRM155R61C104K,EMK105BJ104KV-F,CC0402KRX5R7BB104,CM05X5R104K16AH,C1005X5R1C104K	C1103,C1112,C1301,C1513,C213,C320,C321,C344
650	1	MURATA,PHYCOMP,TDK,TAIYOYUDEN	GRM033R60J224M,CC0201MRX5R5BB224,C0603X5R0J224MT,JMK063BJ224MP-F	C524
660	6	MURATA	GRM033R60J474K	C701,C702,C707,C708,C709,C710
670	7	TAIYOYUDEN,MURATA,PHYCOMP,TDK,AVX KYOCERA	JDK063BJ105MP-FD,GRM033R60J105M,CC0201MRX5R5BB105,C0603X5R0J105MT,CV03X5R105M06AH	C1013,C1015,C1019,C1020,C1701,C1702,C1706
680	21	MURATA,TAIYOYUDEN,AVX KYOCERA,TDK	GRM155R61A105K,LMK105BJ105KV-F,CV05X5R105K10AH,C1005X5R1A105K	C1111,C115,C119,C1207,C140,C1412,C1416,C148,C152,C155,C207,C304,C307,C308,C316,C322,C342,C352,C353,C522,C711
690	2	MURATA	GRM188R61H105K	C1107,C1411
700	15	MURATA,TDK,TAIYOYUDEN,AVX KYOCERA	GRM155R60J225M,C1005X5R0J225MT,JMK105BJ225MV-F,CV05X5R225M06AH	C1101,C1204,C1210,C1215,C1302,C1514,C1515,C1516,C1517,C1704,C313,C537,C539,C540,C542
710	2	MURATA,TAIYOYUDEN,AVX KYOCERA,TDK	GRM188R61A225K,LMK107BJ225KA-T,CM105X5R225K10AT,C1608X5R1A225K	C334,C819
720	3	MURATA	GRM155R60J475ME87D	C1006,C1108,C627
730	8	TDK,TAIYOYUDEN,MURATA	C1608X5R1A475K,LMK107BJ475KA-T,GRM188R61A475K	C301,C302,C303,C309,C536,C538,C646,C825
740	11	AVX KYOCERA,TDK,TAIYOYUDEN,MURATA	CV105X5R106M06AT,C1608X5R0J106MT,JMK107BJ106MA-T,GRM188R60J106M	C1211,C1212,C129,C130,C132,C133,C134,C136,C1401,C306,C311
750	6	TDK	C1608X5R1A226MT	C318,C341,C625,C629,C636,C648
760	1	MURATA,AVX KYOCERA,TAIYOYUDEN	GRM219R60J226M,CT21X5R226M06AT095,JMK212BJ226MD-T	C908
770	1	TDK	MLG0603P1N0BT	R833
780	2	TDK	MLG0603P1N5BT	L707,L708
790	2	MURATA	LQP03TN1N8B04D	L810,L811
810	2	TDK	MLG0603S2N0CT	L711,L826
820	3	MURATA	LQP03TN2N2B02D	L701,L702,L812
830	2	TDK	MLG1005S2N7ST000	C913,R909
842	1	TDK	MLG0603P3N3BT	R831
845	1	MURATA	LQG15HN3N3S02D	R913
850	1	TDK	MLG0603P4N7HT	L814
860	2	TAIYOYUDEN	HK06035N1S-T	L825,R828
870	1	TDK	MLG0603P6N2HT	R1007
880	1	TDK	MLG0603S6N8JT	L710
890	1	TDK	MLG0603S7N5JT	L1007
900	2	TDK	MLK0603L10NJT	L705,L706
910	1	TDK	MLG0603S16NJT	L822
920	1	TAIYOYUDEN	HK060333NJ-T	L709
930	1	TDK	MLG0603S39NJT	L1004
940	1	TAIYOYUDEN	HK100539NJ	L905
950	1	TDK	MLG0603S47NJT	C605

13. REPLACEMENT PART LIST

960	1	TDK	MLG0603S82NJT	L1006
970	2	TDK	VLS252010HBX-R68M	L301,L303
980	1	TDK	MLP2520S1R0M	L1201
990	1	TDK	VLS201610HBX-2R2M-1	L302
1000	1	TDK	VLS3010ET-100M	L1100
1010	4	MURATA	BLM18SG121TN1D	B1401,B301,B302,L824
1020	1	MURATA	BLM15AX221SN1D	B701
1030	1	INPAQ	MHC1608S221NBP	L604
1040	8	MURATA	BLM15AX102SN1D	B1001,B601,B602,B604,B605,B606,B609,B610
1050	2	TDK	MPZ1608S102A	R1117,R1118
1060	1	EPSON TOYOCOM	Q13FC1350004500	X301
1070	1	KDS	1XXB26000CTB	X1001
1080	1	TAI-SAW	TZ1689A	X701
1090	1	PHILIPS	PESD12VS1UA	V1402
1100	3	ICT	IECS0505C040FR	TVS1402,TVS1403,TVS609
1110	7	PRISEMI	PESDHC2FD5VB	C1100,C1102,TVS604,TVS605,TVS610,TVS611,TVS613
1120	1	PHILIPS	PMEG4010BEA	D1100
1130	1	PRISEMI	PZ5D4V2H	D1401
1140	1	EVERLIGHT	ELCHSB-5565J6J7284110-N8(ARI)	LED1201
1150	1	MTK	MT6582V/W	U202
1160	1	MTK	MT6323GA	U301
1170	1	SAMSUNG	KMQ72000SM-B316	U501
1180	1	MTK	MT6166V	U701
1190	1	SKYWORKS	SKY77769-11	U804
1200	1	SKYWORKS	SKY77589-11	U901
1210	1	MTK	MT6627N	U1001
1220	1	INFINEON	BGA524N6	U1002
1230	1	ETEK	ET5120A	U1100
1240	1	SKYWORKS	SKY81294-14-001	U1201
1250	1	ELAN	ePL21822KTWJP	U1701
1260	1	BOSCH	BMA250E	U1702
1270	1	ASAHI KASEI	AK09911C	U1703
1290	1	RICHTEK	RT9536GQW	U1401
1300	1	GETTOP ACOUSTIC	SOM4013SL-Z422-RC-HS	MIC601
1310	1	PANASONIC	AXQ1201	J1500
1320	4	ACRON	CON16-N3-01022	J101,J102,J105,J106
1330	1	JTCONN	AJ017-14346A03	J611
1340	1	MURATA	MM8130-2600RA2	J901
1350	1	TAIWAN SUNCAGEY	BL125-25RL-TAGF	J1103

13. REPLACEMENT PART LIST

1360	1	HIROSE	DF37NB-10DS-0.4V(51)	J1104
1370	1	HIROSE	BM20B(0.8)-30DS-0.4V(51)	J1201
1380	1	JTCONN	TF001-25116A82	J1301
1390	1	HIROSE	DF40C-20DS-0.4V(51)	J1302
1400	1	SMK	CSS5305-9321F	J1401
1410	9	KITAGAWA	OG-320816	J103,J107,J1701,J1702,J601,J602,J902,J903,J904
1415	1	EMI STOP	QP-15G	J104
1420	1	JTCONN	BV028-15313A41	J1402
1430	2	CITIZEN ELECTRONIC	LS12M2E-T	SW1702,SW1703
1440	1	SHANGHAI AWA SEIMITSU	GF-3288	VIB1201
1450	1	INPAQ	HCM1012GH900BP	FL1401
1460	1	WALSIN	RFBLN2012090BM5T25	SAW701
1470	1	EPCOS	B39162B9877P810	SAW1002
1480	11	INPAQ	TCF03025FH90002A	FL1102,FL1103,FL1104,FL1201,FL1202,FL1203,FL1204,FL1205,FL1504,FL1505,FL1506
1490	1	EPCOS	B39212B8575P810	U808
1500	1	MURATA	SAYFH897MHA0F00R05	U805
1510	1	PLIGHT(JIANGSU)	415-55180-0007	J612
1520	1	MPTS	415-55180-0001	SH1
1530	1	MPTS	415-55180-0002	SH2
1540	1	MPTS	415-55180-0006	SH3
1550	1	MPTS	415-55180-0004	SH4
1560	1	MPTS	415-55180-0003	SH5
1760	0	SEN JU	M705-GRN360-K2-VZH	